

**MINISTRY OF EDUCATION AND TRAINING
NGUYEN TAT THANH UNIVERSITY
FACULTY OF INFORMATION TECHNOLOGY**



GRADUATION THESIS

Title:

**BUILDING WEBSITE PROVIDING CHALLENGES
AND RECRUITMENT SUPPORT FOR DEVELOPER
FRONT-ICE**

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Thank you very much.

HCM City, December, 12, 2024

Student name

(Sign and write full name)

NGUYEN TRONG PHUC

INTRODUCTION

The Platform Providing Challenges and Recruitment support for Developer – Front is a robust and user-friendly application designed to create a seamless experience for developers in accessing coding challenges and recruitment opportunities. It combines modern user interface design with powerful backend features to cater to a global community of programmers and recruiters.

With the growing trend of online learning and programming, this website will cater to the needs of a vast community of programmers, making the process of finding and accessing coding resources seamless and convenient. The objective is to create a user-friendly platform that enhances the overall learning experience and supports the professional growth of programmers. Helping administrators easily in decentralizing management, managing customers and their orders.

The platform offers developers the ability to participate in technical challenges, showcase their skills, and gain visibility among top recruiters. Recruiters, in turn, can easily manage challenges, evaluate submissions, and identify potential candidates through a streamlined recruitment pipeline.

This website leverages contemporary technologies to offer real-time interactions, detailed analytics, and a personalized experience for every user, significantly enhancing the value proposition for both developers and recruiters. I propose the project “**BUILDING WEBSITE PROVIDING CHALLENGES AND RECRUITMENT SUPPORT FOR DEVELOPER FRONT-ICE**”

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 - Lên kế hoạch quản lý dự án.
 - Thiết kế UI/UX.
 - Xây dựng cơ sở dữ liệu và API cho Back-end (Spring Boot)
 - Thiết kế Front-end tạo giao diện người dùng (ReactJs).
 - Tích hợp upload file trên Google Drive lưu trữ tài nguyên người dùng.
 - Thiết kế thử thách cho lập trình viên.
 - Kiểm thử chức năng và đánh giá.
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(Ký và ghi rõ họ tên)

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- Hệ thống có tiềm năng phát triển nếu bổ sung các tính năng để kiểm tra tự động các thiết kế.

TPHCM, Ngày tháng năm 2024

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3. Kết luận

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TPHCM, Ngày tháng năm 2024

Giáo viên hướng dẫn
(Ký tên, ghi rõ họ tên)

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CHAPTER 1: OVERVIEW

1.1 Introduction

1.1.1 Overview Front-Ice project

Front-Ice was designed to offer developers an interactive platform where they can participate in **coding challenges** tailored to varying levels of expertise, showcase their solutions, and connect with potential recruiters. By integrating features such as personalized user profiles, challenge submission, and a streamlined recruitment process, Front-Ice serves as a comprehensive solution for enhancing developer skills while facilitating meaningful career opportunities.

The Front-Ice provides users with robust **profile management tools**, enabling developers to present their skills, experience, and completed challenges effectively. Recruiters benefit from tailored features, including **real-time notifications**, a curated **talent marketplace**, and **filter options** to identify candidates best suited for their requirements.

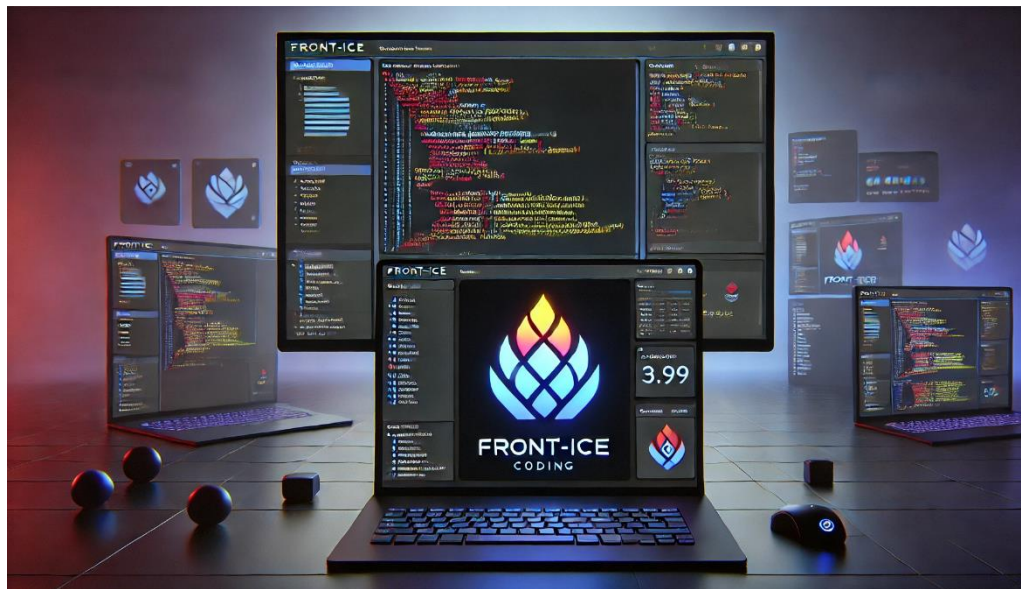


Figure 1 - 1 Front-Ice Overview

The platform further bridges the gap between learning and application by fostering a community-driven environment. Developers gain access to community solutions, **technical feedback**, and detailed insights into their performance through solution evaluations. Front-Ice's end-to-end challenge management and

user engagement capabilities make it a valuable resource for career growth, networking, and skill refinement.

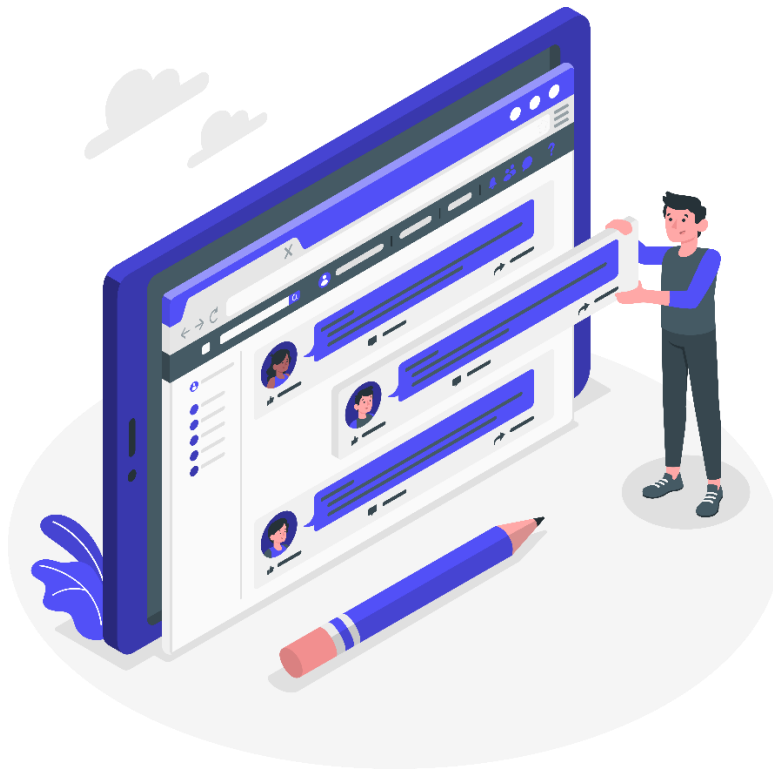


Figure 1 - 2 Comunity in Front-Ice

The Front-Ice platform empowers developers to build meaningful connections through its interactive communication features. Developers can engage with others by utilizing **commenting** and **reaction functions**, fostering collaboration and a sense of community. This two-way interaction allows developers to receive feedback on their submissions, learn from their peers, and showcase their expertise to **recruiters** and **HR professionals**. These features are designed to promote visibility and strengthen professional relationships, providing developers with direct pathways to career growth. By creating a collaborative environment where users can discuss solutions, share insights, and celebrate achievements, Front-Ice ensures that both personal development and networking opportunities are seamlessly integrated into the system.

This focus on connectivity not only enhances the learning experience but also positions developers for long-term success in their careers. The ability to network

effectively with industry professionals and fellow developers sets Front-End apart as a platform tailored for both skill-building and professional advancement.

1.1.2 Market Differentiators

According to the analysis of [Front-end Mentor](https://www.frontendmentor.io/challenges)¹, while the platform excels in providing coding challenges for skill development, it exhibits certain limitations that can hinder a more comprehensive developer experience and professional growth. These limitations primarily stem from its lack of integrated recruitment features and broader functionality

- **No Recruitment Options:** Front-end Mentor does not have features to help developers find jobs. For example, even if a developer finishes a task, they cannot use the platform to show their skills to employers. This makes it harder for developers who are looking for jobs.
- **Hard to Make Connections:** On Front-end Mentor, developers cannot easily connect with others. They must use other websites like LinkedIn or GitHub to share their work. This takes more time and effort, which can make it less interesting for developers to use.
- **Small Range of Tasks:** The tasks on Front-end Mentor are mostly about building websites with front-end skills like HTML and CSS. There are no tasks for backend or other kinds of programming. This means developers cannot practice real-world jobs that need more skills.
- **Not Enough Feedback:** Feedback on Front-end Mentor comes from other users. This feedback is not always detailed or useful. For example, developers might get simple comments but not advice to make their work better. This can slow down their learning.

¹ Front-end Mentor. (n.d.). **Improve your front-end coding skills by building real projects**. Available at: <https://www.frontendmentor.io/challenges> [Accessed 27 Nov. 2024].

1.1.3 Problems and Challenges of Front-Ice Project

The integration of coding challenge platforms into university curricula remains limited. A systematic review published in the "Education and Information Technologies" journal analyzed 47 publications on technology acceptance in education. The findings suggest that e-learning and Learning Management Systems are more commonly adopted than specialized coding challenge platforms, indicating a gap in the integration of such tools in higher education [1].

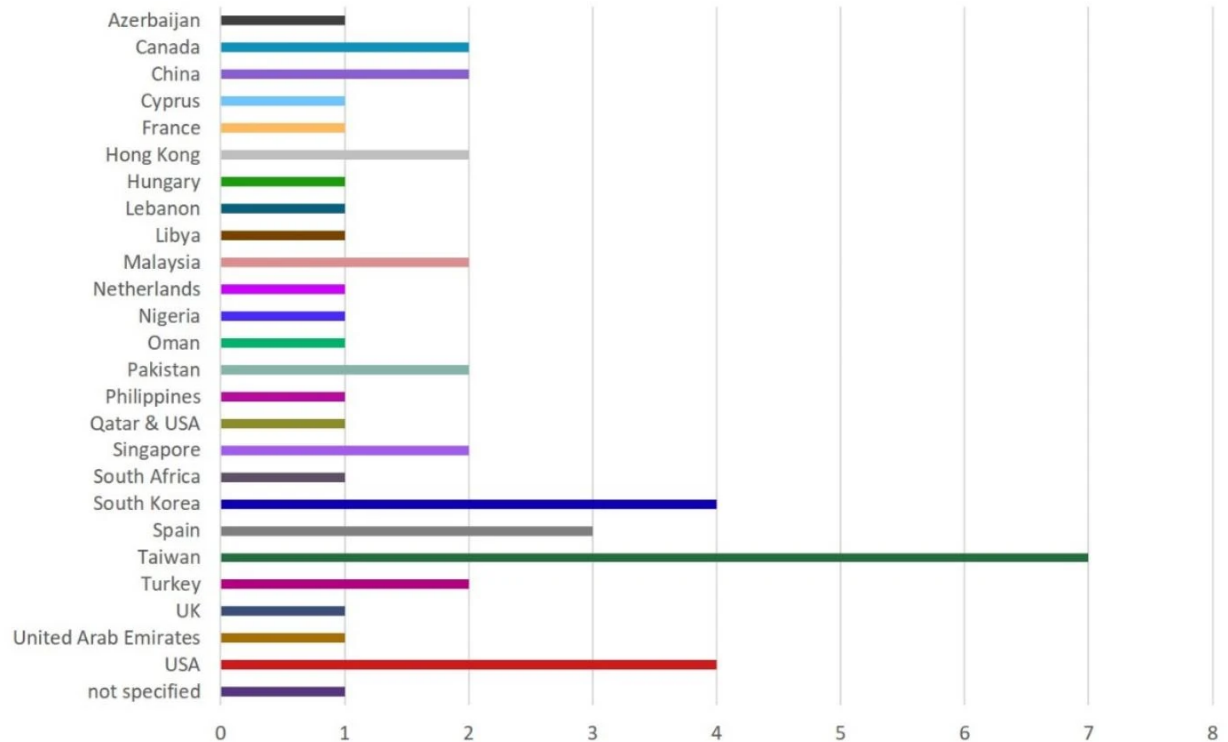


Figure 1 - 3 Distribution of education technology [1]

The visibility of coding challenge platforms like Front-Ice is hindered by the lack of widespread recognition of their educational value. A study by EdTech Hub mapped digital learning platforms across 184 countries, indicating varied adoption rates of such technologies in education. The research highlights that while some countries have embraced digital learning platforms, others lag due to infrastructural and policy challenges.

1.2 Aim of Project

The aim of this project is to create a platform that brings together developers and employers. By practicing challenges and applying the basic principles of programming, developers will improve their skills. Meanwhile, companies will be able to find talented people quickly, saving both time and money during the hiring process. The platform also helps developers follow important coding standards to ensure they write clean and efficient code.

- **Help Developers Practice Basic Programming Skills:** The platform will allow developers to practice the basic principles of programming through various challenges. It will help them improve their skills in real-life situations, making it easier for them to become better at coding.
- **Enable Employers to Find and Filter Potential Candidates:** The platform will give recruiters the tools to find more candidates for their job openings. Employers can use the challenges created by companies to filter and identify talented developers who meet their needs. This makes the hiring process more effective.
- **Optimize the Recruitment Process for Companies:** Companies and businesses can improve their hiring process by using Front-Ice. The platform makes it easier for them to test and evaluate candidates' skills before making any hiring decisions. This helps businesses save time and find the best people for the job.
- **Help Developers Follow Programming Standards:** The platform will help developers learn and apply coding standards. By completing challenges, developers will practice following the rules of good programming, making their code cleaner, more efficient, and professional.

CHAPTER 2: TECHNOLOGIES

2.1 Java Programming Language (Back-end)

2.1.1 Java 8

Java 8 is a revolutionary release of the world's #1 development platform. It includes a huge upgrade to the Java programming model and a coordinated evolution of the JVM, Java language, and libraries. Java 8 includes features for productivity, ease of use, improved polyglot programming, security and improved performance. Welcome to the latest iteration of the largest, open, standards-based, community-driven platform.

On March 18, 2014, Java 8 was released. Even though it was a long time ago, Java 8 is still used in many applications because Java 8 got a significant update with several new features. Let's take a look at all of Java 8 features using sample code.

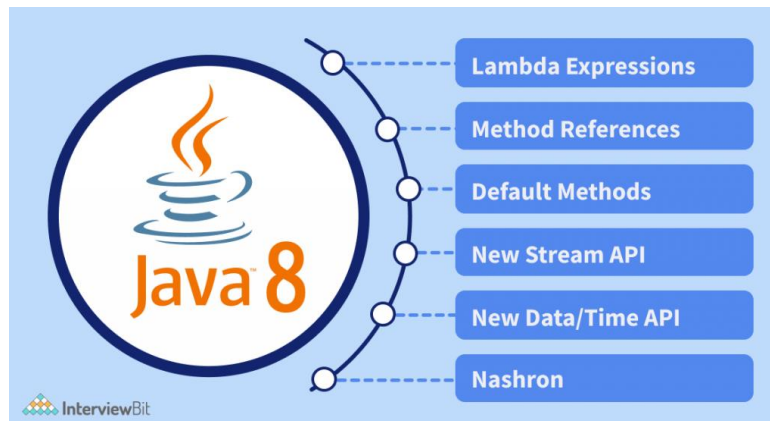


Figure 2 - 1 Java 8 (InterviewBit)

Java ME 8 is now Generally Available-Java ME 8 is an ideal platform for intelligent devices and an optimal foundation for developers to build new services for the IoT.

2.1.2 Spring Framework (Back-end)

Spring Boot is an automated version of the traditional spring framework, simplifying and automating the process. Spring Boot is built on top of the Spring framework and is a layered structure in which every layer communicates with each other layer(from top to bottom in hierarchical order). The Spring Boot framework documentation provides the following definition. Spring Boot makes it possible to create high-quality, production-grade Spring applications that are simple to deploy and run.



Figure 2 - 2 Spring Boot Framework

Spring boot is designed to remove XML and annotations-based configuration settings from the application. Spring boot provides the following advantages, including opinionated (options to later change the configuration), convention over configuration, stand-alone, and production-ready: It provides stand-alone, production-ready, and convention over configuration options.

Spring Boot is a Java framework. It is built on the Spring framework and takes an apertures, opinionated, boilerplate-free approach to Spring applications. Spring Boot, however, does not require you to write excessive boilerplate code. The majority of Spring Boot applications, however, require minimal Spring configuration.

A Java developer who uses the auto-configuration feature can automatically configure his or her Spring application using jar dependencies, thereby making it very convenient for them to develop applications.

2.1.3 Maven

Maven is a popular open-source build tool developed by the Apache Group to build, publish, and deploy several projects at once for better [project management](#). The tool provides allows developers to build and document the lifecycle framework.

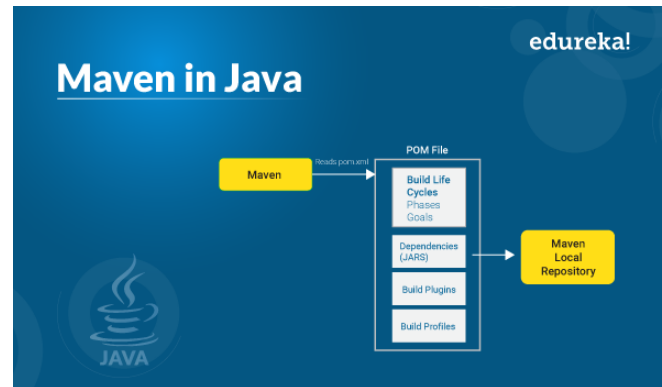


Figure 2 - 3 Maven in Java

Maven is written in [Java](#) and is used to build projects written in [C#](#), Scala, [Ruby](#), etc. Based on the Project Object Model (POM), this tool has made the lives of [Java developers](#) easier while developing reports, checks build and testing automation setups

- Builds
- Documentation
- Dependencies
- Reports
- SCMs
- Distribution
- Releases
- Mailing list

2.1.4 SQL

Structured query language (SQL) is a programming language for storing and processing information in a relational database. A relational database stores information in tabular form, with rows and columns representing different data attributes and the various relationships between the data values. You can use SQL statements to store, update, remove, search, and retrieve information from the database. You can also use SQL to maintain and optimize database performance.



Figure 2 - 4 SQL (Strutured Query Language)

Structured query language (SQL) is a popular query language that is frequently used in all types of applications. Data analysts and developers learn and use SQL because it integrates well with different programming languages. For example, they can embed SQL queries with the Java programming language to build high-performing data processing applications with major SQL database systems such as Oracle or MS SQL Server.

2.2 Html – Css – JavaScript and Framework – Library (Front - End)

2.2.1 What is HTML ?

HTML stands for **HyperText Markup Language**. It is a standard markup language for web page creation. It allows the creation and structure of sections, paragraphs, and links using HTML elements (the building blocks of a web page) such as tags and attributes.



Figure 2 - 5 HTML

It's also worth noting that HTML is not considered a programming language as it can't create dynamic functionality, although it is now considered an official web standard. The [World Wide Web Consortium \(W3C\)](https://www.w3.org/) maintains and develops HTML specifications, along with providing regular updates.

This article will go over the basics of HTML, including how it works, its pros and cons, and how it relates to CSS and JavaScript.

2.2.2 What is CSS ?

[CSS](#) (Cascading Style Sheets) allows you to create great-looking web pages, but how does it work under the hood? This article explains what CSS is with a simple syntax example and also covers some key terms about the language.



Figure 2 - 6 CSS

In the middle of the shield, there is a bold white stylized “3” placed. It has its right part straight, which makes it look like the Russian letter “Э”. The upper horizontal bar of the symbol is also a straight line, as well as the middle one, but the bottom bar repeats the triangular part of the shield, adding sharpness and confidence to the whole image.

2.2.3 What is JavaScript ?

JavaScript is a scripting or programming language that allows you to implement complex features on web pages — every time a web page does more than just sit there and display static information for you to look at — displaying timely content updates, interactive maps, animated 2D/3D graphics, scrolling video jukeboxes, etc. — you can bet that JavaScript is probably involved. It is the third layer of the layer cake of standard web technologies, two of which ([HTML](#) and [CSS](#)) we have covered in much more detail in other parts of the Learning Area.



Figure 2 - 7 JavaScript

What is even more exciting however is the functionality built on top of the client-side JavaScript language. So-called **Application Programming Interfaces (APIs)** provide you with extra superpowers to use in your JavaScript code.

2.2.4 What is Bootstrap ?

Bootstrap makes responsive web design a reality. It makes it possible for a web page or app to detect the visitor's screen size and orientation and automatically adapt the display accordingly. The mobile-first approach assumes [smartphones](#), [tablets](#) and task-specific mobile apps are employees' primary tools for getting work done. Bootstrap addresses the requirements of those technologies in design and includes [UI](#) components, layouts, JavaScript tools and the implementation framework. The software is available precompiled or as [source code](#).



Figure 2 - 8 Bootstrap

Mark Otto and Jacob Thornton developed Bootstrap at Twitter to improve the consistency of tools used on the site and to reduce maintenance. The software was formerly known as Twitter Blueprint and is sometimes referred to as Twitter Bootstrap.

2.2.5 What is React Js ?

The React.js framework is an open-source JavaScript framework and library developed by Facebook. It's used for building interactive user interfaces and web applications quickly and efficiently with significantly less code than you would with vanilla JavaScript.



Figure 2 - 9 ReactJs

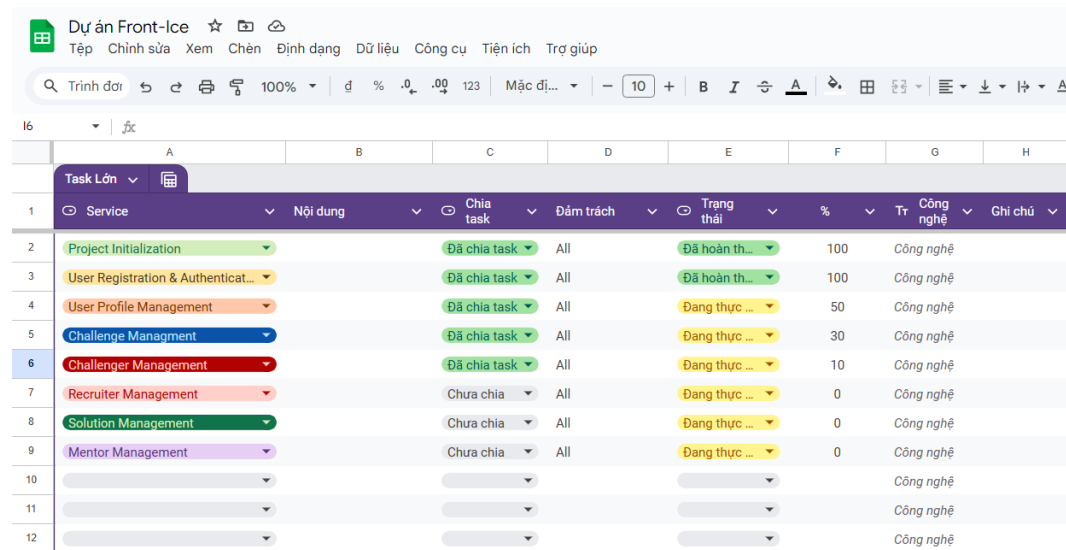
In React, you develop your applications by creating reusable components that you can think of as independent Lego blocks. These components are individual pieces of a final interface, which, when assembled, form the application's entire user interface.

React's primary role in an application is to handle the view layer of that application just like the V in a model-view-controller (MVC) pattern by providing the best and most efficient rendering execution. Rather than dealing with the whole user interface as a single unit, React.js encourages developers to separate these complex UIs into individual reusable components that form the building blocks of the whole UI. In doing so, the ReactJS framework combines the speed and efficiency of JavaScript with a more efficient method of manipulating the DOM to render web pages faster and create highly dynamic and responsive web applications.

2.3 Tools use to plan and manage project

2.3.1 Microsoft Excel

Microsoft Excel is the industry leading spreadsheet software program, a powerful data visualization and analysis tool.



The screenshot shows a Microsoft Excel spreadsheet titled 'Dự án Front-Ice'. The spreadsheet contains a table with the following data:

Task Lớn	Nội dung	Chia task	Đảm trách	Trạng thái	%	Tr Công nghệ	Ghi chú
Service	Project Initialization	Đã chia task	All	Đã hoàn th...	100	Công nghệ	
	User Registration & Authenticat...	Đã chia task	All	Đã hoàn th...	100	Công nghệ	
	User Profile Management	Đã chia task	All	Đang thực ...	50	Công nghệ	
	Challenge Management	Đã chia task	All	Đang thực ...	30	Công nghệ	
	Challenger Management	Đã chia task	All	Đang thực ...	10	Công nghệ	
	Recruiter Management	Chưa chia	All	Đang thực ...	0	Công nghệ	
	Solution Management	Chưa chia	All	Đang thực ...	0	Công nghệ	
	Mentor Management	Chưa chia	All	Đang thực ...	0	Công nghệ	
						Công nghệ	
						Công nghệ	
						Công nghệ	

Figure 2 - 10 Microsoft Excel

Excel is a powerful spreadsheet program developed by Microsoft, part of the Office suite. It allows users to organize, format, and analyze data using various tools, including charts, functions, and pivot tables. Excel is widely used for tasks such as data visualization, financial analysis, and statistical reporting.

2.3.2 StarUML

Open-source modeling software that is used in Unified Modelling Language where system and software modeling is supported with the UML concept to generate code for different languages by providing different types of UML diagrams to develop a fast and flexible UML platform so that requirements and design are given before starting the project where it is called as a big upfront design approach for the users who require the software designs to improve their project and hence to support their UML design diagrams in the system. In this topic, we are going to learn about StarUML.

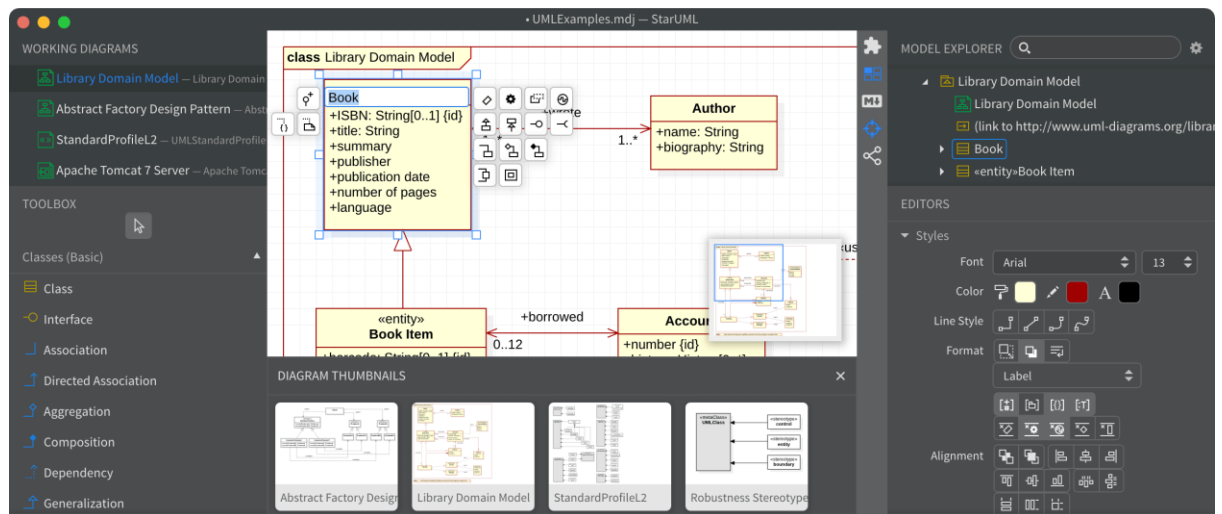


Figure 2 - 11 Start UML

Software architecture is a critical process that can reach 10 years or more into the future. The intention of the OMG (Object Management Group) is to use MDA (Model Driven Architecture) technology to create platform independent models and allow automatic acquisition of platform dependent models or codes from platform independent models. StarUML™ truly complies with UML 1.4 standards, UML 2.0 notation and provides the UML Profile concept, allowing creation of platform independent models. Users can easily obtain their end products through simple template document.

2.3.3 EdrawMax

EdrawMax Pro is an all-in-one diagram application that allows you to create flow charts, mind maps, org charts, network diagrams and floor plans with a rich gallery of examples and templates. EdrawMax enables students, teachers and business professionals to reliably create and publish many kinds of diagrams to represent any ideas. It's an all-in-one graphics software that makes it simple to create professional looking flowcharts, network diagrams, organizational charts, business presentations, building plans, mind maps, fashion designs, UML diagrams, workflows, program structures, web design diagrams, electrical engineering diagrams, directional maps, database diagrams and more.

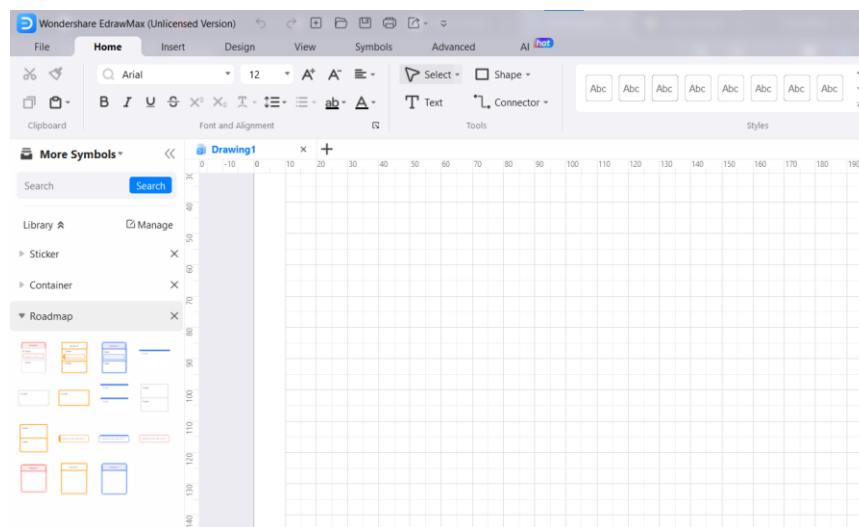


Figure 2 - 12 EdrawMax

EdrawMax Features

- Diagram Types - More than 280 types of diagrams to meet all your needs for business office, strategic analysis, human resources, engineering management, and more.
- Easy to Get Started - In-depth analysis of various office drawing tools, EdrawMax simplifies the user operations and improves the intelligence of drawing.

- Robust File Compatibility - Excellent file compatibility supports users to import and export drawings to a variety of file formats, such as Microsoft Visio, PDF, Microsoft Office, images, etc.
- Cross-Platform Supported - Run all features and templates identically on different platforms, including Microsoft Windows, Apple macOS, Linux, and EdrawMax Online.
- Create More Effortlessly - EdrawMax has 2000 built-in professional templates and 26,000 drag-and-drop symbols waiting for you to get started in no time.
- Visualize More Intuitively - EdrawMax supports the customization of every element to make your ideas and data more intuitive and make your presentation more vivid.
- Communicate More Effectively - EdrawMax keeps everyone on the same page everywhere and at all times. Share and collaborate on your work in any file format and on any channel you want.

2.4 Tools use to Coding Project

2.4.1 IntelliJ IDEA

IntelliJ IDEA is an Integrated Development Environment (IDE) for JVM languages designed to maximize developer productivity. It does the routine and repetitive tasks for you by providing clever code completion, static code analysis, and refactorings, and lets you focus on the bright side of software development, making it not only productive but also an enjoyable experience.



Figure 2 - 13 IntelliJ IDEA

Development of modern applications involves using multiple languages, tools, frameworks, and technologies. IntelliJ IDEA is designed as an IDE for JVM languages but numerous [plugins](#) can extend it to provide a polyglot experience.[15]

IntelliJ IDEA comes in two editions:

- IntelliJ IDEA Ultimate: a full-featured Java and Kotlin IDE. It includes all the features of the Community edition, plus adds support for [languages](#) that other IntelliJ platform-based IDEs focus on, as well as support for a variety of server-side and front-end frameworks, [application servers](#), integration with [database](#) and profiling tools, and more.[15]
- IntelliJ IDEA Community Edition: a free IDE built on open-source code that provides essential features for Java and Kotlin enthusiasts.

See the [IntelliJ IDEA editions comparison matrix](#).

2.4.2 Postman

Postman is one of the most popular software testing tools which is used for API testing. With the help of this tool, developers can easily create, test, share, and document APIs.



Figure 2 - 14 POSTMAN

This tutorial will help in understanding why Postman is so famous and what makes it unique when compared to other API testing tools. All the examples in this tutorial are tested and can be imported in Postman.

- Postman is a standalone software testing API (Application Programming Interface) platform to build, test, design, modify, and document APIs. It is a simple Graphic User Interface for sending and viewing HTTP requests and responses.
- While using Postman, for testing purposes, one doesn't need to write any HTTP client network code. Instead, we build test suites called collections and let Postman interact with the API.
- In this tool, nearly any functionality that any developer may need is embedded. This tool has the ability to make various types of HTTP requests like GET, POST, PUT, PATCH, and convert the API to code for languages like JavaScript and Python.

2.4.3 Dbeaver Comunity

DBeaver Community is a free cross-platform database tool for developers, database administrators, analysts, and everyone working with data. It supports all popular SQL databases like MySQL, MariaDB, PostgreSQL, SQLite, Apache Family, and more.



Figure 2 - 15 Dbeaver

2.4.4 Visual studio

Visual Studio is an Integrated Development Environment(IDE) developed by Microsoft to develop GUI(Graphical User Interface), console, Web applications, web apps, mobile apps, cloud, and web services, etc. With the help of this IDE, you can create managed code as well as native code. It uses the various platforms of Microsoft software development software like Windows store, Microsoft Silverlight, and Windows API, etc. It is not a language-specific IDE as you can use this to write code in C#, C++, VB(Visual Basic), Python, JavaScript, and many more languages.



Figure 2 - 16 Visual Studio Code

It provides support for 36 different programming languages. It is available for Windows as well as for macOS. Evolution of Visual Studio: The first version of VS(Visual Studio) was released in 1997, named as Visual Studio 97 having version number 5.0. The latest version of Visual Studio is 15.0 which was released on March 7, 2017. It is also termed as Visual Studio 2017. The supported .Net Framework Versions in latest Visual Studio is 3.5 to 4.7. Java was supported in old versions of Visual Studio but in the latest version doesn't provide any support for Java language.

2.4.5 JWT.oi

JSON Web Token (JWT) is an open standard ([RFC 7519](#)) that defines a compact and self-contained way for securely transmitting information between parties as a JSON object. This information can be verified and trusted because it is digitally signed. JWTs can be signed using a secret (with the **HMAC** algorithm) or a public/private key pair using **RSA** or **ECDSA**.

Although JWTs can be encrypted to also provide secrecy between parties, we will focus on signed tokens. Signed tokens can verify the integrity of the claims contained within it, while encrypted tokens hide those claims from other parties. When tokens are signed using public/private key pairs, the signature also certifies that only the party holding the private key is the one that signed it.

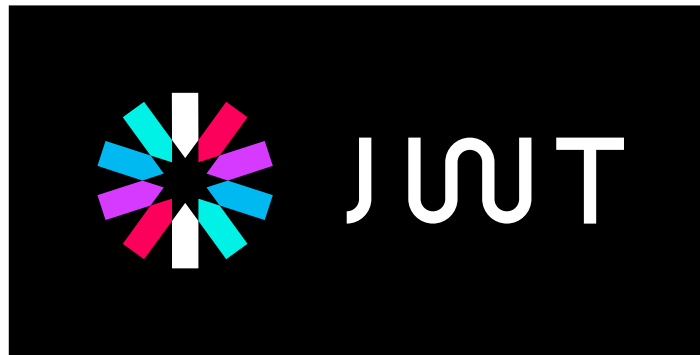


Figure 2 - 17 JSON Web Tokens

Here are some scenarios where JSON Web Tokens are useful:

- **Authorization:** This is the most common scenario for using JWT. Once the user is logged in, each subsequent request will include the JWT, allowing the user to access routes, services, and resources that are permitted with that token. Single Sign On is a feature that widely uses JWT nowadays, because of its small overhead and its ability to be easily used across different domains.

CHAPTER 3: ANALYZING SYSTEM

3.1 System architecture design

The **Front-Ice** platform, which provides both **coding challenges** and **recruitment support** for developers, is designed around three core user roles: **Challenger**, **Recruiter**, and **Challenge Manager**. The **end-user** is the **Challenger**, who interacts with the platform to complete challenges and gain recognition.

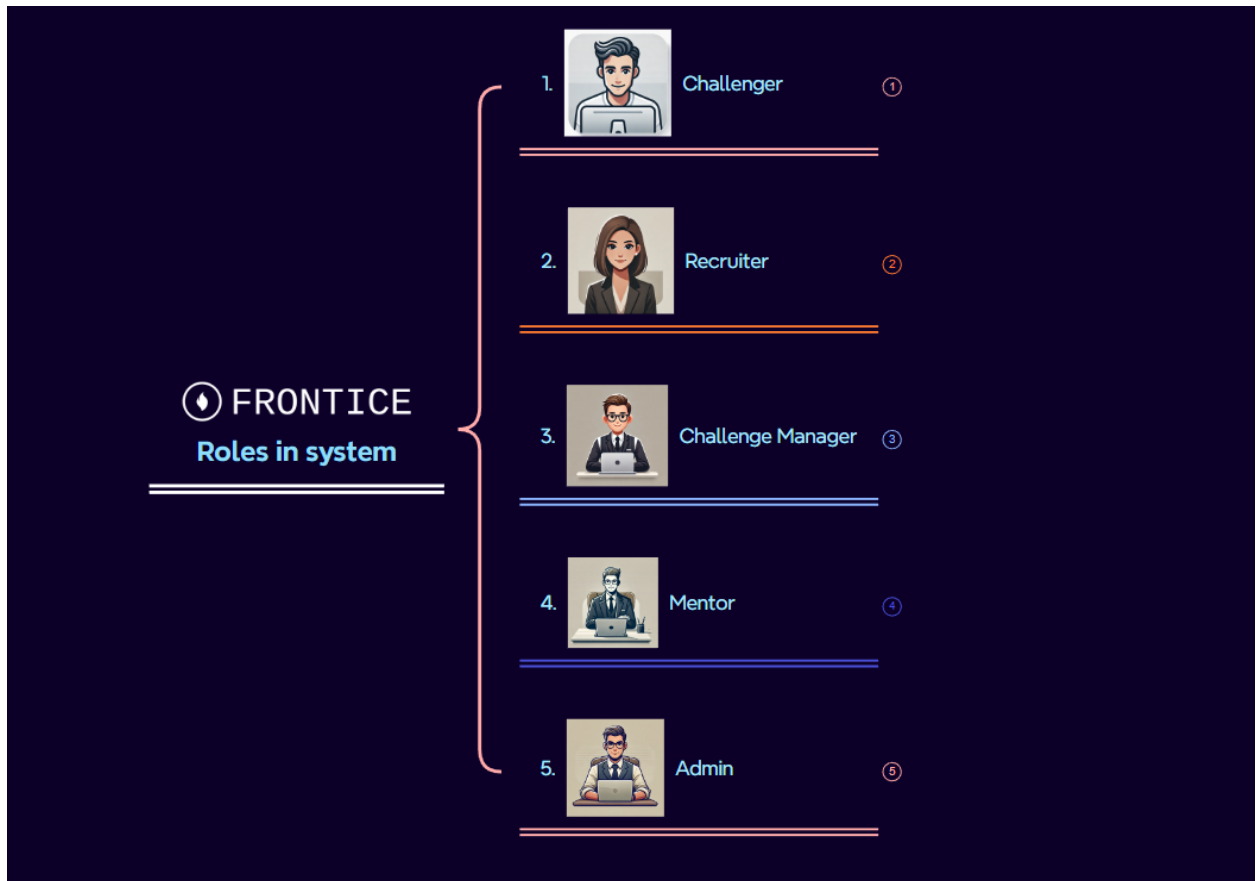


Figure 3 - 1 Role in whole System

Challenger:

- **Definition:** A **Challenger** is a user who logs into the platform and participates in the challenges available on the site. After completing a challenge, they earn points based on the challenge's difficulty level. Users are ranked according to the total points they accumulate by completing

challenges. These rankings are visible on the platform, motivating challengers to improve their performance.

Key Features for Challenger:

- **User Registration & Authentication:** Challengers must sign up by providing their email and password. If they do not have an account, they will be required to register. Authentication is done via email verification to ensure the user's identity.
- **Challenge Management:** After logging in, challengers can view **Detail Challenges**, which include important information such as the **Challenge Title, Brief Description, Assets (e.g., Figma files)**, and relevant images. Once the challenger reviews the details, they can start solving the challenges.
- **Solution Management:** Upon completing a challenge, the **Solution Management** feature allows challengers to submit their solutions and receive feedback. Other challengers can also view and interact with the submitted solutions by commenting and reacting to them. The system provides a percentage of correctness after submission, indicating how accurate their solutions are.
- **User Profile Management:** Challengers can update and manage their profile information, including personal details and skills, through the **User Profile Management** feature.
- **Notification Management:** The platform provides real-time notifications to challengers about important activities, such as challenge updates or feedback on their submissions.

Recruiter:

- **Definition:** A **Recruiter** is a user who has the ability to create challenges or purchase premium challenges within the system. Based on the outcomes of challenges, recruiters can identify and contact candidates whose skills match their job requirements.

Key Features for Recruiter:

- **Challenge Creation & Management:** Recruiters can create new challenges tailored to the skills they are looking for, or they can select from existing challenges in the system.
- **Candidate Evaluation:** Recruiters have the ability to assess challengers based on their performance in these challenges, using the points and rankings to identify potential candidates for recruitment.
- **Purchase Premium Challenges:** Recruiters have the option to purchase **premium challenges** that are more advanced or specialized. These challenges offer a deeper level of testing and are designed to assess higher levels of skill, making them ideal for hiring more experienced developers.

Challenge Manager:

- **Definition:** A **Challenge Manager** is a user responsible for overseeing and managing all activities related to challenges, challengers, and recruiters on the platform. The Challenge Manager ensures that challenges are running smoothly, and can moderate the interactions between challengers and recruiters.

Key Features for Challenge Manager:

- **Challenge Oversight:** The Challenge Manager manages challenge creation, ensures the appropriate challenges are available, and monitors ongoing challenges.
- **User Moderation:** The Challenge Manager has the authority to oversee and manage the activities of both challengers and recruiters. They can intervene in cases of inappropriate behavior and moderate feedback or comments in the system.

- **View Candidate Information:** Challenge Manager can access detailed information about candidates who have participated in challenges. This includes personal profiles, skills, achievements, and past challenge results. This helps recruiters get a complete view of a candidate's capabilities and suitability for a role.
- **Set Challenge Deadlines:** Challenge Manager can set deadlines for challenges they create or manage. This ensures that challengers complete tasks within a specific timeframe, mirroring real-world deadlines that may be required in professional settings.

Mentor:

- **Definition:** A **Mentor** is a user responsible for answer for premium account when they have a question or comment in solution. In this role Mentor must response any question for challenger. And report any solution have fraudulent and notify for **Challenge Manager** and **Admin**

Key Features for Mentor:

- **Solution Oversight:** The Mentor will be showned all of solution of challenger have account premium and the mentor responsible to response the question of challenger
- **Management Profile Mentor:** The mentor can update password and update information in its account and that is convinient for mentor to manage profile

Admin(Root):

- **Definition:** A **Admin** has full power in system for manage everythings in system from manage package member, account, challenge, solutions, task (mission), recruiter.....

Key Features for Admin:

- **Challenge Oversight:** The Admin manages challenge creation, ensures the appropriate challenges are available, and monitors ongoing challenges.
- **User Moderation:** The Admin has the authority to oversee and manage the activities of both challengers and recruiters. They can intervene in cases of inappropriate behavior and moderate feedback or comments in the system.
- **Set Challenge Deadlines:** Admin can set deadlines for challenges they create or manage. This ensures that challengers complete tasks within a specific timeframe, mirroring real-world deadlines that may be required in professional settings.
- **Purchase Premium Challenges:** Admin have the option to purchase **premium challenges** that are more advanced or specialized. These challenges offer a deeper level of testing and are designed to assess higher levels of skill, making them ideal for hiring more experienced developers.
- **Solution Oversight:** The Admin will be showned all of solution of challenger have account premium and the mentor responsible to response the question of challenger
- **Management Profile Mentor:** The admin can update password and update information in its account and that is convinient for admin to manage profile

3.2 Functional and Non-functional Requirements

3.2.1 Functional Requirements

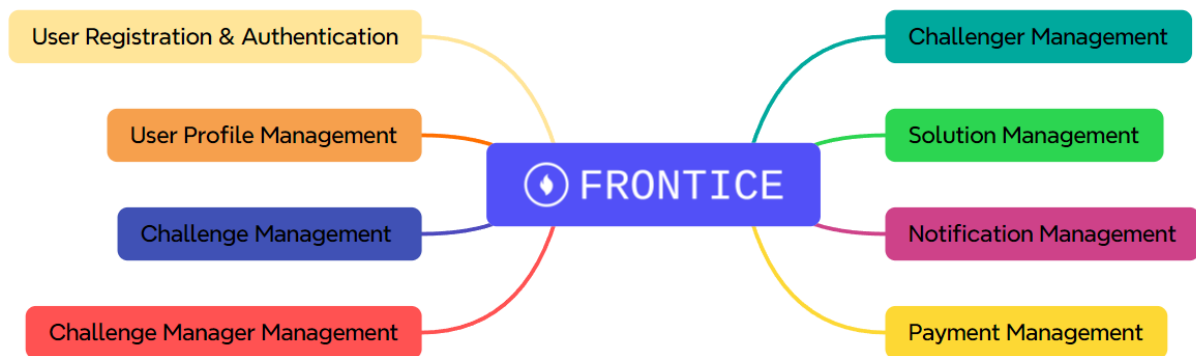


Figure 3 - 2 Functional Requirements

Functional Requirements(FRs) define the specific behaviors, features, and functions that the system must support. They outline what the system should do and describe the system's interaction with users, systems, and other external entities. These requirements are critical for guiding the development process, ensuring that the system meets the needs of stakeholders. Below are the key functional requirements for the system:

- **User Registration & Authentication:**
 - **Login:** Users (Challengers, Recruiters, Challenge Managers) must be able to log in securely using their registered email and password. The system should verify credentials and provide access based on correct authentication.
 - **Register:** New users must be able to register by providing their email address, password, and required personal details. The system must verify the email address to confirm the user's identity before account activation.
 - **Reset Password:** Users should be able to reset their password via email if they forget it. The system must provide a secure password reset process, including email verification.

- **Logout:** The system should allow users to log out, terminating the session securely.
- **User Profile Management:**
 - **User Profile:** Users (Challengers, Recruiters, Challenge Managers) must be able to view and edit their profile, including personal details such as name, email, contact information, and relevant skills. Challengers can also update their skill sets and achievements.
 - **Dashboard:** The system should display a personalized dashboard showing the user's key metrics, such as progress in challenges, total points, rankings, and notifications.
- **Challenge Management:**
 - **List Challenge:** The system must display a list of available challenges. The list should include challenge titles, brief descriptions, and difficulty levels.
 - **Detail Challenge:** Users (Challengers) must be able to view detailed information about a challenge, including the title, description, assets (e.g., Figma files), and other relevant information.
 - **Submit Challenge:** Challengers should be able to submit their solutions to a challenge after completing it. Submissions should be reviewed, and feedback should be provided.
 - **Challenge System:** The system must allow challenges to be created, categorized, and managed. Challenges can be public or premium, with specific requirements set by Recruiters or Challenge Managers.
 - **Recruiter of Challenge Storage:** Recruiters should be able to store challenges in the system, which they can assign to relevant candidates based on their skills.

- **Review challenge of recruiter:** Recruiters should be able to review submitted challenges, evaluate them based on predefined criteria, and leave feedback for challengers.
- **Store challenge:** Challenges should be stored securely within the system, allowing easy retrieval and reuse by recruiters or challengers.
- **Posted Challenge:** Recruiters must have the ability to post and manage their challenges on the platform for challengers to access and complete.
- **Challenge Manager Management:**
 - **Challenge Manager Profile:** Challenge Managers must have a personal profile displaying relevant information such as their role, responsibilities, and activities within the system.
 - **List Challenge Manager:** The system should allow users to view a list of all Challenge Managers and access their profiles for review or management purposes.
- **Challenger Management:**
 - **Challenger Profile:** Each challenger should have a profile displaying their personal information, performance in challenges, points, ranking, and achievements.
 - **List Challenger:** The system should allow Challenge Managers and Recruiters to view a list of challengers who have participated in challenges, along with key performance metrics such as points and rankings.
- **Solution Management:**
 - **List Solution:** After completing a challenge, challengers must be able to view a list of their submitted solutions and those submitted by others.

- **Detail Solution:** The system must allow challengers to view the details of a solution, including the correctness percentage, feedback, and comments.
- **Solutions System:** The system should support the management of solutions, allowing challengers to submit, edit, and view solutions.
- **Report Solutions:** Users (Challengers, Challenge Managers) must be able to report issues or concerns regarding solutions, such as incorrect content or plagiarism.
- **Notification Management:**
 - **Detail Notifications:** Users must be able to view detailed notifications for key actions such as challenge updates, new messages, or feedback on submitted solutions.
 - **List Notifications:** The system should display a list of all notifications for users, which may include activity updates, feedback on challenges, system alerts, and news.
- **Payment Management:**
 - **Detail Payment:** Challenger must be able to view detailed payment information for purchasing premium challenges or accessing premium features. This includes the payment status, amount, and transaction history.
 - **List Payment:** The system should provide a history of all payments made by the user, whether for challenge creation, premium challenges, or other services within the platform.

3.2.2 Non-functional Requirements

Non-functional requirements (NFRs) describe the qualities or characteristics of the system that do not relate directly to its specific functions but significantly impact its performance, stability, and usability. These requirements ensure the system not only operates correctly but also performs efficiently, securely, and is easy to maintain. Below are the key non-functional requirements for the system:

3.2.2.1 Performance

- **Response Time:** The system must provide a response time of no more than 2 seconds for 95% of user interactions (e.g., logging in, accessing challenges, viewing solutions).
- **Scalability:** The system must support up to 10,000 concurrent users without performance degradation. It should be able to scale horizontally by adding additional servers to accommodate growth in users, challenges, and transactions.
- **System Throughput:** The system should handle up to 1000 transactions per minute, including challenge submissions, payments, and profile updates, with no slowdown in processing time.
- **Peak Load:** The system must be able to handle peak loads, particularly during challenge deadlines or major events, with at least 30% extra capacity for sudden traffic spikes.

3.2.2.2 Reliability

- **Uptime:** The system must ensure a minimum of 99.9% uptime, meaning no more than 8 hours of downtime per year.
- **Disaster Recovery:** The system should have a Recovery Time Objective (RTO) of 4 hours, meaning that in the event of a failure, the system should be able to recover within this time frame. The data should be restored to a consistent state within 24 hours.

- **Fault Tolerance:** The system should be designed to continue functioning in the event of hardware or software failures, ensuring that critical services (e.g., user authentication, challenge submission) remain operational.
- **Backup and Redundancy:** Data should be backed up daily, and multiple copies of key services and databases should be maintained for redundancy.

3.2.2.3 Security

- **Authentication and Authorization:** The system must support strong **authentication mechanisms** such as password-based login, with optional multi-factor authentication (MFA) for added security. User access levels (e.g., admin, recruiter, challenger) should be restricted based on roles using **role-based access control (RBAC)**.
- **Data Encryption:** All sensitive data, including user credentials, personal information, and financial transactions, must be encrypted both **in transit (using SSL/TLS)** and **at rest (using AES-256)**.
- **Access Logs:** The system should maintain detailed logs of all user activities, including login attempts, content submissions, and profile updates. These logs should be stored securely and should be accessible to administrators for audit purposes.
- **Compliance:** The system must comply with relevant data protection regulations (e.g., GDPR, CCPA), ensuring that personal data is handled, processed, and stored securely and in compliance with legal requirements.

3.2.2.4 Usability

- **User Interface (UI):** The system must provide a **user-friendly interface** that is intuitive and easy to navigate, with clear instructions

for users on how to participate in challenges, view solutions, and manage their profiles.

- **Mobile Compatibility:** The platform must be fully responsive, ensuring that it is optimized for use on a variety of devices, including desktop computers, smartphones, and tablets. Mobile users should have the same access and functionality as desktop users.
- **Localization and Language Support:** The system must support **multilingual capabilities**, at a minimum offering **English and Vietnamese**. The user interface should adapt to different language settings and display culturally relevant content where applicable.
- **Accessibility:** The platform must adhere to basic accessibility standards (WCAG 2.1 Level AA) to ensure users with disabilities can fully interact with the system (e.g., using screen readers, keyboard navigation).

3.2.2.5 Maintainability

- **Code Quality:** The system's codebase should follow best practices for maintainability and clarity, with **clear documentation** for both developers and administrators. Code should be modular, reusable, and easy to update or extend.
- **Version Control:** All code and configurations must be managed via **version control** (e.g., Git), allowing developers to track changes, roll back versions, and collaborate efficiently.
- **Automated Testing:** The system should include a suite of **automated unit and integration tests** to ensure that new code does not break existing functionality and that updates are tested for reliability.
- **Error Logging and Monitoring:** The system must have error logging mechanisms in place, with real-time monitoring of system health.

Critical errors should trigger notifications to system administrators for prompt resolution.

3.2.2.6 Compatibility

- **Platform Compatibility:** The system must be compatible with major **web browsers** (e.g., Chrome, Firefox, Safari, Edge) and operating systems (Windows, macOS, Linux) without significant performance issues or functionality degradation.
- **Third-Party Integrations:** The system must support integration with third-party services such as **payment gateways** (e.g., PayPal, Stripe), **email services** (e.g., SendGrid, Amazon SES), and **analytics platforms** (e.g., Google Analytics, Mixpanel).
- **APIs:** The system should offer RESTful APIs for external integrations, such as enabling third-party tools to access challenge data, user profiles, or solutions.

3.2.2.7 Availability and Scalability

- **Horizontal Scaling:** The system must be able to scale horizontally, adding more application servers as the number of concurrent users and requests increases, ensuring minimal degradation in performance under high load.
- **Auto-scaling:** The system should support **auto-scaling** to manage traffic spikes, automatically adjusting server capacity to accommodate changing load without human intervention.
- **Load Balancing:** The system should employ load balancing to distribute incoming traffic evenly across available servers, ensuring high availability and optimal performance.

3.3 Functional Specification Document

3.3.1 Challenger



Figure 3 - 3 Challenger in System

Challenger: Will have access to modules such as **User Registration & Authentication module** for registering an account considered as a "Users" Additionally, there are modules like **User Profile Management module** for managing accounts, **Challenge module** for viewing the list of accounts and account details, **Solutions module** for viewing the list of solutions and their details, and **Recruiter module** for viewing recruiter information and notifications in the **Notification module**.

Table 3 - 1 Challenger FSD

Module	Function	Page	Detail in Page
	Login	Login	• Enter Email • Enter Password • Remember password • Forgot password • Login button • Social login button (Github) • Sign up button (If you don't have an account)"

User Registration & Authentication		Login Successful (Notification)	Automatically redirects to the homepage
		Login Failed	- Email does not exist/Incorrect - Incorrect password
	Register	Enter Registration Information	- Enter Email - Enter Password - Re-enter Password - Sign Up button - Login button (If you already have an account)
		Email Does Not Exist/Incorrect	<Error message description>
		Enter Correct OTP	<Notification that OTP has been sent>
		Incorrect OTP	<Error message description>
		Verify OTP	- Enter OTP - Verify button - Resend button
		OTP Expired	<Error message description>
		New Password Does Not Match/Correct Format	<Error message description>
		Registration Successful	Registration success message, automatically redirects to the login screen

	Reset Password	Enter Email	Email address for verification
		Email Does Not Exist/Incorrect	<Error message description>
		Enter OTP	<Notification that OTP has been sent to the email>
		Verify OTP	- Enter OTP - Verify - Resend OTP
		Create New Password	- Enter new password - Re-enter new password
		New Password Created Successfully	<Success message for updating and redirecting to the login screen>
		New Password Does Not Match	<Error message description>
	Logout	Confirm Logout	- Display confirmation dialog - Yes <Automatically log out> - No <Close dialog>"
	User Profile	Home Page	Section Statistics Challenge - Total number of challenges participated - Total number of submissions - Total number of unsubmitted challenges - Total number of recruit challenges participated List of Submitted Solutions - Id - Challenge Name - Interaction Volume - Score

User Profile Management			• Start Time • Submission Time • Level Name List of Challenges You Can Join (Same level as the challenger) • Contains challenges (Swipe format): found in the homepage Figma design List of Company Challenges: • Challenge: ○ Title: Responsive Food Web ○ Expiry Time: 6 days • “See more” button
		Profile	• Banner: Image • Avatar: Image • User: Name, rank, email, GitHub link, social link, bio • View CV button • Follow button • Solution • Solution System: List Solution • Solution company: List Solution company • Company Following: Number of following of company
		Setting Profile	Banner & Avatar • Banner: Image

			• Upload Image button • Avatar: Image • Upload Image button • Contact Information: Gmail, etc. • Add Information button Personal Information • First Name • Last Name • Email • Phone Number • Password • Change Password button • Address • GitHub URL (place this section on top to link GitHub) • Website Link • Bio (Description) Contact & Social Links • CodePen URL • Codewars URL • Dev.to URL • freeCodeCamp URL • GitLab URL • Hashnode URL • LinkedIn URL • Medium URL • StackOverflow URL • Twitch URL • Twitter URL • YouTube URL
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		Challenger Profile	• Banner: Image • Avatar: Image • User: Name, rank, email, GitHub link, social link, bio • View CV button • Follow button • Solution • Solution System: List Solution • Solution company: List Solution company • Company Following: Number of following of company
	List Challenge	List Challenger	Challenge List • Mark as new • Mark as free/premium • Challenge banner • Programming language • Challenge name • Score • Level name • Description • Practice button
		Filter Challenge	• Filter by category: Free, Free+/Premium • Level: Bronze, Silver • Difficulty: Easy, Medium, Hard • Technical: HTML, CSS, Java

Challenge	Detail Challenge	Detail Challenge (Tab Information)	Detail Challenge • Technical challenge • Challenge name • Level • Description • Message (if any) • Join Challenge button • Level name • Image • Responsive options Tab Information Challenge • Note • Assets provided • Message FQA (Frequently Asked Questions)
	Submit Challenge	Form Submit Challenge	Detail Challenge • Challenge Title • Technical Challenge • Level • Description • Message • Challenge Image Send Challenge Information (Below are the input fields to fill in) • Solution Title* • Repository URL* • Live Site URL* • Which tools, libraries, frameworks, or

			methodologies did you use for this project? • Private or Public • What are you most proud of, and what would you do differently next time? • What challenges did you encounter, and how did you overcome them? • What specific areas of your project would you like help with? • Submit Solution button
	List Solution	List Solution	• Action: Filter • Solution: ○ Solution Name ○ Technical ○ Owner ○ Level
		Filter Solution (Pop-up)	Display fields to filter such as: • Filter by challenge type (Free, Free+, Premium) • Filter by Level and difficulty of the challenge (Level + Hard, Easy, Medium, or All) • Filter by required Technicals (HTML, CSS, Javascript, etc.)

Solutions	Detail Solution	Detail Solution	Display challenge information that this solution submits: • Required Technicals of the challenge • Level and difficulty of the challenge • Challenge type (Free, Free+, Premium) • Challenge title • Short description of the challenge • Challenge score • Number of participants in the challenge • Number of challenge submissions • Button (displayed based on status): ○ If the viewer of this solution detail is the author of the solution, this button will be "view other solutions of this challenge" ○ If the viewer is not the author, the button will be "return to my solution."
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			• Preview image of the challenge • Tabs to view responsive preview images of the challenge View source code button View demo preview button Display solution information: • Submission time of the solution • Solution title • Technicals used for the solution • Action buttons for the solution (Like, Dislike, Report, Save) • Solution descriptions in format (title + description or only description) Display comment and feedback section: • Comments for challengers: ○ Avatar ○ Challenger's name ○ Comment time ○ Comment content ○ Interaction buttons for the comment (Like,
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			Dislike, Report, or Reply) - Feedback for mentors: - Mentor's avatar - Mentor's name - Feedback time - Feedback content - Interaction buttons for the feedback (Like, Dislike, Report, or Reply) Display solution author's information - To be added later.
Recruiter	Detail Recruiter	Detail Recruiter	Button Follow Recruiter

3.3.2 Recruiter



Figure 3 - 4 Recruiter In System

Recruiter: Will have access to log into the system and view Challenges like a normal User, and also have the ability to purchase challenges from the exclusive Recruiter Challenge list. The Recruiter can view information about job-seeking candidates within the system. Additionally, the Recruiter can create Challenges through the "Create Challenge Job for Candidate" function. The Recruiter can also view individual Challenger information and make payments through the Payment function in the system.

Table 3 - 2 Recruiter FSD

Module	Function	Page	Detail in Page
	Login	Login	• Enter Email • Enter Password • Remember password • Forgot password • Login button • Social login button (Github) • Sign up button (If you don't have an account)"

User Registration & Authentication		Login Successful (Notification)	Automatically redirects to the homepage
		Login Failed	- Email does not exist/Incorrect - Incorrect password
	Register	Enter Registration Information	- Enter Email - Enter Password - Re-enter Password - Sign Up button - Login button (If you already have an account)
		Email Does Not Exist/Incorrect	<Error message description>
		Enter Correct OTP	<Notification that OTP has been sent>
		Incorrect OTP	<Error message description>
		Verify OTP	- Enter OTP - Verify button - Resend button
		OTP Expired	<Error message description>
		New Password Does Not Match/Correct Format	<Error message description>
		Registration Successful	Registration success message, automatically redirects to the login screen

	Reset Password	Enter Email	Email address for verification
		Email Does Not Exist/Incorrect	<Error message description>
		Enter OTP	<Notification that OTP has been sent to the email>
		Verify OTP	- Enter OTP - Verify - Resend OTP
		Create New Password	- Enter new password - Re-enter new password
		New Password Created Successfully	<Success message for updating and redirecting to the login screen>
		New Password Does Not Match	<Error message description>
	Logout	Confirm Logout	- Display confirmation dialog - Yes <Automatically log out> - No <Close dialog>"
	User Profile	Home Page	Section Statistics Challenge - Total number of challenges participated - Total number of submissions - Total number of unsubmitted challenges - Total number of recruit challenges participated List of Submitted Solutions - Id - Challenge Name - Interaction Volume - Score

Recruiter Profile Management			• Start Time • Submission Time • Level Name List of Challenges You Can Join (Same level as the challenger) • Contains challenges (Swipe format): found in the homepage Figma design List of Company Challenges: • Challenge: ○ Title: Responsive Food Web ○ Expiry Time: 6 days • “See more” button
		Profile	• Banner: Image • Avatar: Image • User: Name, rank, email, GitHub link, social link, bio • View CV button • Follow button • Solution • Solution System: List Solution • Solution company: List Solution company • Company Following: Number of following of company
			Banner & Avatar • Banner: Image

		Setting Profile	• Upload Image button • Avatar: Image • Upload Image button • Contact Information: Gmail, etc. • Add Information button Personal Information • First Name • Last Name • Email • Phone Number • Password • Change Password button • Address • GitHub URL (place this section on top to link GitHub) • Website Link • Bio (Description) Contact & Social Links • CodePen URL • Codewars URL • Dev.to URL • freeCodeCamp URL • GitLab URL • Hashnode URL • LinkedIn URL • Medium URL • StackOverflow URL • Twitch URL • Twitter URL • YouTube URL
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		List Challenger Follower	• Banner: Image • Avatar: Image • User: Name, rank, email, GitHub link, social link, bio • View CV button • Follow button • Solution • Solution System: List Solution • Solution company: List Solution company • Company Following: Number of following of company
	List Challenge	List Challenger	Challenge List • Mark as new • Mark as free/premium • Challenge banner • Programming language • Challenge name • Score • Level name • Description • Practice button
		Filter Challenge	• Filter by category: Free, Free+/Premium • Level: Bronze, Silver • Difficulty: Easy, Medium, Hard • Technical: HTML, CSS, Java

Challenge Managment	Detail Challenge	Detail Challenge (Tab Information)	Detail Challenge • Technical challenge • Challenge name • Level • Description • Message (if any) • Join Challenge button • Level name • Image • Responsive options Tab Information Challenge • Note • Assets provided • Message FQA (Frequently Asked Questions)
	Store challenge	Danh sách challenge đã mua	Detail Challenge • Challenge Title • Technical Challenge • Level • Description • Message • Challenge Image Send Challenge Information (Below are the input fields to fill in) • Solution Title* • Repository URL* • Live Site URL* • Which tools, libraries, frameworks, or

			methodologies did you use for this project? • Private or Public • What are you most proud of, and what would you do differently next time? • What challenges did you encounter, and how did you overcome them? • What specific areas of your project would you like help with? • Submit Solution button
		Filter list challenge was bought	Filter • Level • Price • Technical
		List challenge was created	• Challenge list • Challenge name • Technology • Challenge banner • Number of participants • Number of submissions • Challenge duration • Status (Approved, Pending approval) • Option button (Edit, Hide, Delete) (All the above fields will be sorted)

		Filter list challenge was created	• Filter by technology • Filter by level
	Store Challenge	Create challenge(Pop-up)	• Challenge Name • Technical Challenge (Select Multiple) • Image Challenge (After uploading, the first image will be the main image) • Description • Level (select option: easy, medium, hard) • Duration • Upload Desktop design challenge (sample image for developers to work on) • Upload Tablet design challenge (sample image for developers to work on) • Upload Mobile design challenge (sample image for developers to work on) • Requirement (Text: requirements of the challenge) • Assets Provider (Text: content of the available assets) • Hint (Text: displays hints for challengers)

			• Upload File assets (Zip file with size limit) • Upload Figma File • Create or Cancel button
		Update Information challenge	• Challenge Name • Technical Challenge (Select Multiple) • Image Challenge (After uploading, the first image will be the main image) • Description • Level (select option: easy, medium, hard) • Duration • Upload Desktop design challenge (sample image for developers to work on) • Upload Tablet design challenge (sample image for developers to work on) • Upload Mobile design challenge (sample image for developers to work on) • Requirement (Text: requirements of the challenge) • Assets Provider (Text: content of the available assets)

			• Hint (Text: displays hints for challengers) • Upload File assets (Zip file with size limit) • Upload Figma File • Update or Cancel button
	Posted Challenge	List challenger (submitted)	Challenger list • Challenger name • Challenge name • Challenge level • Number of submissions • Technology • Submission link • Submission time • Challenger level
		List Challenge	Challenge list • Challenge banner • Programming language • Challenge name • Points • Level name • Description
		Details Challenge	Challenge Detail • Technical challenge • Challenge name • Level • Description • Message (if any) • Level name • Image

			Responsive options
		Detail Challenger	Challenger Detail - Banner: image - Avatar: image - User: Name, rank, email, GitHub link, social link, bio - View CV button - Solution: (Default tab) system solution: - System's solution (Tab 2) company solution: - Solution of the company's challenge (other users can view the company's challenge of the user but cannot see the challenge's solution details)

3.3.3 Challenge Manager



Figure 3 - 5 Challenge Manager in Systems

Challenge Manager: Has the right to log in to the Admin page of the system and perform administrative tasks such as **Challenge Management**, including creating, editing, deleting challenges, approving challenges created by **Recruiters**, and viewing challenger information. In addition to the above functions, the **Challenge Manager** also has the authority to modify certain features and view user information and transaction status for each batch of users. In **Solution Management**, the **Challenge Manager** can view uploaded solution information, as well as manage personal information and handle user comments, such as reporting solutions.

Table 3 - 3 Challenge Manager FSD

Module	Function	Page	Detail in Page
	Login	Login	• Enter Email • Enter Password • Remember password • Forgot password • Login button • Social login button (Github) • Sign up button (If you don't have an account)"

User Registration & Authentication		Login Successful (Notification)	Automatically redirects to the homepage
		Login Failed	- Email does not exist/Incorrect - Incorrect password
	Register	Enter Registration Information	- Enter Email - Enter Password - Re-enter Password - Sign Up button - Login button (If you already have an account)
		Email Does Not Exist/Incorrect	<Error message description>
		Enter Correct OTP	<Notification that OTP has been sent>
		Incorrect OTP	<Error message description>
		Verify OTP	- Enter OTP - Verify button - Resend button
		OTP Expired	<Error message description>
		New Password Does Not Match/Correct Format	<Error message description>
		Registration Successful	Registration success message, automatically redirects to the login screen

	Reset Password	Enter Email	Email address for verification
		Email Does Not Exist/Incorrect	<Error message description>
		Enter OTP	<Notification that OTP has been sent to the email>
		Verify OTP	- Enter OTP - Verify - Resend OTP
		Create New Password	- Enter new password - Re-enter new password
		New Password Created Successfully	<Success message for updating and redirecting to the login screen>
		New Password Does Not Match	<Error message description>
	Logout	Confirm Logout	- Display confirmation dialog - Yes <Automatically log out> - No <Close dialog>"
	User Profile	Home Page	Section Statistics Challenge - Total number of challenges participated - Total number of submissions - Total number of unsubmitted challenges - Total number of recruit challenges participated List of Submitted Solutions - Id - Challenge Name - Interaction Volume - Score

User Profile Management			• Start Time • Submission Time • Level Name List of Challenges You Can Join (Same level as the challenger) • Contains challenges (Swipe format): found in the homepage Figma design List of Company Challenges: • Challenge: ○ Title: Responsive Food Web ○ Expiry Time: 6 days • “See more” button
		Profile	• Banner: Image • Avatar: Image • User: Name, rank, email, GitHub link, social link, bio • View CV button • Follow button • Solution • Solution System: List Solution • Solution company: List Solution company • Company Following: Number of following of company
		Setting Profile	Banner & Avatar • Banner: Image

			• Upload Image button • Avatar: Image • Upload Image button • Contact Information: Gmail, etc. • Add Information button Personal Information • First Name • Last Name • Email • Phone Number • Password • Change Password button • Address • GitHub URL (place this section on top to link GitHub) • Website Link • Bio (Description) Contact & Social Links • CodePen URL • Codewars URL • Dev.to URL • freeCodeCamp URL • GitLab URL • Hashnode URL • LinkedIn URL • Medium URL • StackOverflow URL • Twitch URL • Twitter URL • YouTube URL
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		Challenger Profile	• Banner: Image • Avatar: Image • User: Name, rank, email, GitHub link, social link, bio • View CV button • Follow button • Solution • Solution System: List Solution • Solution company: List Solution company • Company Following: Number of following of company
	List Challenge	List Challenger + Premium Challenge	Challenge List • Mark as new • Mark as free/premium • Challenge banner • Programming language • Challenge name • Score • Level name • Description • Practice button
		Filter Challenge	• Filter by category: Free, Free+/Premium • Level: Bronze, Silver • Difficulty: Easy, Medium, Hard • Technical: HTML, CSS, Java

Challenge Management	Detail Challenge	Detail Challenge (Tab Information)	Detail Challenge • Technical challenge • Challenge name • Level • Description • Message (if any) • Join Challenge button • Level name • Image • Responsive options Tab Information Challenge • Note • Assets provided • Message FQA (Frequently Asked Questions)
	Submit Challenge	Form Submit Challenge	Detail Challenge • Challenge Title • Technical Challenge • Level • Description • Message • Challenge Image Send Challenge Information (Below are the input fields to fill in) • Solution Title* • Repository URL* • Live Site URL* • Which tools, libraries, frameworks, or

			methodologies did you use for this project? • Private or Public • What are you most proud of, and what would you do differently next time? • What challenges did you encounter, and how did you overcome them? • What specific areas of your project would you like help with? • Submit Solution button
	Challenge system	Create Challenge (Pop-up)	• Challenge Name • Technical Challenge (Select Multiple) • Image Challenge (After uploading, the first image will be the main image) • Description • Level (select option: easy, medium, hard) • Duration • Upload Desktop design challenge (sample image for developers to work on) • Upload Tablet design challenge (sample image for developers to work on)

			• Upload Mobile design challenge (sample image for developers to work on) • Requirement (Text: requirements of the challenge) • Assets Provider (Text: content of the available assets) • Hint (Text: displays hints for challengers) • Upload File assets (Zip file with size limit) • Upload Figma File Create or Cancel button
		Update Information Challenge	• Challenge Name • Technical Challenge (Select Multiple) • Image Challenge (After uploading, the first image will be the main image) • Description • Level (select option: easy, medium, hard) • Duration • Upload Desktop design challenge (sample image for developers to work on) • Upload Tablet design challenge (sample image

			for developers to work on) • Upload Mobile design challenge (sample image for developers to work on) • Requirement (Text: requirements of the challenge) • Assets Provider (Text: content of the available assets) • Hint (Text: displays hints for challengers) • Upload File assets (Zip file with size limit) • Upload Figma File Update or Cancel button
	List Solution	List Solution	• Action: Filter • Solution: ○ Solution Name ○ Technical ○ Owner ○ Level
		Filter Solution (Pop-up)	Display fields to filter such as: • Filter by challenge type (Free, Free+, Premium) • Filter by Level and difficulty of the challenge (Level + Hard, Easy, Medium, or All)

Solutions Managment			Filter by required Technicals (HTML, CSS, Javascript, etc.)
	Solution System	Detail Solution	Display challenge information that this solution submits: - Required Technicals of the challenge - Level and difficulty of the challenge - Challenge type (Free, Free+, Premium) - Challenge title - Short description of the challenge - Challenge score - Number of participants in the challenge - Number of challenge submissions - Button (displayed based on status): - If the viewer of this solution detail is the author of the solution, this button will be "view other solutions of this challenge" - If the viewer is not the author,

			the button will be "return to my solution." • Preview image of the challenge • Tabs to view responsive preview images of the challenge View source code button View demo preview button Display solution information: • Submission time of the solution • Solution title • Technicals used for the solution • Action buttons for the solution (Like, Dislike, Report, Save) • Solution descriptions in format (title + description or only description) Display comment and feedback section: • Comments for challengers: ○ Avatar ○ Challenger's name ○ Comment time ○ Comment content
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			○ Interaction buttons for the comment (Like, Dislike, Report, or Reply) • Feedback for mentors: ○ Mentor's avatar ○ Mentor's name ○ Feedback time ○ Feedback content ○ Interaction buttons for the feedback (Like, Dislike, Report, or Reply) Display solution author's information • To be added later.
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3.3.4 Admin



Figure 3 - 6 Admin in System

Admin: Has full authority to manage all modules in the system, including **Challenger Management**, **Recruiter Management**, **Challenge Management**, and **Solution Management**. The Admin holds the highest power in the system and has the right to make decisions on all matters within the system. The **Admin** does not need to register an account and is assigned as a single member by default.

Table 3 - 4 Admin FSD

Module	Function	Page	Detail in Page
	Login	Login	• Enter Email • Enter Password • Remember password • Forgot password • Login button • Social login button (Github) • Sign up button (If you don't have an account)"
		Login Successful (Notification)	• Automatically redirects to the homepage
		Login Failed	• Email does not exist/Incorrect

User Registration & Authentication			• Incorrect password
	Reset Password	Enter Email	• Email address for verification
		Email Does Not Exist/Incorrect	• <Error message description>
		Enter OTP	• <Notification that OTP has been sent to the email>
		Verify OTP	• Enter OTP • Verify • Resend OTP
		Create New Password	• Enter new password • Re-enter new password
		New Password Created Successfully	• <Success message for updating and redirecting to the login screen>
		New Password Does Not Match	• <Error message description>
	Logout	Confirm Logout	• Display confirmation dialog • Yes <Automatically log out> • No <Close dialog>"
	User Profile	Home Page	Section Statistics Challenge • Total number of challenges participated • Total number of submissions • Total number of unsubmitted challenges • Total number of recruit challenges participated List of Submitted Solutions • Id • Challenge Name • Interaction Volume

User Profile Management			• Score • Start Time • Submission Time • Level Name List of Challenges You Can Join (Same level as the challenger) • Contains challenges (Swipe format): found in the homepage Figma design List of Company Challenges: • Challenge: ○ Title: Responsive Food Web ○ Expiry Time: 6 days • “See more” button
		Profile	• Banner: Image • Avatar: Image • User: Name, rank, email, GitHub link, social link, bio • View CV button • Follow button • Solution • Solution System: List Solution • Solution company: List Solution company • Company Following: Number of following of company
		Setting Profile	Banner & Avatar

			• Banner: Image • Upload Image button • Avatar: Image • Upload Image button • Contact Information: Gmail, etc. • Add Information button Personal Information • First Name • Last Name • Email • Phone Number • Password • Change Password button • Address • GitHub URL (place this section on top to link GitHub) • Website Link • Bio (Description) Contact & Social Links • CodePen URL • Codewars URL • Dev.to URL • freeCodeCamp URL • GitLab URL • Hashnode URL • LinkedIn URL • Medium URL • StackOverflow URL • Twitch URL • Twitter URL
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			• YouTube URL
		Challenger Profile	• Banner: Image • Avatar: Image • User: Name, rank, email, GitHub link, social link, bio • View CV button • Follow button • Solution • Solution System: List Solution • Solution company: List Solution company • Company Following: Number of following of company
	List Challenge	List Challenger + Premium Challenge	Challenge List • Mark as new • Mark as free/premium • Challenge banner • Programming language • Challenge name • Score • Level name • Description • Practice button
		Filter Challenge	• Filter by category: Free, Free+/Premium • Level: Bronze, Silver • Difficulty: Easy, Medium, Hard

Challenge Management			• Technical: HTML, CSS, Java
	Detail Challenge	Detail Challenge (Tab Information)	Detail Challenge • Technical challenge • Challenge name • Level • Description • Message (if any) • Join Challenge button • Level name • Image • Responsive options Tab Information Challenge • Note • Assets provided • Message FQA (Frequently Asked Questions)
	Submit Challenge	Form Submit Challenge	Detail Challenge • Challenge Title • Technical Challenge • Level • Description • Message • Challenge Image Send Challenge Information (Below are the input fields to fill in) • Solution Title* • Repository URL* • Live Site URL*

			• Which tools, libraries, frameworks, or methodologies did you use for this project? • Private or Public • What are you most proud of, and what would you do differently next time? • What challenges did you encounter, and how did you overcome them? • What specific areas of your project would you like help with? • Submit Solution button
	Challenge system	Create Challenge (Pop-up)	• Challenge Name • Technical Challenge (Select Multiple) • Image Challenge (After uploading, the first image will be the main image) • Description • Level (select option: easy, medium, hard) • Duration • Upload Desktop design challenge (sample image for developers to work on) • Upload Tablet design challenge (sample image

			for developers to work on) • Upload Mobile design challenge (sample image for developers to work on) • Requirement (Text: requirements of the challenge) • Assets Provider (Text: content of the available assets) • Hint (Text: displays hints for challengers) • Upload File assets (Zip file with size limit) • Upload Figma File Create or Cancel button
		Update Information Challenge	• Challenge Name • Technical Challenge (Select Multiple) • Image Challenge (After uploading, the first image will be the main image) • Description • Level (select option: easy, medium, hard) • Duration • Upload Desktop design challenge (sample image for developers to work on)

			• Upload Tablet design challenge (sample image for developers to work on) • Upload Mobile design challenge (sample image for developers to work on) • Requirement (Text: requirements of the challenge) • Assets Provider (Text: content of the available assets) • Hint (Text: displays hints for challengers) • Upload File assets (Zip file with size limit) • Upload Figma File - Update or Cancel button
	List Solution	List Solution	• Action: Filter • Solution: ○ Solution Name ○ Technical ○ Owner ○ Level
		Filter Solution (Pop-up)	Display fields to filter such as: • Filter by challenge type (Free, Free+, Premium) • Filter by Level and difficulty of the challenge (Level + Hard, Easy, Medium, or All)

Solutions Managment			Filter by required Technicals (HTML, CSS, Javascript, etc.)
	Solution System	Detail Solution	Display challenge information that this solution submits: - Required Technicals of the challenge - Level and difficulty of the challenge - Challenge type (Free, Free+, Premium) - Challenge title - Short description of the challenge - Challenge score - Number of participants in the challenge - Number of challenge submissions - Button (displayed based on status): - If the viewer of this solution detail is the author of the solution, this button will be "view other solutions of this challenge" - If the viewer is not the author,

			the button will be "return to my solution." • Preview image of the challenge • Tabs to view responsive preview images of the challenge View source code button View demo preview button Display solution information: • Submission time of the solution • Solution title • Technicals used for the solution • Action buttons for the solution (Like, Dislike, Report, Save) • Solution descriptions in format (title + description or only description) Display comment and feedback section: • Comments for challengers: ○ Avatar ○ Challenger's name ○ Comment time ○ Comment content
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			○ Interaction buttons for the comment (Like, Dislike, Report, or Reply) • Feedback for mentors: ○ Mentor's avatar ○ Mentor's name ○ Feedback time ○ Feedback content ○ Interaction buttons for the feedback (Like, Dislike, Report, or Reply) Display solution author's information • To be added later.
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3.4 Use-case in system

Table 3. 1 Use-case in system

ID USE CASE	USE CASE	DESCRIPTION
UC01	User Registration & Authentication	Allows users to register, log in, and reset passwords with features like OTP verification, and social login.
UC02	User Profile Management	Users can manage their profile, including personal information, viewing solutions, and following companies.
UC03	Challenge Listing	Users can view, filter, and explore available challenges with details like level, technical requirements, and point values.
UC04	Challenge Detail	Users can view details of a selected challenge, including technical requirements and options for participating.
UC05	Submit Challenge	Users can submit solutions to challenges, including providing URLs, descriptions, and related resources.
UC06	Solution Management	Allows users to manage submitted solutions, including filtering, viewing solution details, and interacting via likes or comments.
UC07	Notification Management	Users receive notifications regarding challenge updates, solution interactions, and recruiter activities.

UC08	Recruiter Management	Recruiters can manage challenges, receive notifications about participants, and interact with the submitted solutions.
UC9	Admin Management	Admins manage user authentication, challenge creation, and receive notifications about system events and errors.

CHAPTER 4: DESIGNING SYSTEM

4.1 Overall Use-Case diagram

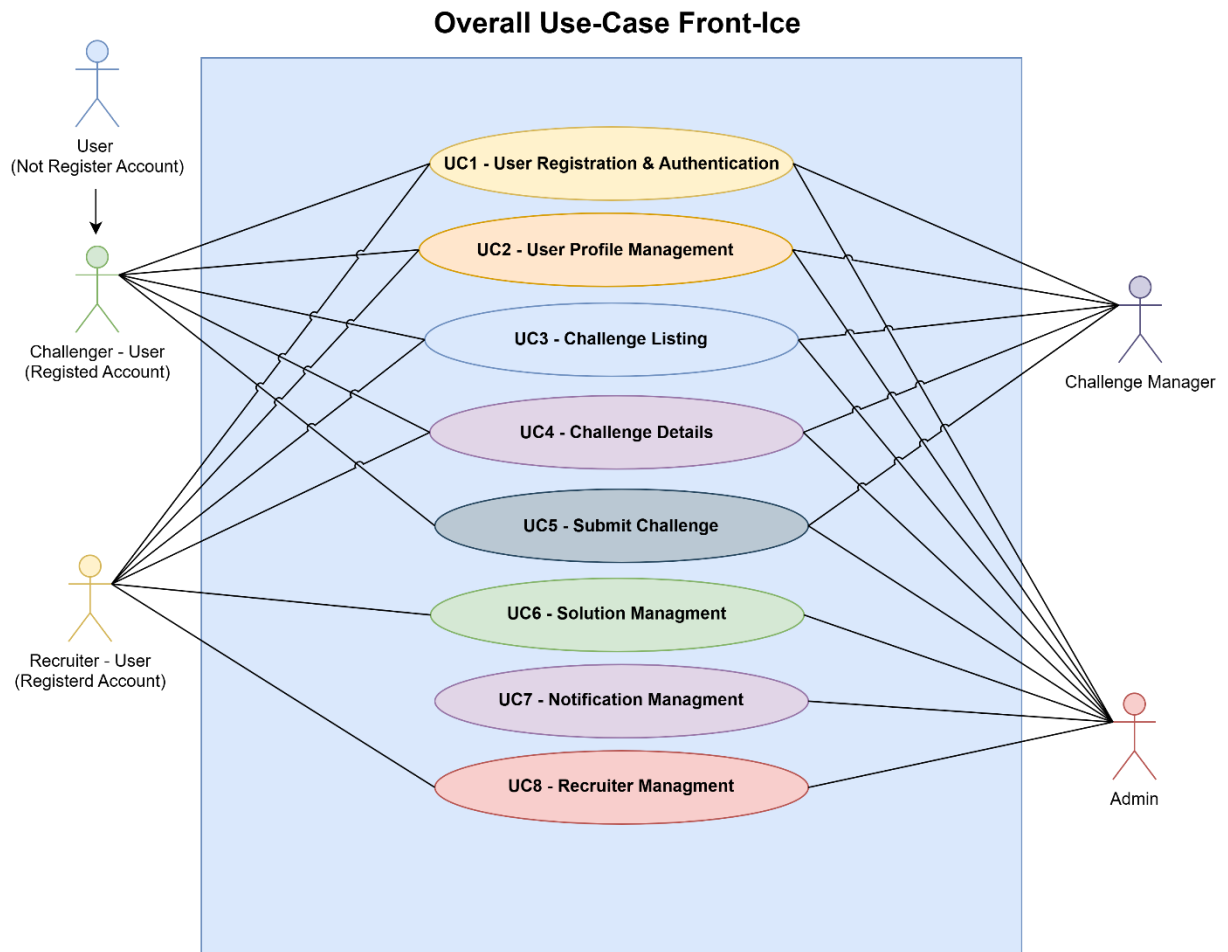


Figure 4 - 1 Overall Use-case Front-Ice

4.2 Detail Use-Case diagram

4.2.1 User Registration & Authentication Use-case Diagram

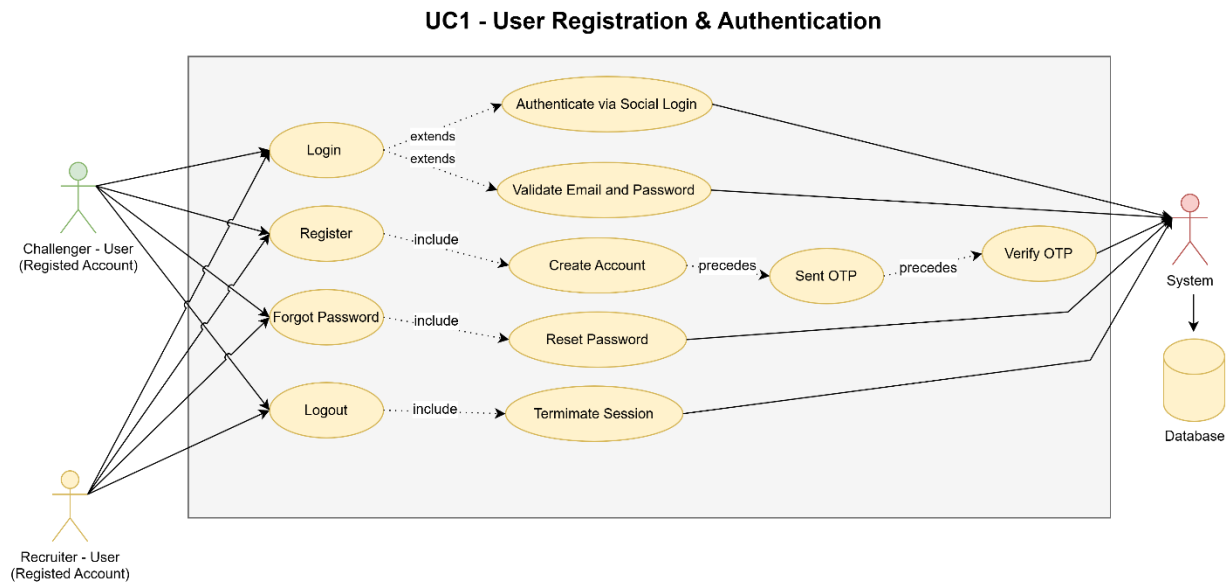


Figure 4 - 2 User Registration & Authentication Use-case

Table 4 - 1 User Registration & Authentication

Headline	Content
ID	UC1
Name	User Registration & Authentication
Requirements	The user must navigate to the login page or click the login button in the navigation bar. For registration, the user must provide a valid email address, password, and confirm the OTP sent to the email.
Description	This use case describes the process for users to register or log in to the system to access features corresponding to their account.
Actors	Challenger, Recruiter, System
Preconditions	The user must have a valid email address. For login, the user must already have an account registered in the system. For

	registration, the user must provide a valid email address, a password, and successfully verify the OTP sent to their email.
Postconditions	Login Success: The user is redirected to the homepage with account-specific functionality available. Registration Success: The user is redirected to the login page after successful account creation. Login Failure: The system displays an error message indicating incorrect credentials. Registration Failure: The system displays an error message for invalid input or OTP issues.
Main Event Stream	User Actions: 1. Open the login or registration page. 2. Enter the required information: - For login: Enter email and password. - For registration: Enter email, password, confirm password, and OTP. 3. Click the login or register button. System Actions: 1. The system displays input fields for the user to fill in. 2. When the user clicks submit: - For login: The system sends a hashed email and password to the server. - For registration: The system sends the user's data and OTP to the server. 3. The system performs the following actions: - For login: 3.1. Queries the database to verify the email and password. 3.2. If valid, fetches the corresponding account details and permissions. 3.3. If invalid, returns a failed login response. - For registration: 3.4. Validates the OTP. 3.5. If valid, stores the user's data and creates the account. 3.6. If invalid, returns an error response for OTP issues. 4. The system returns the login/registration results. 5. Based on the result: - Login Success: Redirect the user to the homepage with

	granted permissions. - Login Failure: Display an error message indicating incorrect credentials. - Registration Success: Redirect the user to the login page after successful account creation. - Registration Failure: Display error messages for invalid input or OTP-related issues.
Exceptions	- Invalid email or password format. - OTP entered is incorrect or expired. - Email already exists during registration. - Account not found in the database during login.

4.2.2 User Profile Management Use-case Diagram

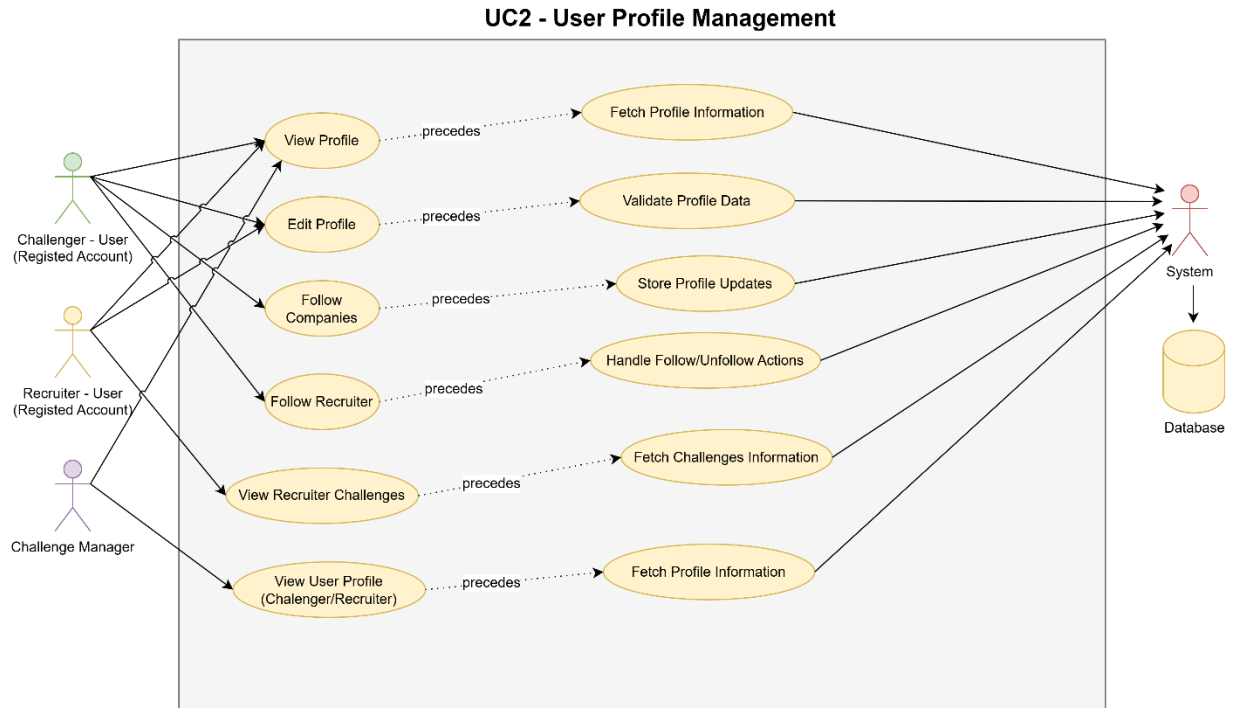


Figure 4 - 3 User Profile Management Use-case Diagram

Table 4 - 2 User Profile Management Use-case

Headline	Content
ID	UC2
Name	User Profile Management
Requirements	The user must have a valid account and be logged into the system. Access to the profile management page is available via the navigation menu or direct URL.
Description	This use case describes the process of managing user profile information, including viewing, editing, and updating personal details, linked accounts, and profile images.
Actors	Challenger, Recruiter, System
Preconditions	The user must have access to their account and have logged in successfully.
Postconditions	Updates to the profile are successfully saved, and changes are reflected immediately on the user's profile page. The user

	receives feedback for any errors encountered during the update process.
Main Event Stream	User Actions: 1. Navigate to the Profile Management page. 2. Perform desired actions: - View profile details, including personal information, contact details, and linked accounts. - Edit personal information such as name, email, phone number, and bio. - Update profile image or avatar. - Manage linked accounts (e.g., GitHub, LinkedIn). 3. Submit changes. System Actions: 1. The system retrieves and displays the current user profile data. 2. When the user makes updates: - Validate the input data (e.g., email format, required fields). - Save valid updates to the database. 3. Provide feedback: - If successful, confirm that changes have been saved. - If validation fails, display error messages for the specific fields requiring correction. 4. Reflect updated profile data on the user's profile page.
Exceptions	- Invalid input format (e.g., email, phone number). - Failure to upload profile images due to size or format restrictions. -Database connection error or save operation failure. - Unauthorized access to profile management features.

4.2.3 Challenge Listing Use-case Diagram

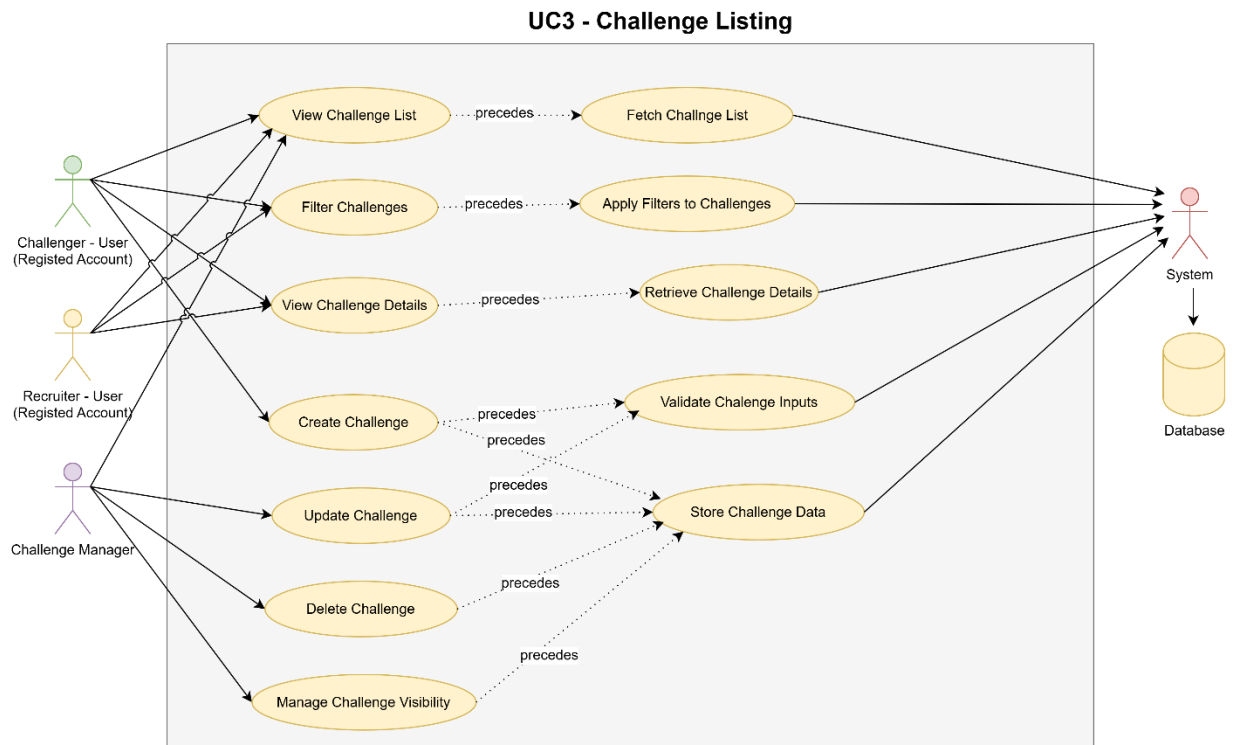


Figure 4 - 4 Challenge Listing Use-case Diagram

Table 4 - 3 Challenge Listing Use-case

Headline	Content
ID	UC3
Name	Challenge Listing
Requirements	The user must be logged in and navigate to the Challenges page via the navigation menu or direct URL.
Description	This use case describes the process of viewing, filtering, and accessing detailed information about challenges listed on the platform.
Actors	Challenger, System
Preconditions	The system must have a list of challenges available in the database. The user must have a valid account and successfully logged into the system.

Postconditions	The user successfully views challenges, applies filters, and accesses detailed information about a selected challenge. The system updates the user's view based on the selected filters and displays the relevant challenges.
Main Event Stream	User Actions: 1. Navigate to the Challenges page. 2. View the list of challenges. 3. Optionally, apply filters to refine the displayed challenges (e.g., by category, difficulty level, or technology). 4. Select a specific challenge to view its details. System Actions: 1. Fetch and display the list of challenges. 2. Provide filtering options to the user (e.g., dropdowns or checkboxes). 3. When filters are applied: - Validate the filter criteria. - Fetch and display the filtered list of challenges. 4. When a challenge is selected: - Retrieve detailed information about the selected challenge. - Display the challenge details, including description, technical requirements, and participation options.
Exceptions	- No challenges available in the system. - Invalid filter criteria applied by the user. - Database error when fetching challenges or their details. - Unauthorized access attempts to view challenges or their details.

4.2.4 Challenge Detail Use-case Diagram

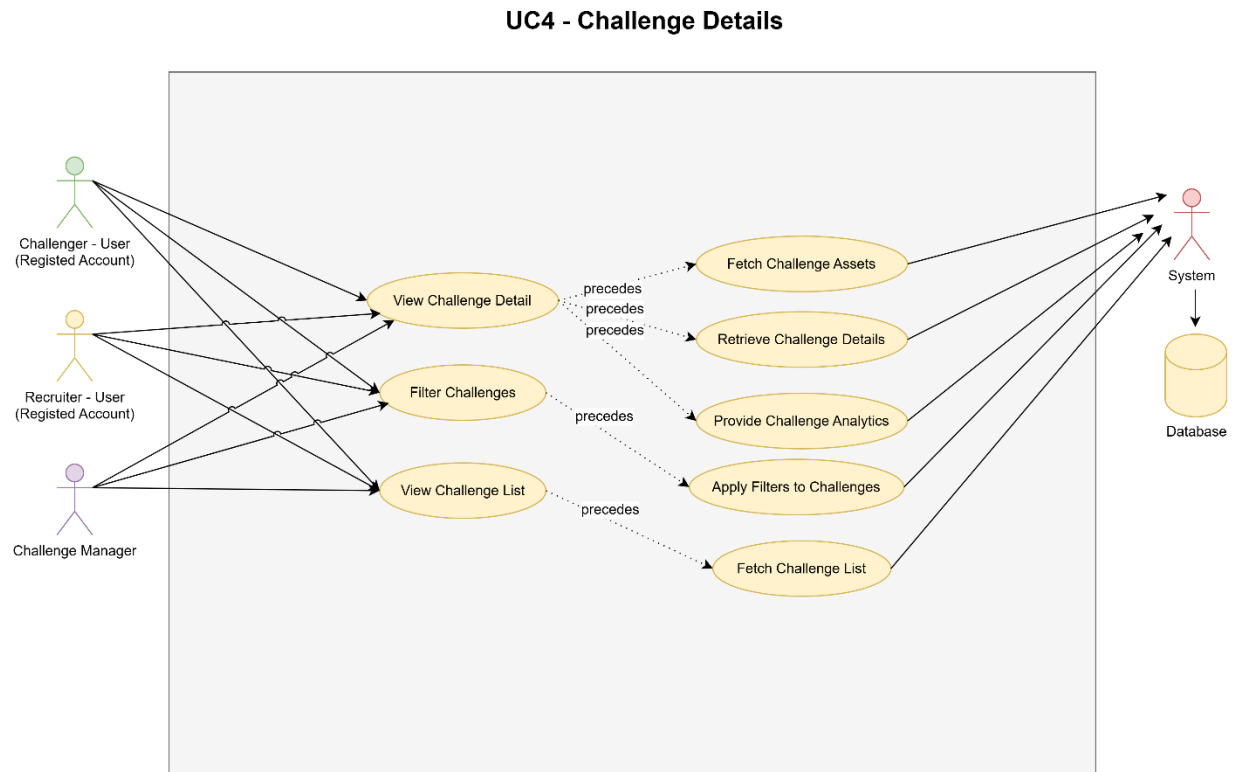


Figure 4 - 5 Challenge Detail Use-case Diagram

Table 4 - 4 Challenge Detail Use-case

Headline	Content
ID	UC4
Name	Challenge Detail
Requirements	The user must navigate to a specific challenge from the Challenge Listing page or a direct link.
Description	This use case describes the process of viewing detailed information about a specific challenge, including its requirements, level, technical details, assets, and options for participation.
Actors	Challenger, System
Preconditions	A list of challenges must exist in the system. The user must have a valid account and be logged in.

Postconditions	The user successfully views the details of the selected challenge, including its description, technical requirements, assets, and participation options.
Main Event Stream	User Actions: 1. Navigate to the Challenge Listing page. 2. Select a specific challenge from the list. 3. View the detailed information about the challenge, including: - Challenge name. - Description and requirements. - Level and difficulty. - Technical stack required. - Assets provided (e.g., images, files). - FAQs or additional guidance. System Actions: 1. Fetch and display the basic information about the challenge. 2. Retrieve the challenge details from the database, including description, technical requirements, and participation options. 3. Provide links or buttons for the user to download assets (e.g., zip files) and participate in the challenge. 4. Display error messages if the challenge details cannot be retrieved.
Exceptions	- Challenge not found or unavailable. - Database error while retrieving challenge details. - Unauthorized access attempts for restricted challenges (e.g., premium or private).

4.2.5 Submit Challenge Use-case Diagram

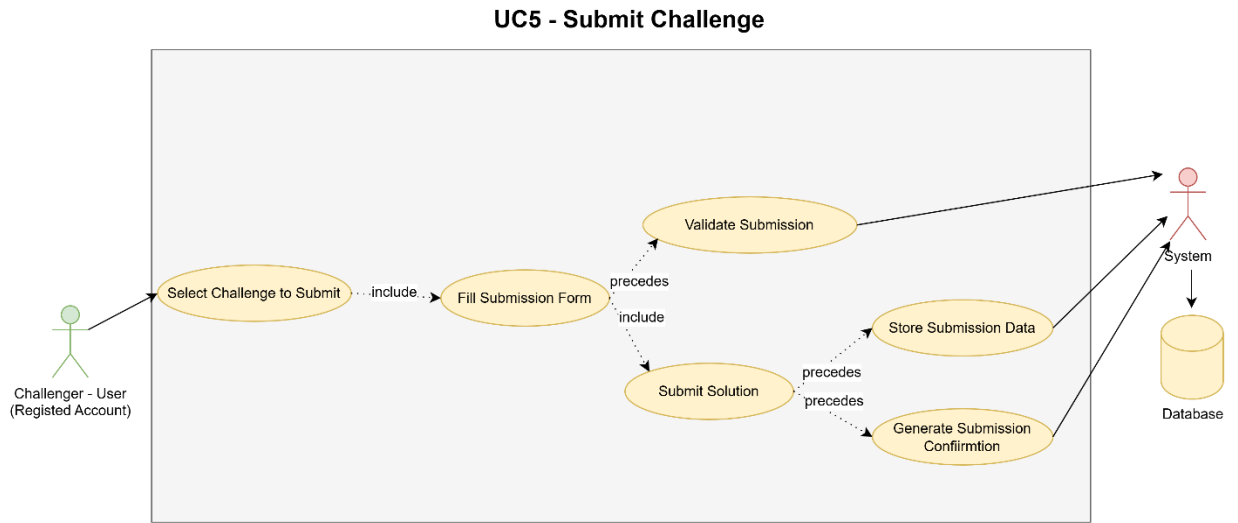


Figure 4 - 6 Submit Challenge Use-case Diagram

Table 4 - 5 Submit Challenge Use-case

Headline	Content
ID	UC5
Name	Submit Challenge
Requirements	The user must navigate to a specific challenge, complete their solution, and access the submission form.
Description	This use case describes the process of submitting a solution for a challenge, including providing necessary details, uploading required files, and confirming the submission.
Actors	Challenger, System
Preconditions	The user must have an account and be logged in. A solution must be prepared to meet the requirements of the selected challenge. The challenge must allow submissions.

Postconditions	The user successfully submits a solution, and the system validates and stores the submission. A confirmation is provided to the user, and the Challenge Manager is notified about the submission.
Main Event Stream	User Actions: 1. Navigate to the details of the challenge to be submitted. 2. Click the "Submit Solution" button. 3. Fill in the submission form, including: - Solution name. - Description of the solution. - Links to the source code (e.g., GitHub repo) or demo. - Upload any required files. 4. Review the submission and confirm. System Actions: 1. Validate the submission form inputs (e.g., file format, required fields). 2. Verify that the user meets the eligibility to submit (e.g., challenge deadline, prerequisites). 3. Store the submission data, including files and metadata, in the database. 4. Notify the Challenge Manager about the new submission. 5. Provide a confirmation message to the user, indicating successful submission or errors to resolve.
Exceptions	- Invalid or incomplete submission form inputs. - File upload issues (e.g., size, format restrictions). - Challenge submission deadline passed. - Database error while storing the submission. - Unauthorized attempts to submit a challenge.

4.2.6 Solution Management Use-case Diagram

UC6 - Solution Management

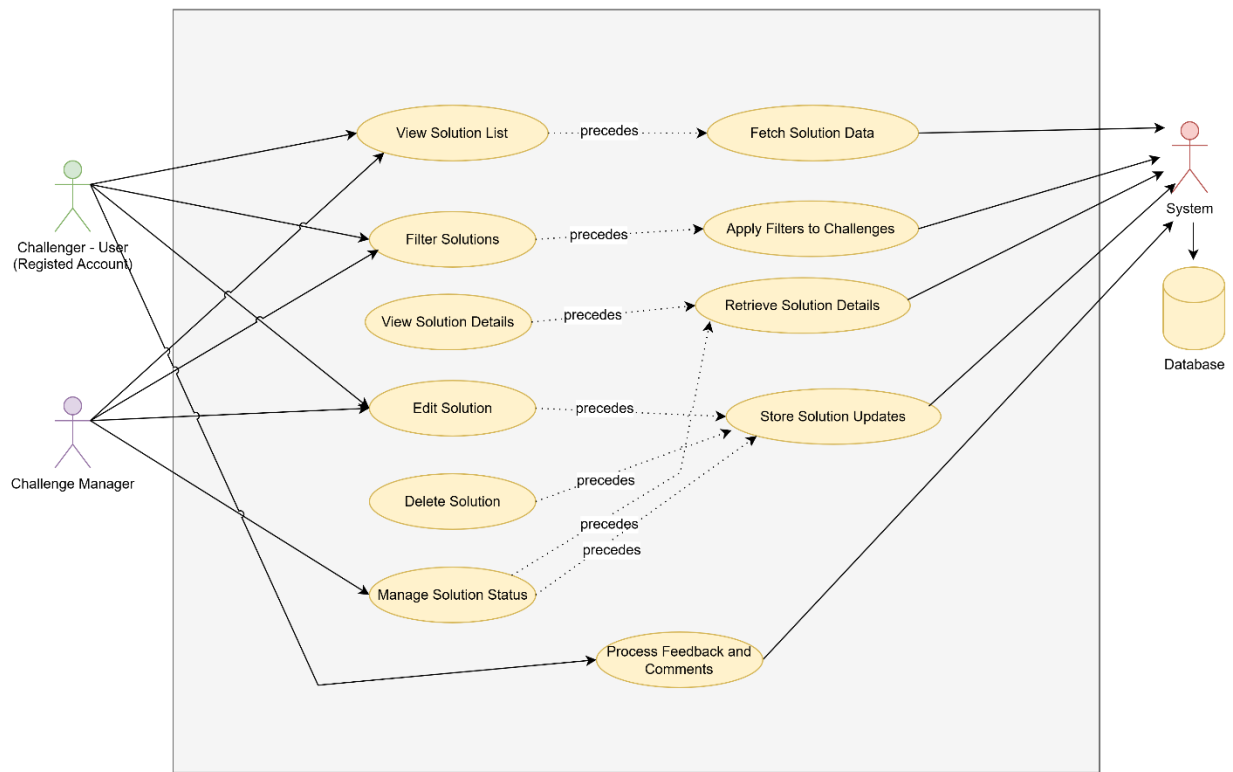


Figure 4 - 7 Solution Management Use-case Diagram

Table 4 - 6 Solution Management Use-case

Headline	Content
ID	UC6
Name	Solution Management
Requirements	The user must navigate to the solution management section via the dashboard or menu.
Description	This use case describes the process of managing submitted solutions, including viewing, updating, deleting, and monitoring feedback and comments.
Actors	Challenger, System
Preconditions	The user must have an account and be logged in. Submitted solutions must exist in the system.

Postconditions	The user successfully views, updates, or deletes a solution. The changes are reflected in the system, and feedback is appropriately managed.
Main Event Stream	User Actions: 1. Navigate to the Solution Management section. 2. View the list of submitted solutions. 3. Perform the desired action: - View Solution Details: Select a solution to view details such as name, description, and submission status. - Edit Solution: Update solution details (e.g., description, links, or uploaded files). - Delete Solution: Remove a solution permanently. - Monitor Feedback and Comments: View feedback provided by reviewers, mentors, or peers. System Actions: 1. Retrieve and display the list of submitted solutions. 2. For each action: - View Solution Details: Fetch and display all related data from the database. - Edit Solution: Validate updated data and save changes to the database. - Delete Solution: Remove solution data and related metadata from the database. - Monitor Feedback and Comments: Fetch and display feedback and comments, including timestamps and authors. 3. Provide feedback on each action, indicating success or failure.

Exceptions	- Solution not found in the system. - Invalid data during the editing process. - Unauthorized attempts to edit or delete solutions. - Database error while fetching or saving solution data. - Feedback or comments unavailable due to system errors.
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4.2.7 Notification Management Use-case Diagram

UC7 - Notification Management



Figure 4 - 8 Notification Management Use-case Diagram

Table 4 - 7 Notification management use-case

Headline	Content
ID	UC7
Name	Notification Management
Requirements	The user must be logged into the system and navigate to the notification center via the dashboard or menu.
Description	This use case describes the process of managing notifications, including viewing, marking as read, and responding to system-generated alerts or messages sent by Challenge Managers.
Actors	Challenger, Challenge Manager, System

Preconditions	Notifications must exist in the system. The user must have an active account and be logged in.
Postconditions	Notifications are successfully delivered, viewed, and managed by the user. Actions triggered by notifications (e.g., opening a challenge detail page) are executed successfully.
Main Event Stream	User Actions: 1. Navigate to the Notification Center. 2. View the list of notifications, including: - Notification title. - Date and time received. - Status (read/unread). 3. Perform the desired action: - Mark as Read: Update the status of a notification. - Open Notification: Access linked content (e.g., challenge details, solutions, or messages). Challenge Manager Actions: 1. Trigger notifications by: - Adding, updating, or deleting challenges. - Sending custom messages to users. System Actions: 1. Generate notifications based on events triggered by the Challenge Manager or system processes. 2. Deliver notifications to the appropriate users. 3. Update notification status (e.g., read/unread) in the database. 4. Execute actions linked to notifications (e.g., redirecting to a specific page). 5. Provide feedback to users about notification-related actions (e.g., error messages for invalid actions).

Exceptions	- Notifications not delivered due to a system error. - Invalid or unauthorized attempts to access notifications. - Database error when updating notification status or fetching notification details. - Notification links leading to unavailable or deleted content.
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4.2.8 Recruiter Management Use-case Diagram

UC8 - Recruiter Management

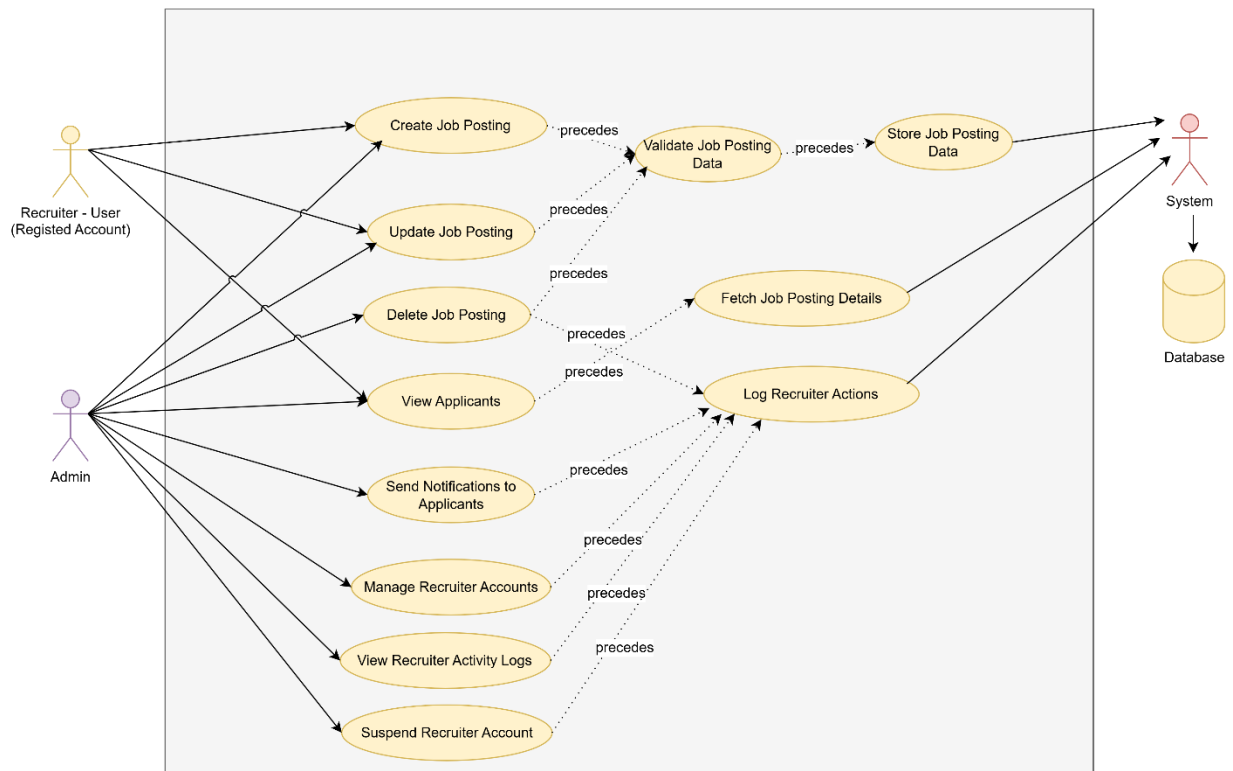


Figure 4 - 9 Recruiter Management Use-case Diagram

Table 4 - 8 Recruiter Management Use-case

Headline	Content
ID	UC8
Name	Recruiter Management
Requirements	Recruiters must have valid accounts and be logged into the system. Admin must have system-level privileges to manage recruiters.
Description	This use case describes the process of managing recruiters, including creating, updating, and deleting recruiter accounts, as well as monitoring their activities and performance metrics.
Actors	Recruiter, Admin, System

Preconditions	Recruiter accounts must exist in the system. Admin must have access to system management tools.
Postconditions	Recruiter accounts are successfully created, updated, or deleted. Admin actions are logged, and recruiters' activities are monitored and reported accurately.
Main Event Stream	Admin Actions: 1. Navigate to the Recruiter Management section. 2. Perform one of the following actions: - Create Recruiter: Enter recruiter details (e.g., name, email, organization) and submit. - Update Recruiter: Modify details of an existing recruiter account. - Delete Recruiter: Remove a recruiter account from the system. - Monitor Recruiter Activity: View activity logs and performance metrics. 3. Confirm the action and review the system's feedback. Recruiter Actions: 1. Log into the system to access recruiter-specific functionalities. 2. Perform tasks such as posting challenges, reviewing applicants, and managing challenge details. 3. Receive feedback or notifications about their actions from the system. System Actions: 1. Validate data entered by Admin during create or update actions. 2. Store recruiter information or changes in the database. 3. Log actions performed by Admin and Recruiter. 4. Generate activity reports and performance metrics for Admin to review.

	5. Send system notifications to Recruiters about updates or actions affecting them.
Exceptions	- Duplicate recruiter email during account creation. - Unauthorized access to the recruiter management system. - Database error when storing or updating recruiter details. - Failure to generate activity reports due to system errors. - Deleting a recruiter account associated with active challenges or pending tasks.

4.3 Class Diagram

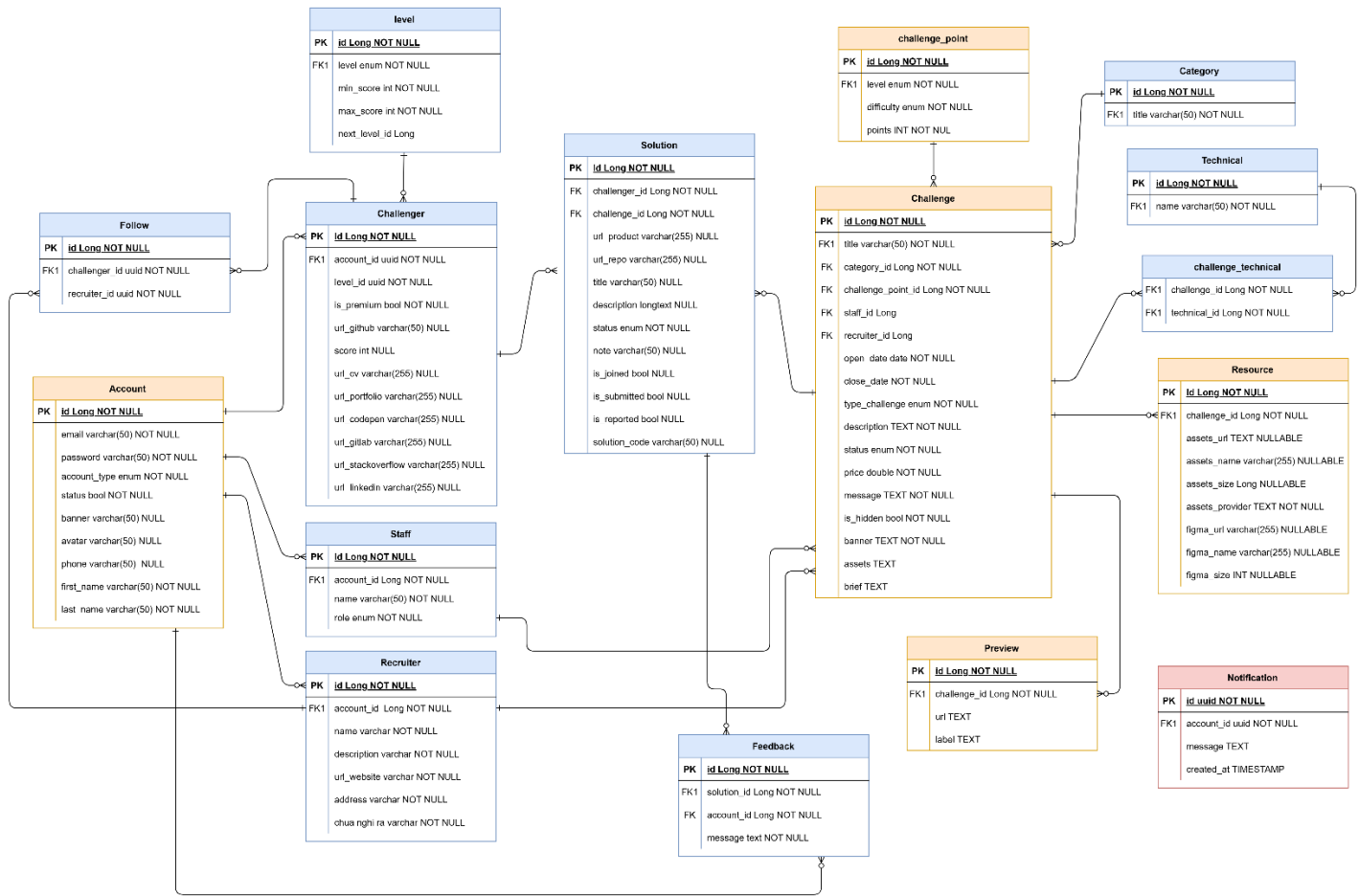


Figure 4 - 10 Class Diagram

4.4 Relational database schema

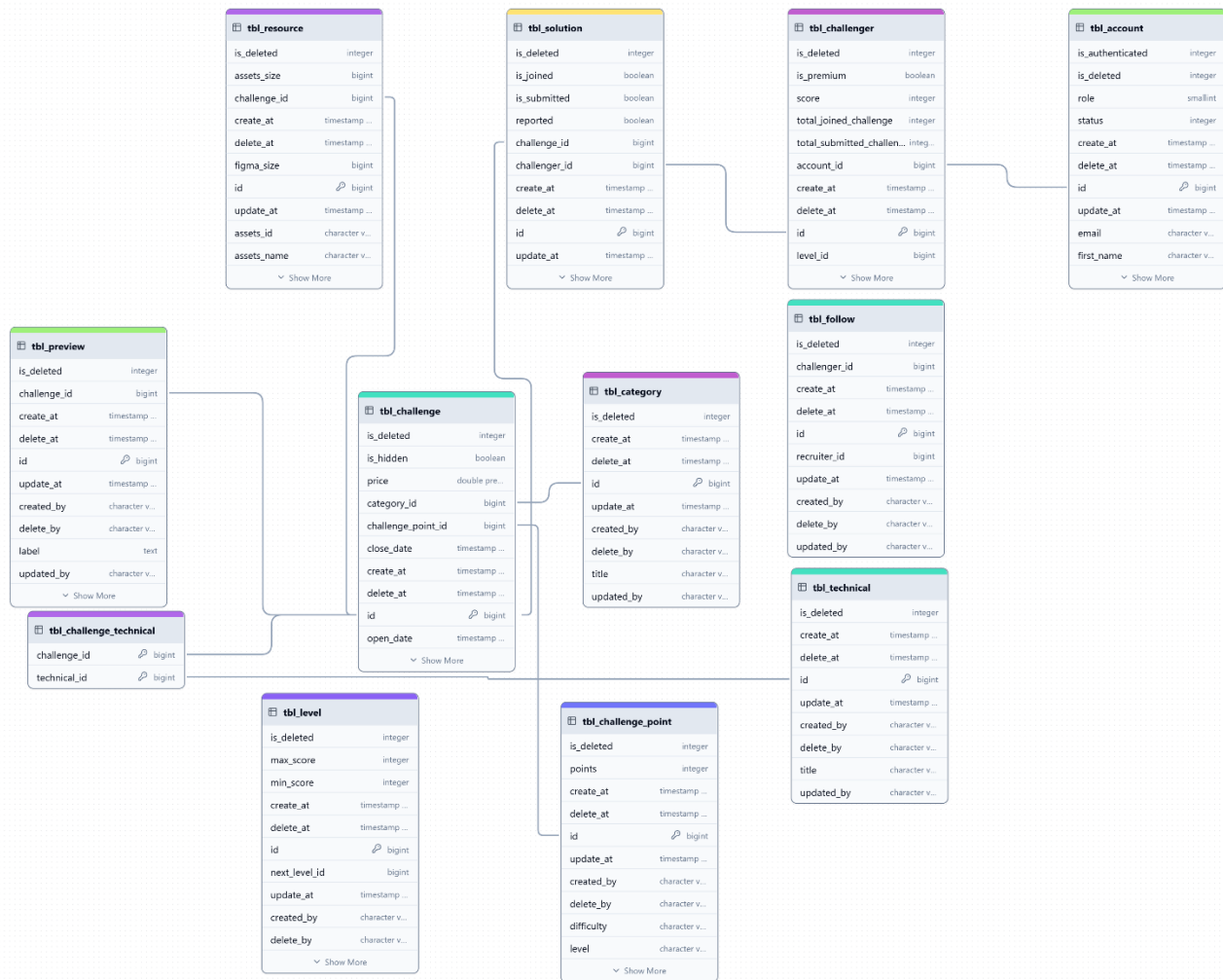


Figure 4 - 11 Relational Database Schema

4.5 Describing Details Database

4.5.1 Table *tbl_account*

Table 4 - 9 Account

No.	Attribute	Data Type	Null	Key	Description
1	id	ObjectID	Not Null	PK	Primary key, unique account ID
2	email	String	Not Null	Unique	User's email

3	password	String	Not Null		Encrypted password
4	first_name	String	Not Null		User's first name
5	last_name	String	Not Null		User's last name
6	phone	String	Not Null		User's phone number
7	role	String	Not Null		User's role
8	status	Boolean	Not Null		Account status
9	is_authenticated	Boolean	Not Null		Authentication status
10	is_deleted	Boolean	Not Null		Indicates if the account is deleted
11	create_at	timestamps	Not Null		Account creation timestamp
12	update_at	timestamps	Not Null		Account update timestamp
13	delete_at	timestamps	Null		Account deletion timestamp
14	avatar	String	Null		URL for the user's avatar

15	banner	String	Null		URL for the user's banner
16	created_by	ObjectID	Null	FK	User who created the account
17	delete_by	ObjectID	Null	FK	User who deleted the account
18	updated_by	ObjectID	Null	FK	User who last updated the account
19	refresh_token	String	Null		Token used for refreshing authentication

4.5.2 Table *tbl_challenge*

Table 4 - 10 Challenge

No.	Attribute	Data Type	Null	Key	Description
1	id	ObjectID	Not Null	PK	Primary key, unique identifier for the challenge
2	title	String	Not Null		Title of the challenge
3	description	String	Null		Detailed description of the challenge

4	category_id	ObjectID	Not Null	FK	Foreign key, ID of the related category
5	challenge_point_id	ObjectID	Not Null	FK	Foreign key, ID of the related challenge point
6	price	Number	Not Null		Price associated with the challenge
7	type_challenge	String	Not Null		Type or format of the challenge
8	open_date	timestamps	Not Null		Date and time when the challenge opens
9	close_date	timestamps	Not Null		Date and time when the challenge closes
10	is_deleted	Boolean	Not Null		Indicates whether the challenge is deleted

11	is_hidden	Boolean	Not Null		Indicates whether the challenge is hidden
12	status	Boolean	Not Null		Current status of the challenge
13	assets	Array	Null		List of assets related to the challenge
14	banner	String	Null		URL for the challenge banner image
15	brief	String	Null		Brief description or overview
16	message	String	Null		Additional message or note about the challenge
17	challenge_code	String	Null		Unique code for the challenge
18	create_at	timestamps	Not Null		Timestamp for when the challenge was created

19	update_at	timestamps	Not Null		Timestamp for the last update
20	delete_at	timestamps	Null		Timestamp for when the challenge was deleted
21	created_by	ObjectID	Null	FK	ID of the user who created the challenge
22	delete_by	ObjectID	Null	FK	ID of the user who deleted the challenge
23	updated_by	ObjectID	Null	FK	ID of the user who last updated the challenge

4.5.3 Table *tbl_challenger*

Table 4 - 11 Challenger

No.	Attribute	Data Type	Null	Key	Description
1	id	ObjectID	Not Null	PK	Primary key, unique identifier for the profile
2	account_id	ObjectID	Not Null	FK	Foreign key, ID of the

					associated account
3	level_id	ObjectID	Not Null	FK	Foreign key, ID of the associated level
4	is_premium	Boolean	Not Null		Indicates if the user has a premium profile
5	score	Number	Not Null		Total score earned by the user
6	total_joined_challenge	Number	Not Null		Total number of challenges joined
7	total_submitted_challenge	Number	Not Null		Total number of challenges submitted
8	bio	String	Null		Short biography or introduction
9	birthday	timestamps	Null		User's date of birth

10	gender	String	Null		Gender of the user
11	url_codepen	String	Null		URL to the user's CodePen profile
12	url_cv	String	Null		URL to the user's CV
13	url_git_lab	String	Null		URL to the user's GitLab profile
14	url_github	String	Null		URL to the user's GitHub profile
15	url_linked_in	String	Null		URL to the user's LinkedIn profile
16	url_portfolio	String	Null		URL to the user's portfolio
17	url_stack_overflow	String	Null		URL to the user's Stack Overflow profile

18	is_deleted	Boolean	Not Null		Indicates if the profile is deleted
19	create_at	timestamps	Not Null		Timestamp for when the profile was created
20	update_at	timestamps	Not Null		Timestamp for the last update
21	delete_at	timestamps	Null		Timestamp for when the profile was deleted
22	created_by	ObjectID	Null	FK	ID of the user who created the profile
23	updated_by	ObjectID	Null	FK	ID of the user who last updated the profile
24	delete_by	ObjectID	Null	FK	ID of the user who deleted the profile

4.5.4 Table *tbl_solution*

Table 4 - 12 Solution

No.	Attribute	Data Type	Null	Key	Description
1	id	ObjectID	Not Null	PK	Primary key, unique identifier for the solution
2	challenge_id	ObjectID	Not Null	FK	Foreign key, ID of the associated challenge
3	challenger_id	ObjectID	Not Null	FK	Foreign key, ID of the user who submitted the solution
4	solution_code	String	Not Null	Unique	Unique code for the solution
5	title	String	Not Null		Title of the solution
6	description	String	Null		Detailed description of the solution
7	note	String	Null		Additional notes about the solution
8	status_solution	String	Not Null		Current status of the solution (e.g., active, inactive)
9	url_product	String	Null		URL to the solution product

10	url_repository	String	Null		URL to the solution repository
11	is_joined	Boolean	Not Null		Indicates if the user has joined the challenge
12	is_submitted	Boolean	Not Null		Indicates if the solution has been submitted
13	reported	Boolean	Not Null		Indicates if the solution has been reported
14	is_deleted	Boolean	Not Null		Indicates if the solution is deleted
15	create_at	timestamps	Not Null		Timestamp for when the solution was created
16	update_at	timestamps	Not Null		Timestamp for the last update
17	delete_at	timestamps	Null		Timestamp for when the solution was deleted
18	created_by	ObjectID	Null	FK	ID of the user who created the solution
19	updated_by	ObjectID	Null	FK	ID of the user who last updated the solution

20	delete_by	ObjectID	Null	FK	ID of the user who deleted the solution
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4.5.5 Table *tbl_preview*

Table 4 - 13 Preview

No.	Attribute	Data Type	Null	Key	Description
1	id	ObjectID	Not Null	PK	Primary key, unique identifier for the label
2	challenge_id	ObjectID	Not Null	FK	Foreign key, ID of the associated challenge
3	label	String	Not Null	Unique	Text or identifier for the label
4	url	String	Null		URL associated with the label
5	is_deleted	Boolean	Not Null		Indicates if the label is deleted
6	create_at	timestamps	Not Null		Timestamp for when the label was created
7	update_at	timestamps	Not Null		Timestamp for the last update
8	delete_at	timestamps	Null		Timestamp for when the label was deleted

9	created_by	ObjectID	Null	FK	ID of the user who created the label
10	updated_by	ObjectID	Null	FK	ID of the user who last updated the label
11	delete_by	ObjectID	Null	FK	ID of the user who deleted the label

4.5.6 Table *tbl_resource*

Table 4 - 14 Resource

No.	Attribute	Data Type	Null	Key	Description
1	id	ObjectID	Not Null	PK	Primary key, unique identifier for the asset
2	challenge_id	ObjectID	Not Null	FK	Foreign key, ID of the associated challenge
3	assets_id	ObjectID	Not Null	Unique	ID of the asset
4	assets_name	String	Not Null		Name of the asset
5	assets_url	String	Not Null		URL to access the asset
6	assets_size	Number	Null		Size of the asset (in KB, MB, etc.)
7	figma_id	ObjectID	Null	Unique	ID of the Figma file

8	figma_name	String	Null		Name of the Figma file
9	figma_url	String	Null		URL to access the Figma file
10	figma_size	Number	Null		Size of the Figma file (in KB, MB, etc.)
11	is_deleted	Boolean	Not Null		Indicates if the asset is deleted
12	create_at	timestamps	Not Null		Timestamp for when the asset was created
13	update_at	timestamps	Not Null		Timestamp for the last update
14	delete_at	timestamps	Null		Timestamp for when the asset was deleted
15	created_by	ObjectID	Null	FK	ID of the user who created the asset
16	updated_by	ObjectID	Null	FK	ID of the user who last updated the asset
17	delete_by	ObjectID	Null	FK	ID of the user who deleted the asset

4.5.7 Table *tbl_technical*

Table 4 - 15 Technical

No.	Attribute	Data Type	Null	Key	Description
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1	id	ObjectID	Not Null	PK	Primary key, unique identifier for the record
2	title	String	Not Null		Title of the record
3	is_deleted	Boolean	Not Null		Indicates if the record is deleted
4	create_at	timestamps	Not Null		Timestamp for when the record was created
5	update_at	timestamps	Not Null		Timestamp for the last update
6	delete_at	timestamps	Null		Timestamp for when the record was deleted
7	created_by	ObjectID	Null	FK	ID of the user who created the record
8	updated_by	ObjectID	Null	FK	ID of the user who last updated the record
9	delete_by	ObjectID	Null	FK	ID of the user who deleted the record

4.5.8 Table *tbl_challenge_point*

Table 4 - 16 Challenge Point

No.	Attribute	Data Type	Null	Key	Description
1	challenge_id	ObjectID	Not Null	FK	Foreign key, ID of the associated challenge
2	technical_id	ObjectID	Not Null	FK	Foreign key, ID of the associated technical requirement

4.5.9 Table *tbl_level*

Table 4 - 17 Level

No.	Attribute	Data Type	Null	Key	Description
1	id	ObjectID	Not Null	PK	Primary key, unique identifier for the level
2	level	String	Not Null		Name or title of the level
3	min_score	Number	Not Null		Minimum score required for this level
4	max_score	Number	Not Null		Maximum score allowed for this level
5	next_level_id	ObjectID	Null	FK	Foreign key, ID of the next level

6	is_deleted	Boolean	Not Null		Indicates if the level is deleted
7	create_at	timestamps	Not Null		Timestamp for when the level was created
8	update_at	timestamps	Not Null		Timestamp for the last update
9	delete_at	timestamps	Null		Timestamp for when the level was deleted
10	created_by	ObjectID	Null	FK	ID of the user who created the level
11	updated_by	ObjectID	Null	FK	ID of the user who last updated the level
12	delete_by	ObjectID	Null	FK	ID of the user who deleted the level

CHAPTER 5: PROJECT MANAGEMENT

5.1 Project Analysis and Planning

The Front-Ice project is clearly planned with specific objectives and timelines. The versions are outlined clearly and executed consistently throughout the implementation process. Below are the goals to be achieved and the allocated timelines for the project.

Deadline on below including 7 sprint and have 11 weeks to completed the project



Figure 5 - 1 Plan Overview Project

Each sprint is designed with specific goals and deliverables to ensure steady progress:

- **Sprint 1:** Project setup, requirement gathering, and initial prototyping.
- **Sprint 2-3:** Core functionality development and foundational implementation.
- **Sprint 4-5:** Advanced functionalities and system integrations.
- **Sprint 6:** Final touches, performance optimization, and pre-launch checks.

- **Sprint 7:** Deployment, documentation, and project handover.

Functions	#	FE	#	BE	Tr	Details Functions	Status
Project Initialization	24,00	24,00				Project Analysis	Finished
	12,00	12,00				Draft FSD for Front-Ice Project	Finished
	1,00	7,00				Database Design	Finished
	1,50					UI/UX Design	Finished
	3,00	3,00				Back-end Configuration, Front-end Configuration	Finished
	9,00	9,00				Create API Documentation on Postman	Finished
	4,00	6,00				Set Up Environment for Project Deployment	Finished
User Registration & Authentication	1,00	1,00				Sign Up	Finished
	8,00	8,00				Log In	Finished
	2,00	6,00				Forgot Password	Finished
	3,00	5,00				Log Out	Finished
User Profile Management	5,00	8,00				User Profile	Finished
	8,00	12,00				Dashboard	Finished
						Details Functions	Finished
Challenge Marketplace Management	8,00	14,00				List challenge	Finished
	8,00	16,00				Details Challenge	Finished
	8,00	15,00				Store challenge	Finished
	8,00	15,00				Store challenge	Finished
						Details Functions	Finished
Challenge Managment	8,00	14,00				List Challenge	Finished
	8,00	14,00				Detail Challenge	Finished
	10,00	18,00				Submit Challenge	Finished
	8,00	16,00				Challenge system	Finished
	10,00	8,00				Recruiter of Challenge Storage	Finished
	10,00	8,00				Review challenge of recruiter	Finished
	8,00	8,00				Store challenge	Finished
	10,00	8,00				Posted Challenge	Finished
Challenger Management	10,00	8,00				List Challenger	Finished
Recruiter Management	10,00	8,00				Detail Recruiter	Finished
	10,00	8,00				List Recruiter	Finished
Solution Management	10,00	10,00				Solution System	Finished
	8,00	8,00				Report Solution	Finished
	9,00	8,00				List Solutions	Finished

Figure 5 - 2 List Function of Project

5.2 Task Delegation and Sprint Management

5.2.1 Identifying and Grouping Tasks by Service

Tasks are divided using the Agile method. Tasks are based on project features and split into BE tasks and FE tasks. The team gave ideas, so the task splitting was easy.

Below is the table for task splitting by progress and sprint, with details.

Service	Delegation	Who ?	Status	%
Project Initialization	Đã chia task	Phuc	Đã hoàn th...	100
User Registration & Authenticat...	Đã chia task	Phuc	Đã hoàn th...	100
User Profile Management	Đã chia task	Phuc	Đã hoàn th...	100
Challenge Managment	Đã chia task	Phuc	Đã hoàn th...	100
Challenger Management	Đã chia task	Phuc	Đã hoàn th...	100
Recruiter Management	Đã chia task	Phuc	Đã hoàn th...	100
Solution Management	Đã chia task	Phuc	Đã hoàn th...	100
Mentor Management	Bị chặn	Phuc	Bị chặn	0

Figure 5 - 3 Delegation Sprint

5.2.2 Sprint 1

This is the first important week of the project, the week to determine the direction for the whole project, this week, the whole team needs to understand the process and professional logic of the whole project, so it requires team members to meet together and find problems together in the implementation process.

- Here are the goals to achieve in **Sprint 1**:
 - Meeting to divide tasks for the whole team.
 - Database Design.
 - Plan for UX/UI.
 - Create a Document API that is used to communicate with the front-end.
 - Deploy Database.
 - Do the Registration and Login Functions.

Here's the task breakdown for the Back-end and Front-end in the first Sprint:1

2	Sprint 1				
3	Họp chia task, nghiên cứu	2,00	3,00		Meeting
4	Thiết kế Database	7,00	6,00		Project Initialization
5	Cấu hình dự án Source BE	5,00	8,00		Project Initialization
6	Phân Role trong source BE	2,00	2,00		Project Initialization
7	Tạo Document trên API Dog	2,00	2,00		Project Initialization
8	Deploy Database và Back-End	2,00	2,00		Project Initialization
9	API Login	3,00	3,00	Challenger Recruiter	User Registration & Authenticat...
11	API Login	2,00	2,00	Root Mentor Challenge Manager	User Registration & Authenticat...
12	API SignUp	4,50	4,00	Challenger Recruiter	User Registration & Authenticat...
13	API SentOTP	0,50	0,50	Challenger	User Registration & Authenticat...
14	API Verify-otp	0,50	0,50	Challenger Recruiter	User Registration & Authenticat...
15	API Forgot Password	2,00	1,00	Challenger Recruiter	User Registration & Authenticat...
16	API SentOTP	0,50	1,00	Challenger Recruiter	User Registration & Authenticat...
17	API Verify-otp	0,50	0,50	Challenger Recruiter	User Registration & Authenticat...
25	API Logout	2,00	0,50	Recruiter Root Mentor Challenge Manager Challenger	User Registration & Authenticat...

Figure 5 - 4 Sprint 1 Back-end

PRINT 1					
FE-01	Họp chia task, nghiên cứu	1,00	1,00		Meeting
FE-02	Init source	8,00	16,00		Project Initialization
FE-03	Vẽ figma	16,00	16,00		Design
FE-04	Tạo các components: button, input, alert	8,00	18,00		Project Initialization
FE-05	trang đăng nhập	3,20	8,00	Challenger	User Registration & Authenticat...
FE-06	trang đăng ký	3,20	1,00	Challenger	User Registration & Authenticat...
FE-07	trang quên mật khẩu	3,20	0,50	Challenger	User Registration & Authenticat...
FE-09	trang otp	3,20	1,50	Challenger	User Registration & Authenticat...
FE-11	làm layout	4,00	10,00	Challenger	User Profile Management
FE-12	private route	4,00	2,00		Project Initialization
FE-13	trang profile	16,00	16,00	Challenger Challenge Manager Mentor Recruiter Root	User Profile Management
FE-14	Refactor field component	2,00	2,00	Challenger	Project Initialization
FE-15	Install assets	1,00	1,00	Challenger	Project Initialization
FE-17	Checkbox component	1,00	1,00	Component	Component

Figure 5 - 5 Sprint 1 Front-end

5.2.3 Sprint 2

Sprint 2: This is an important stage in the whole project, initially to get to the big functions, this week it is mandatory to complete the big functions that are considered as the backbone of the whole project, so the whole team has worked very hard when they have completed the main thread functions in the client page.

- Here are the goals to achieve in Sprint 2:
 - Profile Functions
 - System Challenge Functionality
 - File Processing and File Download Functions
 - Complete Components for Sprint 3
 - Search function, filter.

Sprint 2					
Họp Kick off Sprint 2	2,00	3,00			Meeting
Kết Nối API với Front-end	4,00	1,00			Meeting
API Detail Profile	5,00	5,00	Challenger Challenge Manager Mentor Recruiter		User Profile Management
API ChangePassword	1,00	2,00	Challenger Challenge Manager Mentor Recruiter		User Profile Management
API Update Profile	5,00		Challenger Challenge Manager Mentor Recruiter Root		User Profile Management
API Upload CV (File)	2,00	4,00	Challenger		User Profile Management
API Danh sách Challenge + Premium Challenge	4,00		Challenger Challenge Manager Mentor Recruiter Root		Challenge Managment
API Filter Danh sách Challenge	2,00		Challenger Challenge Manager Recruiter Mentor		Challenge Managment
API List Challenge Recruiter	3,00		Challenge Manager		Challenge Managment
API Filter Danh sách Challenge Recruiter	3,00		Challenge Manager		Challenge Managment
API List Challenge Review	3,00		Challenge Manager		Challenge Managment
API List Challenge có thể mua	3,00	3,00	Recruiter		Challenge Managment
API Filter danh sách challenge có thể mua (Pop-up)	2,00		Recruiter		Challenge Managment
API Danh sách challenge đã mua	3,00		Recruiter		Challenge Managment
API Filter danh sách challenge đã mua	2,00		Recruiter		Challenge Managment

Figure 5 - 6 Sprint 2 Back-end

PRINT 2							
Challenge Component	1,50	1,50		Challenge Management	Challenges Page	Statistic Page	
Solution Component				Solution Management	Profile Page	Solutions Page	
FilterWrapper Component				Component	Challenges Page	Challenges Page	
RecruiterCompany Component				Component	Profile Page		
ChallengeOverview Component	2,00	2,00		Component	Challenge Details Page	Submit Challenge Page	
NoteOfChallenge Component	2,00	2,00		Component	Solutions Details Page		
AssetsProvidedOfChallenge Component	2,00	1,00		Component	Challenge Details Page		
MessageOfChallenge Component	2,00	1,00		Component	Challenge Details Page		
DownloadAssets Component	2,00	1,00		Component	Challenge Details Page		
DownloadFigma Component	2,00	2,00		Component	Challenge Details Page		
FaqComponent	2,00	2,00		Component	Challenge Details Page		
DescriptionPreviewCard Component	2,00	2,00		Component	Submit Challenge Page		
RecruiterProfileHeader Component	2,00	1,50		Component	Recruiter Company Page		
RecruiterInformation Component	2,00	0,50		Component	Recruiter Company Page		
RecruiterDescription component	2,00	0,50		Component	Recruiter Company Page		
SolutionDetails Component				Component	Solutions Details Page		
Feedback Component				Component	Solutions Details Page	Common	
Comment Component				Component	Solutions Details Page	Common	

Figure 5 - 7 Sprint 2 Front-end

5.2.4 Sprint 3

Sprint 3: This is a Sprint that has completed the core functions in the project of the Client section such as the submission process of the Challenge and the functions of the comment section related to the project, at this time the Back-end must complete important APIs to pair with the Front-end while at this time there is a bug, the team meeting is still continued so that it can be discussed with the members in the team about the way and operation of the project. At the same time, the API must be combined to give the front-end time to handle the data from the back-end.

- Here are the goals that need to be completed in Sprint 3:
 - Solution Functions
 - Comment Function
 - Client overview reaches 80%
 - Dashboard Interface

Sprint 3						
API Update Profile			Challenger Challenge Manager Mentor Recruiter Root		User Profile Management	
API Danh sách Challenge + Premium Challenge	8,00	16,00	Challenger Challenge Manager Mentor Recruiter Root		Challenge Managment	
API Login Google	3,00	3,00	Challenger Recruiter		User Registration & Authenticat...	
API Get Detail Challenge	16,00		Challenger		Challenge Managment	
API Download Assets	5,00	5,00	Challenger Challenge Manager		Challenge Managment	
API Download Figma File	2,00	2,00	Challenger Challenge Manager		Challenge Managment	
Method Tính điểm và trả về rank của challenger	8,00	8,00	Challenger		User Profile Management	
API Tìm kiếm Challenge bằng Elasticsearch	5,00		Challenge Manager Challenger Recruiter		Challenge Managment	
API Upload Image Desktop	1,00	1,00	Challenge Manager		Challenge Managment	
API Upload Image Mobile	1,00	0,50	Challenge Manager		Challenge Managment	
API Upload Image Tablet	1,00	0,25	Challenge Manager		Challenge Managment	
API Create Challenge (Challenge system)			Challenge Manager		Challenge Managment	
API Submit Challenge + Join Challenge	16,00	16,00	Challenger		Challenge Managment	
Refactor BaseService	0,50	0,50			Project Initialization	
Refactor API SignUp + Tạo 2 record solution cho challenge mẫu	6,00	5,00	Challenger		User Registration & Authenticat...	
Refactor Entity + Liên kết table	0,50	0,50			Project Initialization	

Figure 5 - 8 Sprint 3 Back-end

SPRINT 3

Home Page	1,00	2,00	Challenger	Page	Home Page
Statistic Page	1,00	2,00	Challenger	Page	Statistic Page
Fake Api based structure response define f	1,00	1,00	Challenger	Logic	Other
Challenge skeleton	1,00	1,00	Challenger	Component	Common
Solution skeleton	1,00	1,00	Challenger	Component	Common
Text skeleton	1,00	1,00	Challenger	Component	Common
Box skeleton	1,00	1,00	Challenger	Component	Common
Axios handle interceptor	1,00	1,00	Challenger	Project Initialization	Common
Handler Error response in axios	2,00	1,00	Challenger	Project Initialization	Common
UI Empty challenge	2,00	1,00	Challenger	Logic	Challenges Page
UI Empty solution	2,00	1,00	Challenger	Logic	Solutions Page
Section title component	2,00	1,00	Component	Component	Common
Button Back previous page authentication	2,00	1,00	Component	Logic	Common
Options select component	2,00	1,00	Component	Component	Common
Options select multiple	2,00	1,00	Component	Component	Common
Component Empty			Component	Component	Common

Figure 5 - 9 Sprint 3 Front-end

5.2.5 Sprint 4

Sprint 4: This is a sprint that must complete the remaining functions in the client page such as the payment section and fix other bugs that are being implemented in the project, so it is necessary to be careful in each part to implement the process and at the same time increase the project progress. Prepare the interface components for the Admin page and related sections to look like Sprint 5 6.

- Here are the goals that need to be completed in **Sprint 4**:
 - Complete the payment function
 - Fixed bugs that could occur in Sprints 1, 2, 3
 - Complete 100% Client interface.
 - Component Preparation for Sprint 5.6.

Sprint 4					
Họp phân tích Sprint 4	4,00	4,00		Meeting	
Họp chia task Spring 4	4,00	3,00		Meeting	
API Get Detail Challenge	16,00		Challenger	Challenge Management	
API Create Challenge (Challenge system)	16,00		Challenge Manager	Challenge Management	
API Submit Challenge + Join Challenge	16,00	16,00	Challenger	Challenge Management	
API Danh Sách My Solution	8,00		Challenger Recruiter	Solution Management	
API Filter Solution	8,00		Challenger Recruiter	Solution Management	
API Detail Solution	8,00		Challenger Recruiter Challenge Manager	Solution Management	
API Get List Challenge đang Join	0,50		Challenger Challenge Manager Recruiter	Challenge Management	
API Get List Solution của challenger khác (bản thân đã submit)	16,00	16,00	Challenger Recruiter	Solution Management	
Fix API get Profile	2,00	3,00	Challenger	User Profile Management	
Chức năng tính totalChallengeSubmitted	1,00	1,00	Challenger Recruiter	User Profile Management	
Chức năng tính totalChallengeJoined	1,00	1,00	Challenger Recruiter	User Profile Management	
NextLevel	3,00	2,00	Challenger	User Profile Management	
Fix API Register : Gán lại cho tài khoản mới là newbie	1,00	0,50	Challenger Recruiter	User Profile Management	
Set up soft delete	0,50	0,50		Project Initialization	
refactor Entity	0,50	0,50		Project Initialization	
Thêm Data vào Database	0,50	0,50		Project Initialization	

Figure 5 - 10 Sprint 4 Back-end

SPRINT 4						
Logout logic	2,00	2,00	Challenger	Logic	Other	
Setup react query with zustand	2,00	2,00	Challenger	Logic	Other	
Modifie setup axios	1,00	1,00		Project Initialization	Other	
Fix Bug UI OTP Page	0,30	0,30		Bug	Other	
Refactor routes import	0,30	0,30		Refactor	Other	
Recruiter Company Page			Challenger	Page	Recruiter Company Page	
Profile Challenger Page			Challenger	Page	Profile Page	
Solution List Page	3,00	3,00	Challenger	Page	Solutions Page	
Challenge List Page	2,00	2,00	Challenger	Page	Challenges Page	
Challenge Details Page	1,00	1,00	Challenger	Page	Challenges Page	
Solution Details Page			Challenger	Page	Solutions Page	
Submit Solution Page	1,00	1,00	Challenger	Page	Solutions Page	
Modifie import component solution			Challenger	Refactor	Other	

Figure 5 - 11 Sprint 4 Front-end

5.2.6 Sprint 5

Sprint 5: This is the stage to start implementing the Admin page of the project, the roles that need to be defined and completed as Challenge Manager, Mentor, Root. The functions that need to be performed are. The CRUD of the Manage page and its execution need to be time-optimized.

- Here are the goals to Achieve:
 - Challenge Manager Function
 - Mentor Function
 - Root Function
 - Edit the management interface to be compatible with the project.

Sprint 5						
BE-130	Fix Bug Miss retur Resource	0,50	0,25	Challenger	Hot Fix	
BE-131	API Download Resource	2,00	2,00	Challenger	Challenge Managment	
BE-132	Fix API check Challenger đã join hay chưa để download Assecs và Figma Bug	1,00	1,00	Challenger	Challenge Managment	
BE-133	API Get Solution Detail	4,00	4,00	Challenger	Challenge Managment	
BE-134	Hot Fix can't signup because challenger not found	0,25	0,75	Challenger	User Registration & Authenticat...	
BE-135	Tên Task					

Figure 5 - 12 Sprint 5 Back-end

SPRINT 5					
InformationAuthorComponent	1,50	1,50	Challenger	▼	Design
RecruiterProfileHeader Component	1,50	1,50	Challenger	▼	Design
RecruiterInformation Component	1,50	1,50	Challenger	▼	Design
RecruiterContact component	1,50	1,50	Challenger	▼	Design
Challenge Component	1,00	1,00	Challenger	▼	Design
Paginations Component	1,00	1,00	Challenger	▼	Design
ChallengeOverview Component	1,50	1,50	Challenger	▼	Component
Profile Page	1,50	1,50	Challenger	▼	Page
Solutions Page	1,50	1,50	Challenger	▼	Page
Challenges Page	1,50	1,50	Challenger	▼	Page
Recruiter Company Page	1,50	1,50	Challenger	▼	Page
Solution Details Page	1,50	1,50	Challenger	▼	Page
Submit Solution Page	1,50	1,50	Challenger	▼	Page
Challenge Details Page	1,50	1,50	Challenger	▼	Page

Figure 5 - 13 Sprint 5 Front-end

5.2.7 Sprint 6

Sprint 6: This is the stage where everything must be completed urgently and at the same time must be checked for errors in the management pages because there are bugs that occur during the implementation of the above sections.

- Here are the goals to be achieved
 - Complete 80% of the Mentor, Root, and Challenge Manager pages.
 - Retest old functions and prepare to deploy for Sprint 7

Sprint 6					
Hoàn thành Quy Trình Tham Gia Challenge					
BE-137	API Download Resource	3,00	2,00	Challenger	Challenge Management
BE-137	Fix API check Challenger đã join hay chưa để download Assecs và Figma Bug	2,00	2,00	Challenger	Hot Fix
BE-139	API Get Solution Detail	3,00	3,00	Challenger	Solution Management
BE-140	Fix API Get Detail Challenge hiển thị People Participated	3,00	3,00	Challenger	Challenge Management
BE-141	API Kết Nối API Join Và Submit Challenge và Test thử	3,00	3,00	Challenger	Challenge Management
BE-142	API List challenge recruiter	3,00	3,00	Challenger	Challenge Management
BE-143	API Filter challenger recruiter	3,00	3,00	Challenger	Challenge Management
BE-144	API Hiển thị thông tin solution	4,00	44,00	Challenger	Solution Management
BE-145	API Get List Solution của challenger khác có cùng loại challenge (Những challenge khác)	5,00		Challenger	Solution Management
BE-145 - 5	API: Get list Solution của challenger và các challenger khác (Cùng loại Challenge)	5,00	5,00	Challenger	Solution Management
BE-146	API Danh Sách My Solution	3,00	3,00	Challenger	Solution Management
BE-147	API Filter Solution	3,00	3,00	Challenger	Solution Management
Statistic Page					
Overview					
Phan Nguyen phannguyen@gmail.com					
Account Level Score Challenge Score					
Normal NEWBIE 0 0 Solution					

Figure 5 - 14 Sprint 6 Back-end

5.2.8 Sprint 7

Sprint 7: This is the last week of the project where you have to check everything and try to achieve 100% of the project progress and at the same time write reports when working on the parts of the project.

- Here are the goals that need to be achieved
 - 100% completion of the project
 - Check all functions again
 - Deploy lên Production

Sprints	Start	End	Trạng thái	Thống (FE)	Ý (FE)	Trí (FE)	Phúc (BE)	Nghĩa (BE)	Nhưt (BE)	Tr	Ghi chú
Sprint 1	30/09/2024	05/10/2024	Done		40	40	40	40	40	40	Ghi chú
Sprint 2	07/10/2024	12/10/2024	Done		40	40	40	40	40	40	Ghi chú
Sprint 3	14/10/2024	19/10/2024	Done		40	40	40	40	40	40	Ghi chú
Sprint 4	21/10/2024	19/10/2024	Done			60	60	60	60	60	Ghi chú
Completed Version 1											Ghi chú
Sprint 5	28/10/2024	02/11/2024	Done			30	30	30	30	30	Ghi chú
Sprint 6	04/11/2024	23/11/2024	Done				30	30		30	Ghi chú
Sprint 7	25/11/2024	14/12/2024	Done				60	60		60	Ghi chú
Completed Version 2											Ghi chú

Figure 5 - 15 Include Sprint Before

Sprints	Start	End	Trạng thái	# Thống (FE)	# Ý (FE)	# Trí (FE)	# Phúc (BE)	Nghĩa (BE)	Nhưt (BE)	Tr	Ghi chú
Sprint 1	30/9/24	5/10/24	Done		31,0	35,0	48,0	31,50	23	30	Ghi chú
Sprint 2	7/10/24	12/10/24	Done		17,3	23,3	42,1	15,50	12	8	Ghi chú
Sprint 3	14/10/24	19/10/24	Done		7,3	15,5	17,0	17,75	21	37	Ghi chú
Sprint 4	21/10/24	19/10/24	Done			25,2	14,0	25,75	31	8,5	Ghi chú
Completed Version 1											Ghi chú
Sprint 5	28/10/24	2/11/24	Done				15,0	10,25		9,75	Ghi chú
Sprint 6	4/11/24	23/11/24	Done				30,0	28,50		25	Ghi chú
Completed Version 2											Ghi chú

Figure 5 - 16 Include Sprint After

CHAPTER 6: EXPERIMENT AND DEPLOYING SYSTEM

6.1 Setup environment

6.1.1 Setup develop environment

React Development Tools

Use **Visual Studio** to develop **Javascript** and **TypeScript**:

- + Free IDE that supports multiple programming languages
- + Has built-in support for **TypeScript and JavaScript**
- + IDE features such as IntelliSense, Debugging, etc....

```
"packages": {  
  "": {  
    "name": "frontice-panel",  
    "version": "0.0.0",  
    "dependencies": {  
      "@ant-design/icons": "^5.5.1",  
      "@emotion/react": "^11.13.3",  
      "@emotion/styled": "^11.13.0",  
      "@tanstack/react-query": "^5.59.0",  
      "@types/react-router-dom": "^5.3.3",  
      "antd": "^5.21.6",  
      "antd-img-crop": "^4.23.0",  
      "axios": "^1.7.7",  
      "dayjs": "^1.11.13",  
      "jwt-decode": "^4.0.0",  
      "react": "^18.3.1",  
      "react-dom": "^18.3.1",  
      "react-router-dom": "^6.28.0",  
      "react-toastify": "^10.0.5",  
      "sass": "^1.79.4",  
      "zustand": "^5.0.0-rc.2"  
    },  
  },  
}
```

Figure 6 - 1 Set Up React in Visua studio

Command-Line Tool

Table 6 - 1 Command – Line Tool in visual Studio

Tool	Purpose
Node	For running JavaScript code from command-line
npm	Node Package Manager Download new node package and features similar to Maven
tsc	TypeScript Compiler

Create a React application by

```
> npx create-react-app my-app
```

Running your new React application

Creates a brand new

```
> cd my-app
```

Jump into your new React

```
> npm start
```

1. Builds the app (compile / transpile)
2. Starts the server
3. Watches the source files

```
> set PORT=5100 && npm start
```

Server listens on port 5100
<http://localhost:5100>

Spring Development Tools

Use <http://start.spring.io> to setup environment for Spring project



Initializer project by this.
After download and open

Project

☒ Gradle - Groovy ☐ Gradle - Kotlin ☐ Maven

Language

☒ Java ☐ Kotlin ☐ Groovy

Spring Boot

☐ 3.2.0 (SNAPSHOT) ☐ 3.2.0 (M1) ☐ 3.1.3 (SNAPSHOT) ☒ 3.1.2 ☐ 3.0.10 (SNAPSHOT) ☐ 3.0.9 ☐ 2.7.15 (SNAPSHOT) ☐ 2.7.14

Figure 6 - 2 Spring initializr

Dependencies ADD DEPENDENCIES... CTRL + B

Spring Web WEB	☐
Build web, including RESTful, applications using Spring MVC. Uses Apache Tomcat as the default embedded container.	
Lombok DEVELOPER TOOLS	☐
Java annotation library which helps to reduce boilerplate code.	
Spring Boot DevTools DEVELOPER TOOLS	☐
Provides fast application restarts, LiveReload, and configurations for enhanced development experience.	
MySQL Driver SQL	☐
MySQL JDBC driver.	
Spring Security SECURITY	☐
Highly customizable authentication and access-control framework for Spring applications.	
Spring Data JPA SQL	☐
Persist data in SQL stores with Java Persistence API using Spring Data and Hibernate.	

Figure 6 - 3 Spring Dependencies

Figure 6. 1 Spring Dependencies

Import Project from file .zip be download from <http://start.spring.io>

Use IntelliJ to to run project and dependency

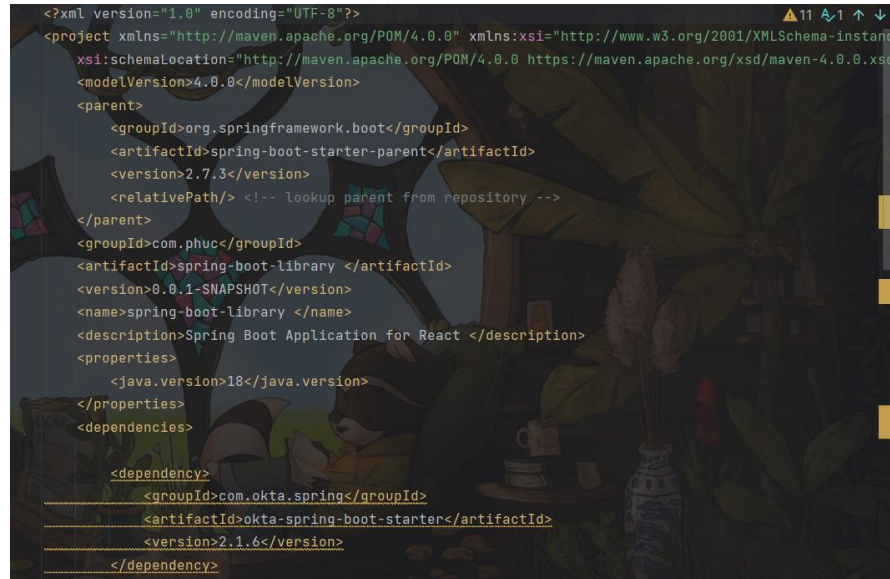


Figure 6 - 4 Dependencies in IntelliJ (pom.xml)

Test API Development Tools

Use PostMan to test URL when connect with server and use Method like GET,PUT,POST,DELETE

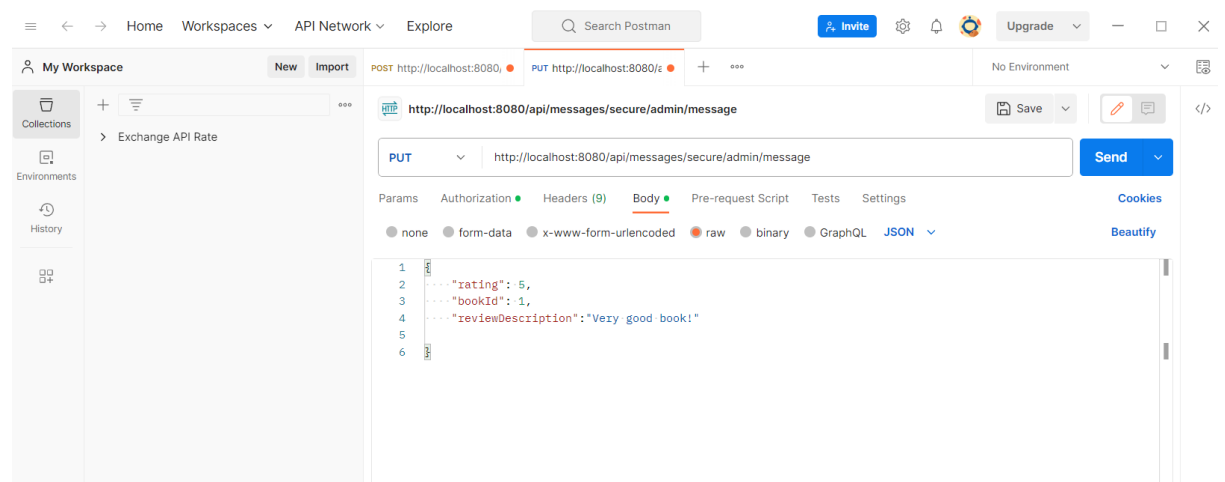


Figure 6 - 5 POSTMAN to use Mehtod GET,PUT,POST,DELETE

6.1.2 Database project Set-up

The project leverages PostgreSQL as the primary database management system to store and manage the core data of the application. This database serves as the backbone for handling essential entities such as Users, Challenges, and Solutions, ensuring data integrity, scalability, and efficient querying

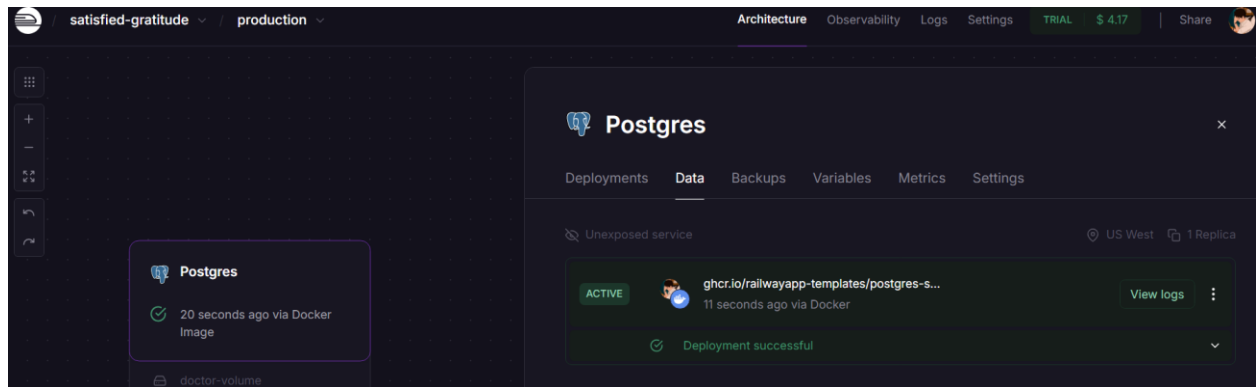


Figure 6 - 6 Database Set-Up

Using [Railway](#) to deploy databases to PostgreSQL using CI/CD technology, this makes it possible to deploy databases quickly.

After deplpoy, you can view the information necessary to use the database and

connect to the database management system through the **Variables** Here is the

inside information **Variables**:

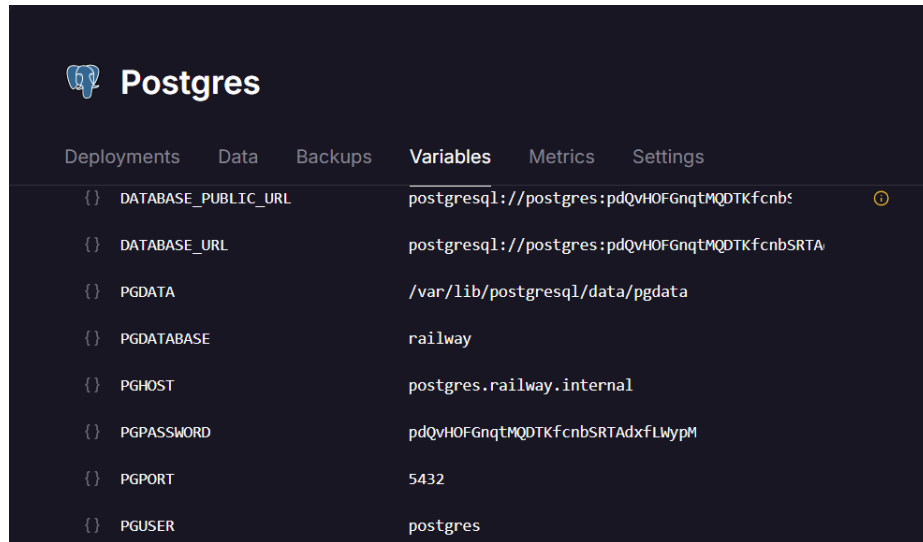


Figure 6 - 7 Variable Postgres

This is the details table in postgresSQL

Table 6 - 2 Variable in Railway

VARIABLE_NAME	VALUES
DATABASE_PUBLIC_URL	postgresql://postgres:pdQvHOFGnqtMQDTKfcnbSRTAdxflWypM@junction.proxy.rlwy.net:31426/railway
DATABASE_URL	jdbc:postgresql://junction.proxy.rlwy.net:31232/railway
PGDATA	/var/lib/postgresql/data/pgdata
PGHOST	postgres.railway.internal
PGPASSWORD	wEnMXuhjoHeZwNvnPkGPTGIEEwFvdZWY
PGPORT	5432
PGUSER	postgres
POSTGRES_DB	railway
POSTGRES_PASSWORD	pdQvHOFGnqtMQDTKfcnbSRTAdxflWypM
POSTGRES_USER	postgres

It will connect with server in IntelliJ via configuration in application.properties

```

1  spring.application.name=front-ice
2
3  #config database
4  ###
5  spring.datasource.driver-class-name= org.postgresql.Driver
6  spring.datasource.url=jdbc:postgresql://junction.proxy.rlwy.net:31232/railway
7  spring.datasource.username=postgres
8  spring.datasource.password=wEnMXuhjoHeZwNvnPkGPTG1EEwFvdZWY
9  spring.jpa.database-platform=org.hibernate.dialect.PostgreSQLDialect
10
11  spring.jpa.hibernate.ddl-auto=none
12  spring.jpa.properties.hibernate.format_sql=true
13
14  spring.jpa.show-sql=true
15  spring.jpa.open-in-view=true

```

Figure 6 - 8 application/properties

Using **Dbeaver** to open schemas in a database

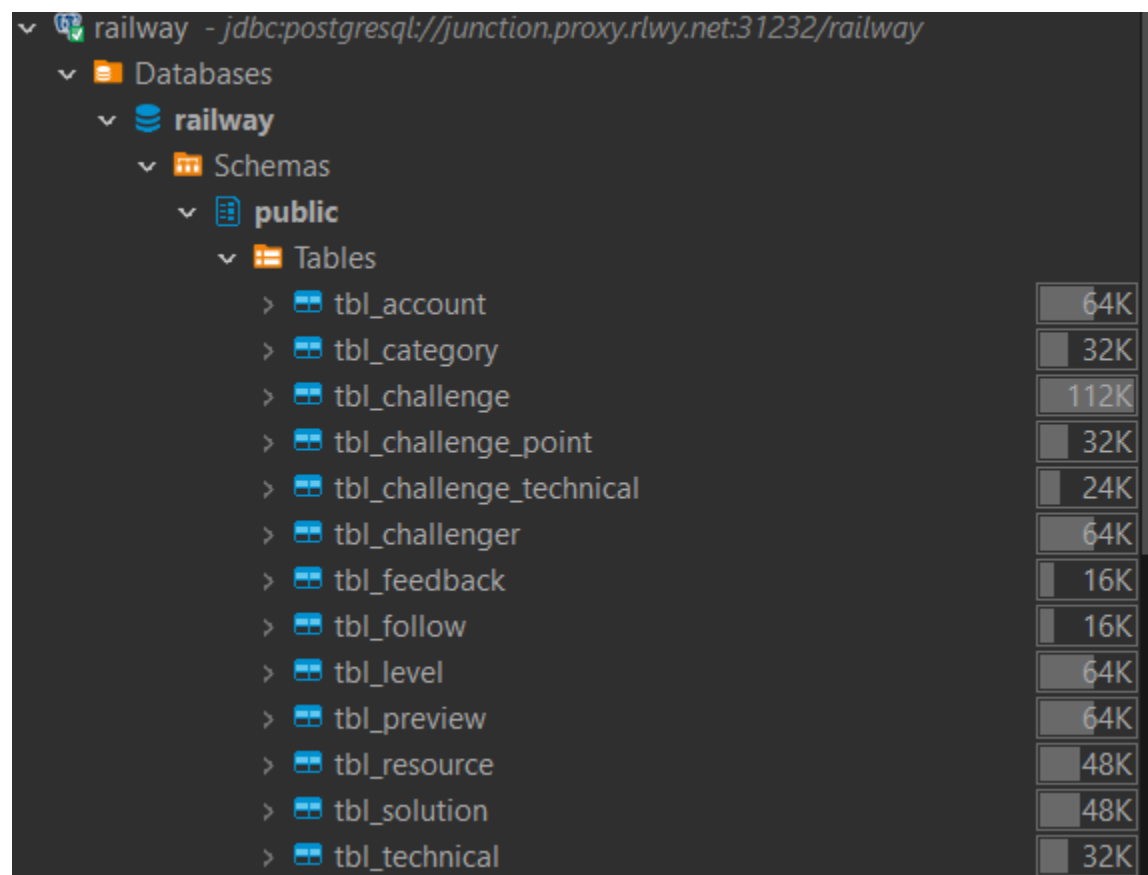


Figure 6 - 9 Dbeaver database

6.2 Client Page Interface

6.2.1 Register Page

The screenshot displays the Register Page interface. On the left, a dark blue gradient panel contains a 'Go back' button at the top left. Below it, the 'FRONTICE' logo is centered, followed by the heading 'Get Started With Us' and the text 'Complete these easy steps to login your account'. Two buttons are listed: '1 Enter Your Account Email' (highlighted) and '2 Go To Application'. On the right, a white 'Sign Up Account' form is shown. It includes a 'Github' login option, an 'OR' separator, and input fields for 'First Name', 'Last Name', 'E-Mail', 'Password', and 'Confirm Password'. A blue 'Register' button is at the bottom of the form, with a link to 'Login' for existing users.

Figure 6 - 10 Register Page

Detailed description of the Register page interface:

Table 6 - 3 Description Register Page

STT	SIGN	EVENT	DESCRIPTION
1	Window	Initialize	Display the registration page with the main layouts
2	[Back]	Click	Back to previous page
3	[Github]	Click	Log in with GitHub
4	[Sign Up]	Click	Register an account

5	[Language]	Select	Change the language
6	[Login]	Click	Go to the login page

API Sign-up for this Page

Sign up

Run

Generate Code

Delete

POST

/api/v1/auth/signup

Released

Created

October 14, 2024

Updated

2 months ago

Updated by

Front Ice

Creator

Front Ice

Maintainer

Not configured

Folder Auth

Mock >

Request

Body Params

application/json

Example

firstName

string

required

lastName

string

required

email

string

required

password

string

required

confirmPassword

string

required

role

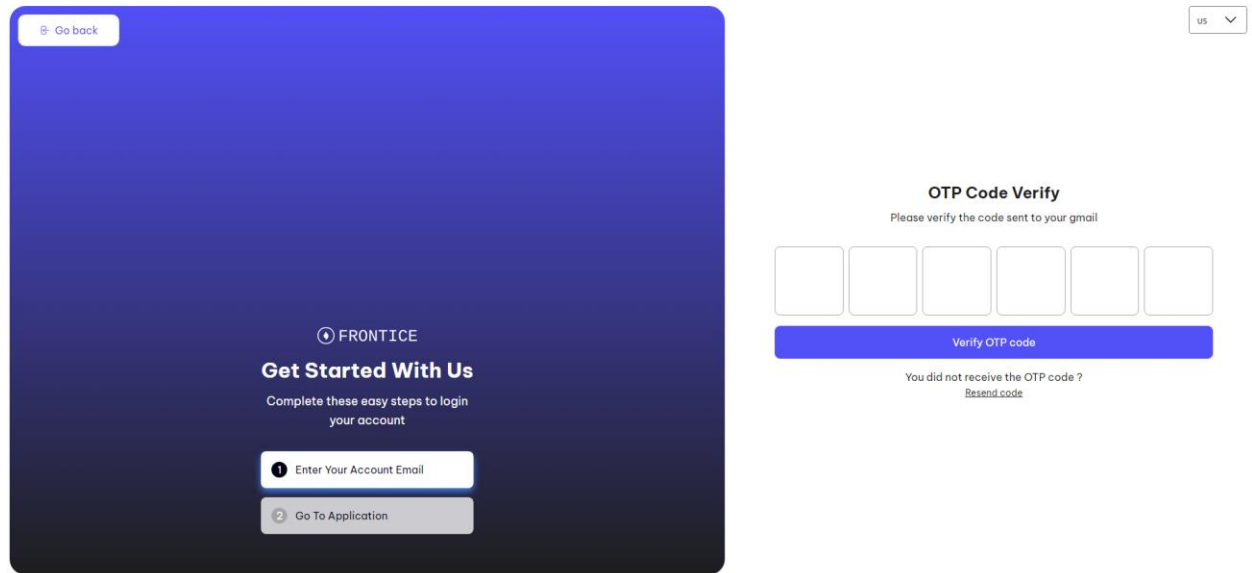
string

required

```
{
  "firstName": "firstName_3ec8e45a6c25",
  "lastName": "lastName_30e9d1f3d6c2",
  "email": "phucksk098@gmail.com",
  "password": "1",
  "confirmPassword": "1",
  "role": "CHALLENGER"
}
```

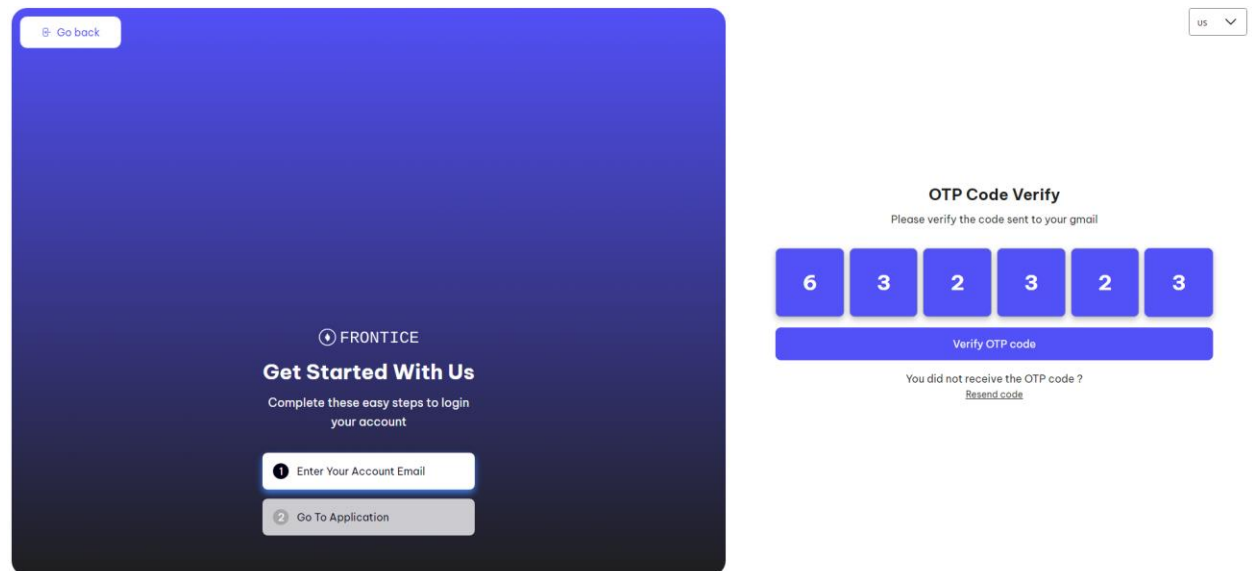
Figure 6 - 11 API Sign-up

6.2.2 Verify OTP Page



The screenshot shows the 'Verify OTP Page' interface. On the left, a dark blue gradient box contains the 'Get Started With Us' section, which includes a 'Go back' button, the 'FRONTICE' logo, and two steps: '1 Enter Your Account Email' and '2 Go To Application'. On the right, the 'OTP Code Verify' section is displayed, featuring a title, a subtitle 'Please verify the code sent to your gmail', a row of six empty input boxes, a 'Verify OTP code' button, and a 'Resend code' link.

Figure 6 - 12 Verify OTP Page not enter OTP



This screenshot is identical to the previous one, but the 'OTP Code Verify' section now shows the OTP code '632323' entered into the six input boxes. The 'Verify OTP code' button remains visible below the input boxes.

Figure 6 - 13 Verify OTP Page entered OTP

Detailed description of the Verify OTP page interface:

Table 6 - 4 Description Verify OTP Page

STT	SIGN	EVENT	DESCRIPTION
-----	------	-------	-------------

1	Window	Initialize	Display the registration page with the main layouts
2	[Back]	Click	Back to previous page
3	[Resend Code]	Click	Resend OTP

API send OTP for this Page

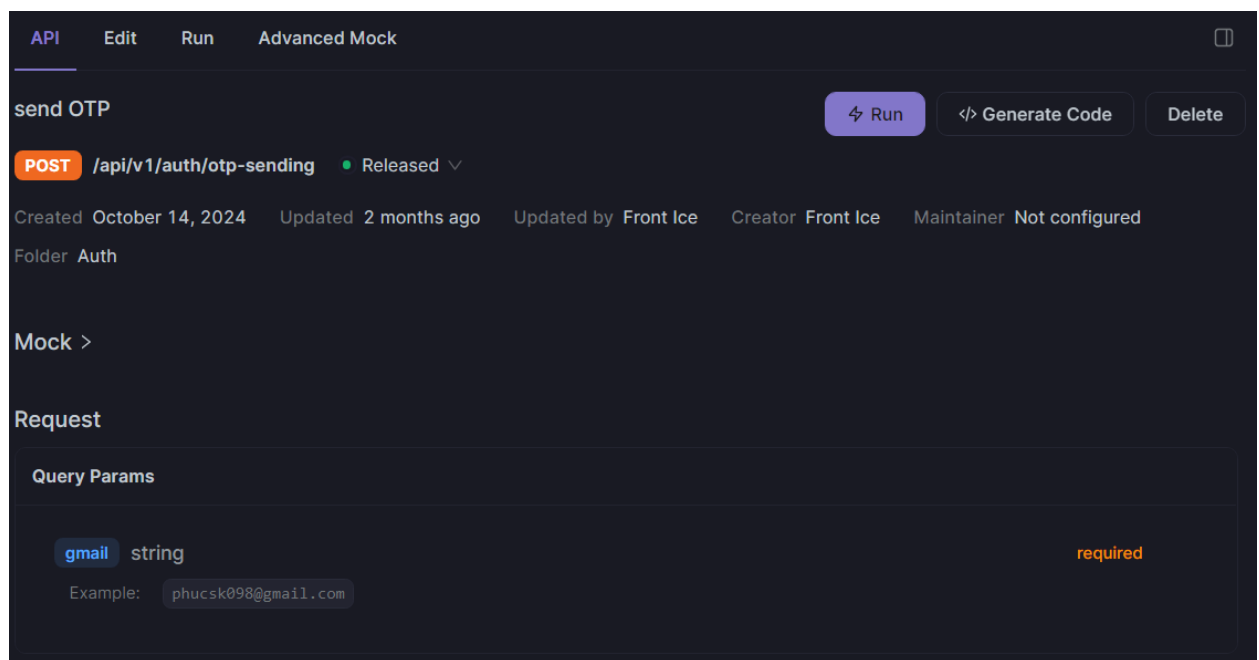


Figure 6 - 14 Sent OTP API

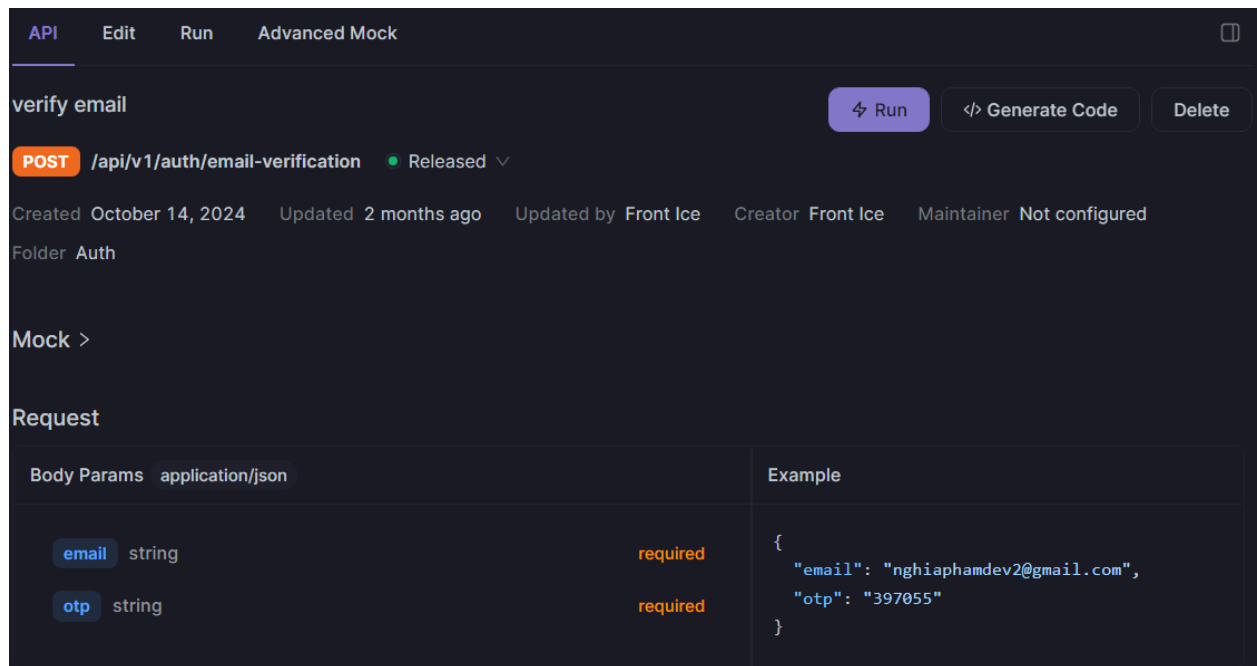


Figure 6 - 15 Verify Email

6.2.3 Login Page

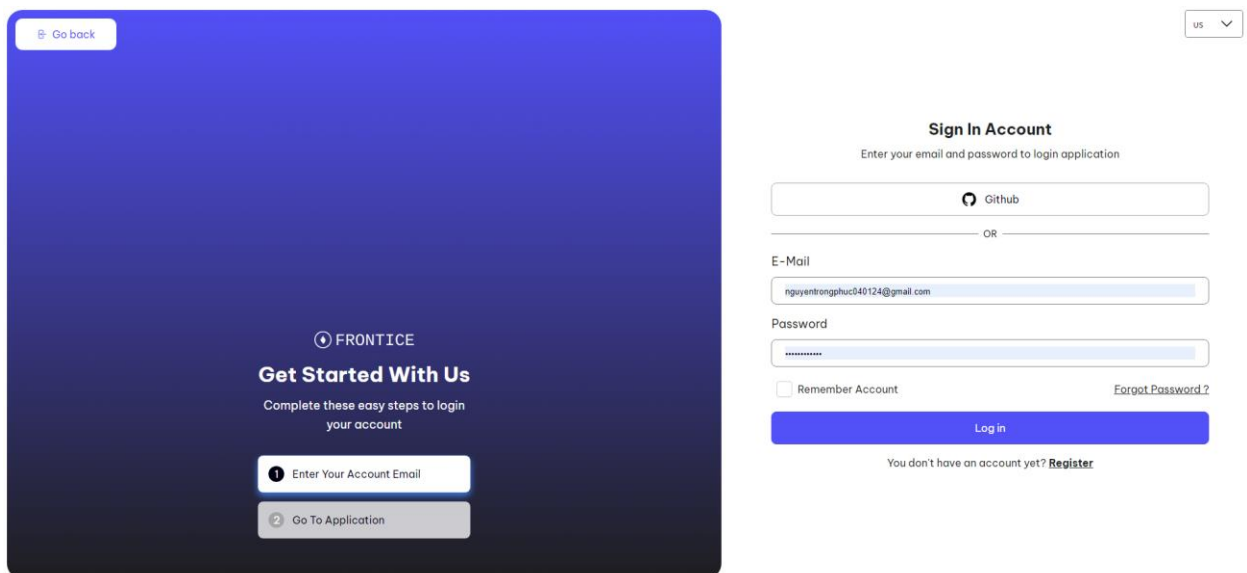


Figure 6 - 16 Login Page

Detailed description of the Login page interface:

Table 6 - 5 Description Login Page

STT	SIGN	EVENT	DESCRIPTION
1	Window	Initialize	Display the registration page with the main layouts
2	[Back]	Click	Back to previous page
3	[Github]	Click	Log in with GitHub
5	[Language]	Select	Change the language
6	[Login]	Click	Go to the login page

Login
Run
Generate Code
Delete

POST /api/v1/auth/login
Released

Created October 14, 2024
Updated 2 months ago
Updated by Front Ice
Creator Front Ice
Maintainer Front Ice

Folder Auth

Mock >

Request

Body Params	application/json	Example
email string email password string mật khẩu	required required	<pre>{ "email": "lenhut0929@gmail.com", "password": "123" }</pre>

Figure 6 - 17 Login API

6.2.4 Reset Password Page

Go back

US

Forgot Password
Enter your email to find account

E-Mail

Forgot password

< Return to previous page

Figure 6 - 18 Reset Password Page

Go back

US

OTP Code Verify
Please verify the code sent to your gmail

1 3 1 6 0

Verify OTP code

You did not receive the OTP code ?
[Resend code](#)

Figure 6 - 19 Reset Password Page Entered OTP

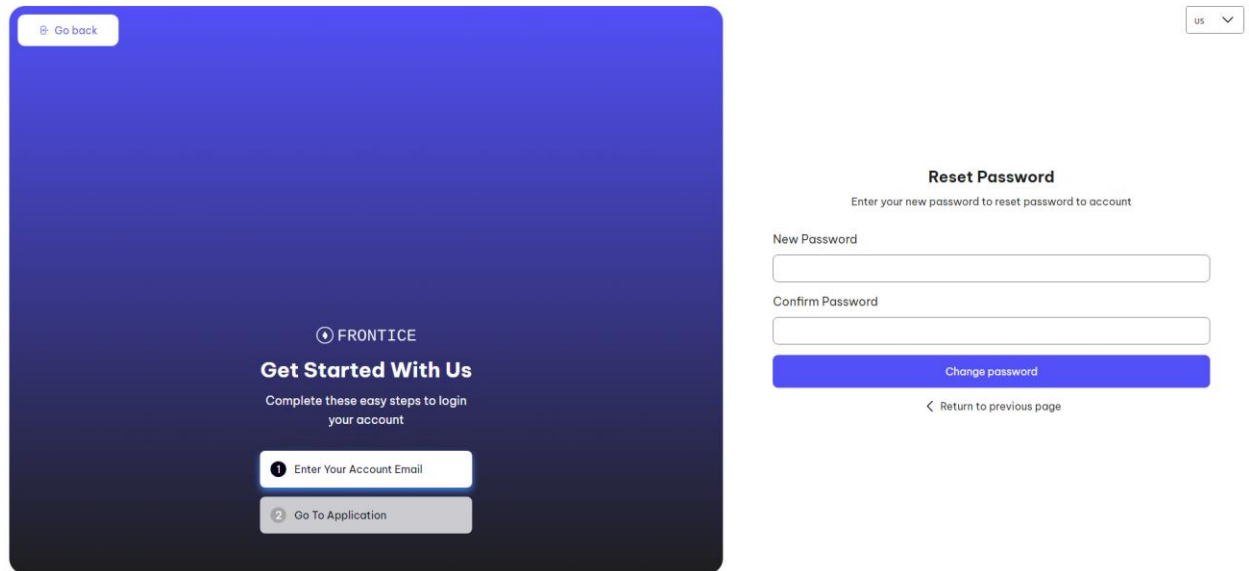


Figure 6 - 20 Reset Password Page

Detailed description of the Reset Password page interface:

Table 6 - 6 Description Reset Password Page

STT	SIGN	EVENT	DESCRIPTION
1	Window	Initialize	Display the registration page with the main layouts
2	[Back]	Click	Back to previous page
3	[Submit]	Click	Sent OTP to gmail of current account
5	[Language]	Select	Change the language
6	[Return to previous page]	Click	Back to previous page

API
Edit
Run
Advanced Mock

Forgot-Password

Run
Generate Code
Delete

POST
/api/v1/auth/forgot-password
Released

Created: October 14, 2024
Updated: a month ago
Updated by: Front Ice
Creator: Front Ice
Maintainer: Not configured

Folder: Auth

Mock >

Request

Body Params	application/json	Example
email string required	<pre>{ "email": "lenhut0929@gmail.com" }</pre>	

Figure 6 - 21 Forgot Password API

reset password

Run
Generate Code
Delete

POST
/api/v1/auth/reset-password
Released

Created: October 14, 2024
Updated: 2 months ago
Updated by: Front Ice
Creator: Front Ice
Maintainer: Not configured

Folder: Auth

Mock >

Request

Body Params	application/json	Example
resetToken string required	<pre>{ "resetToken": "c5295350-cb08-4d33-b6ff-651626462220", "newPassword": "123", "confirmPassword": "123" }</pre>	
newPassword string required		
confirmPassword string required		

Figure 6 - 22 Reset Password API

6.2.5 Home Page

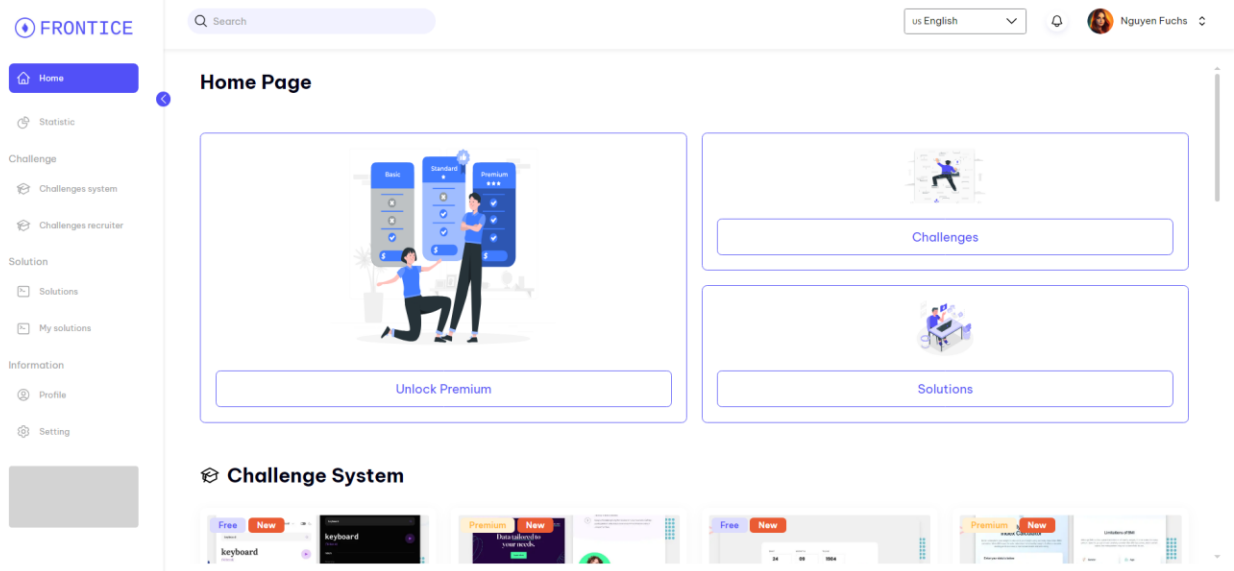


Figure 6 - 23 Home Page

Detailed description of the Home page interface:

Table 6 - 7 Description Home Page

STT	SIGN	EVENT	DESCRIPTION
1	Window	Initialize	Display the registration page with the main layouts
2	[Challenges	Click	Go to challenge page
3	[Solution]	Click	Go to solution page
4	[Unlock Premium]	Click	Go to page unlock account
5	[View Details]	Click	Go to details challenge

6	[View More]	Click	Expand list challenge or solution
7	[Slide Bar]	Click	Go to on the side bar

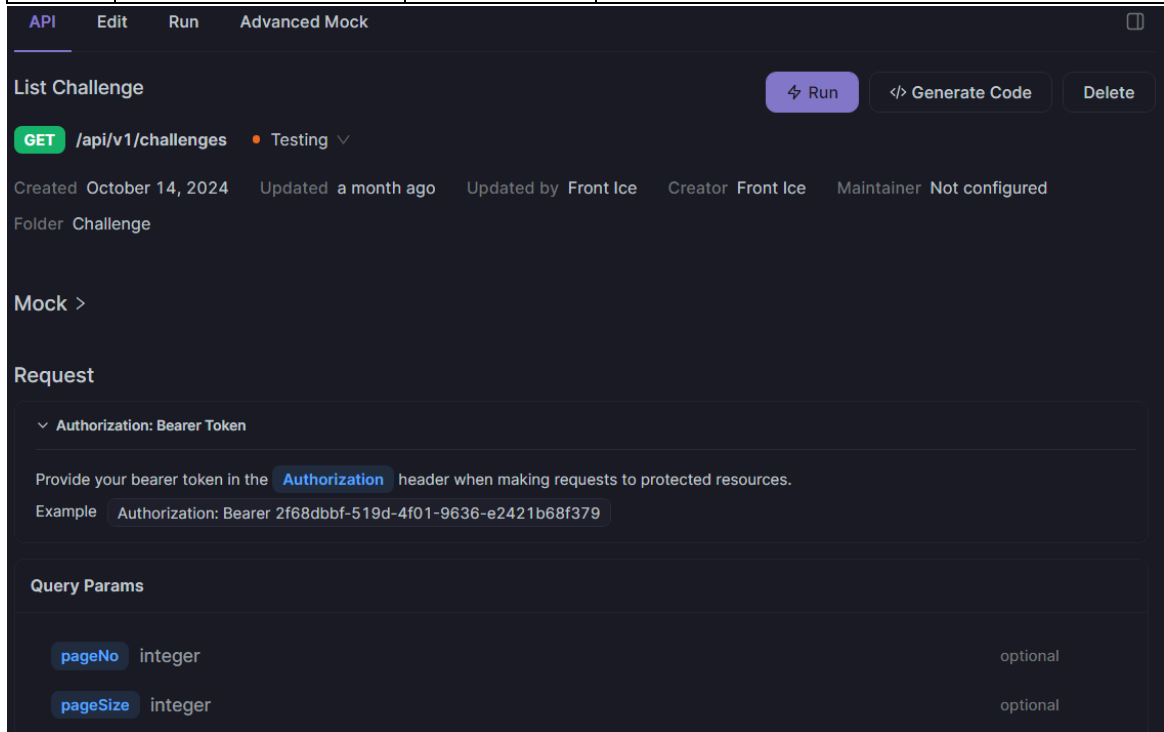


Figure 6 - 24 List Challenge API

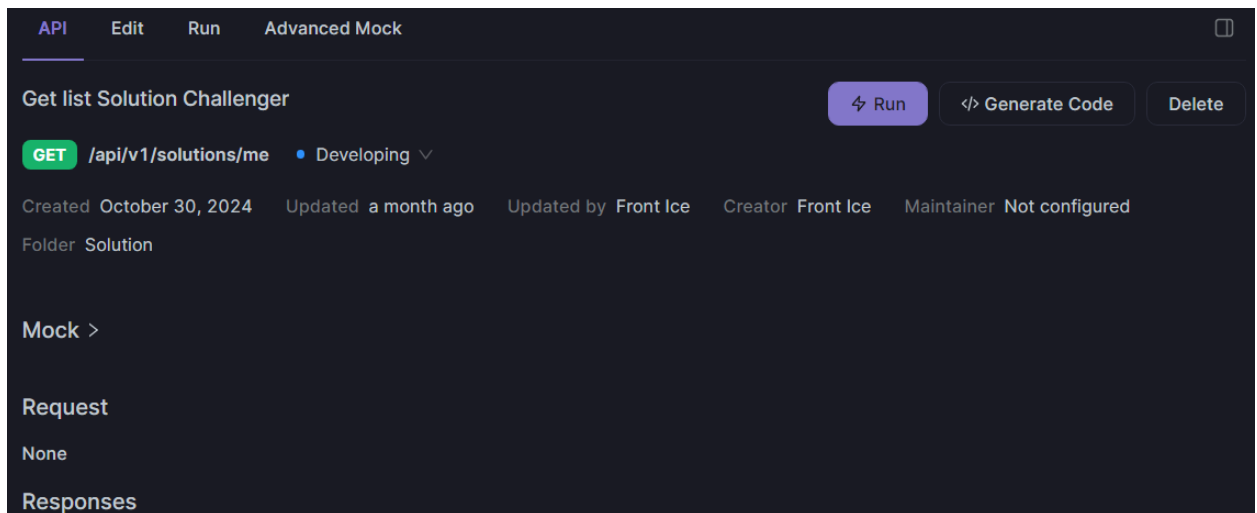


Figure 6 - 25 List Solution API

6.2.6 Statistic Page

FRONTICE Search us English Nguyen Fuchs

Statistic Page

Overview

Account	Level	Score	Challenge Submit
Nguyen Fuchs phuck098@gmail.com	Normal	NEWBIE	0

[View Details](#) [Challenges](#)

Incomplete Challenges

HTML CSS Link-sharing app High Diamond 125 Score

HTML CSS Link-sharing app High Diamond 125 Score

HTML CSS Link-sharing app High Diamond 125 Score

HTML CSS Link-sharing app High Diamond 125 Score

Nguyen Fuchs
NEWBIE

Tham Gia **Hoàn thành**
0 Challenges 0 Challenges

[View Profile](#)

Figure 6 - 26 Statistic Page

FRONTICE Search us English Nguyen Fuchs

My Solution

About 5 hours ago
Contact form
HTML CSS JAVASCRIPT
123 Score
High Diamond

About 5 hours ago
Contact form
HTML CSS JAVASCRIPT
123 Score
High Diamond

About 5 hours ago
Contact form
HTML CSS JAVASCRIPT
123 Score
High Diamond

About 5 hours ago
Contact form
HTML CSS JAVASCRIPT
123 Score
High Diamond

About 5 hours ago
Contact form
HTML CSS JAVASCRIPT
123 Score
High Diamond

About 5 hours ago
Contact form
HTML CSS JAVASCRIPT
123 Score
High Diamond

About 5 hours ago
Contact form
HTML CSS JAVASCRIPT
123 Score
High Diamond

About 5 hours ago
Contact form
HTML CSS JAVASCRIPT
123 Score
High Diamond

Figure 6 - 27 Statistic Page (My Solution)

Detailed description of the Statistic Page interface:

Table 6 - 8 Description Statistic Page

STT	SIGN	EVENT	DESCRIPTION
1	Window	Initialize	Display the registration page with the main layouts
2	[Challenges	Click	Go to challenge page
3	[Solution]	Click	Go to solution page
4	[Unlock Premium]	Click	Go to page unlock account
5	[Challenge Card]	Click	Go to details challenge
6	[Solution Card]	Click	Expand solution
7	[Slide Bar]	Click	Go to on the side bar

Get Challenger Overview

Run
Generate Code
Delete

GET /api/v1/challenger/overview
Testing

Created November 22, 2024
Updated 17 days ago
Updated by Front Ice
Creator Front Ice
Maintainer Not configured

Folder Challenger

Mock >

Request

Authorization: Bearer Token

Provide your bearer token in the Authorization header when making requests to protected resources.

Example Authorization: Bearer 2f68dbbf-519d-4f01-9636-e2421b68f379

Figure 6 - 28 Overview API

Get Joined Challenge but not submit

Run
Generate Code
Delete

GET /api/v1/challenges/joined
Testing

Created October 30, 2024
Updated 17 days ago
Updated by Front Ice
Creator Front Ice
Maintainer Not configured

Folder Challenge

Mock >

Request

Authorization: Bearer Token

Provide your bearer token in the Authorization header when making requests to protected resources.

Example Authorization: Bearer 2f68dbbf-519d-4f01-9636-e2421b68f379

Query Params

pageNo integer optional

pageSize integer optional

Figure 6 - 29 Get Joined Challenge API

6.2.7 Challenge System List Page

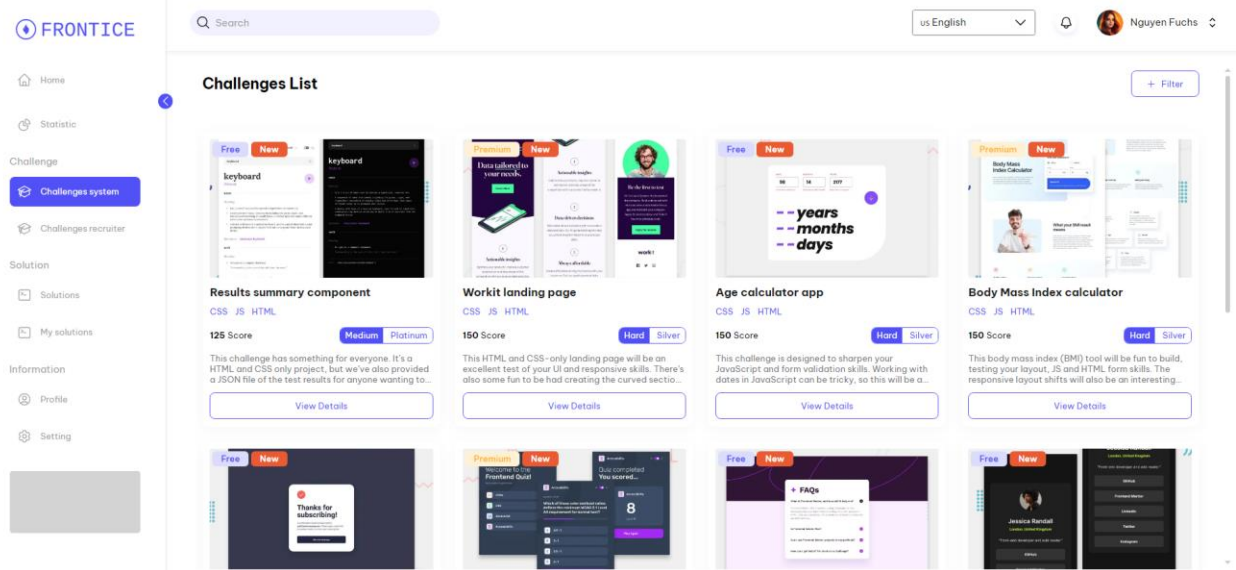


Figure 6 - 30 Challenges List Page

Detailed description of the Challenge System List Page interface:

Table 6 - 9 Description Challenge System List Page

STT	SIGN	EVENT	DESCRIPTION
1	Window	Initialize	Display the registration page with the main layouts
2	[View Details]	Click	Go to challenge details
3	[Filter]	Click	Show Filter to use
4	[Pagination]	Click	Paginate page

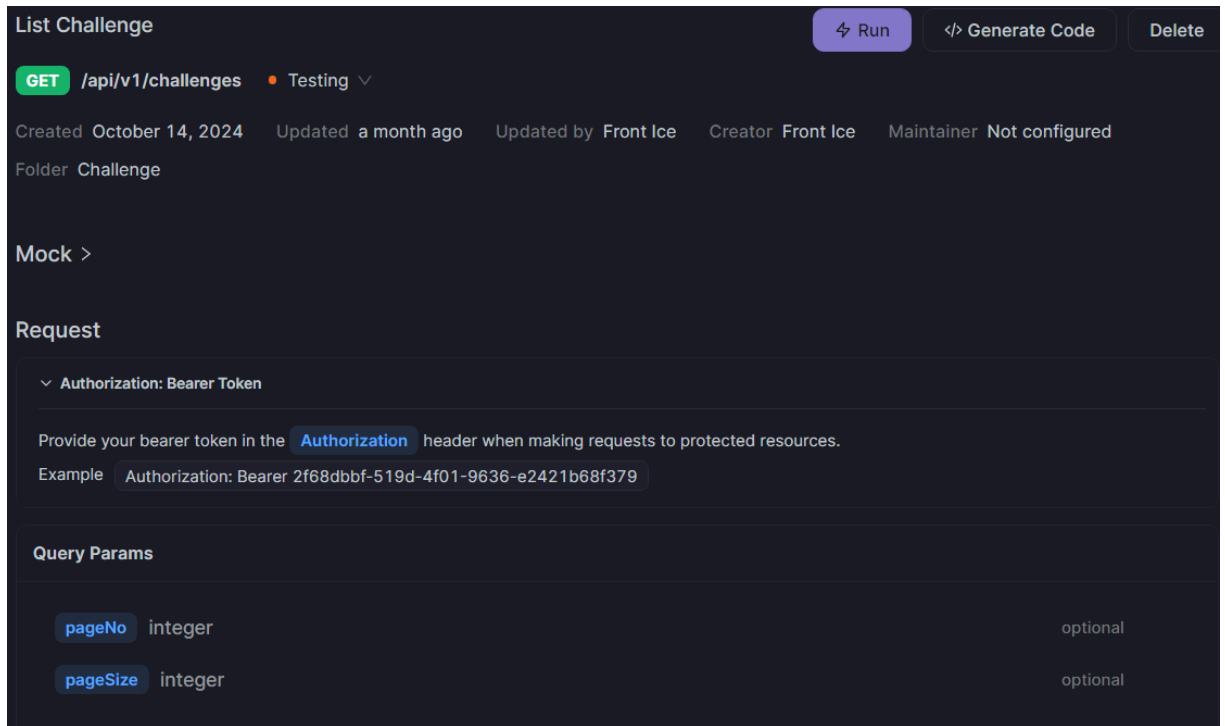


Figure 6 - 31 List Challenge API

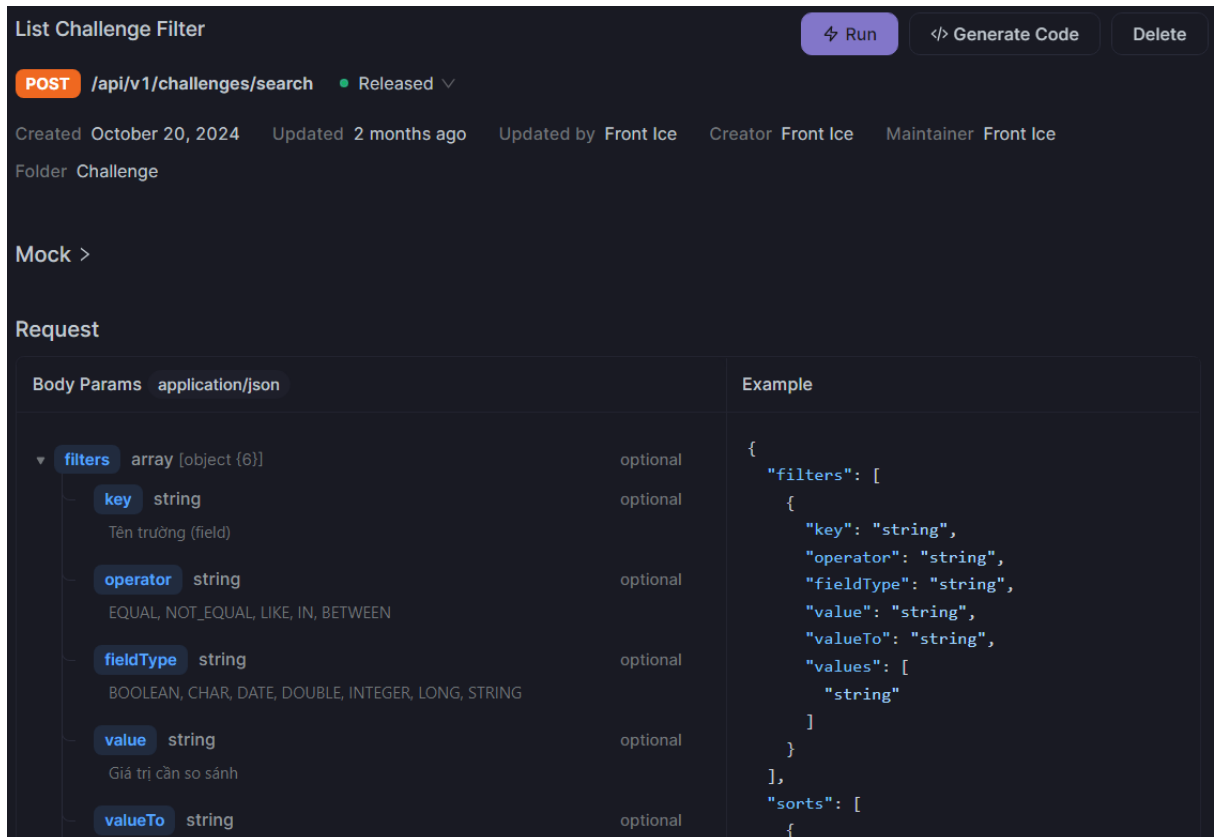


Figure 6 - 32 List Challenge Filter

6.2.8 Solution List Page

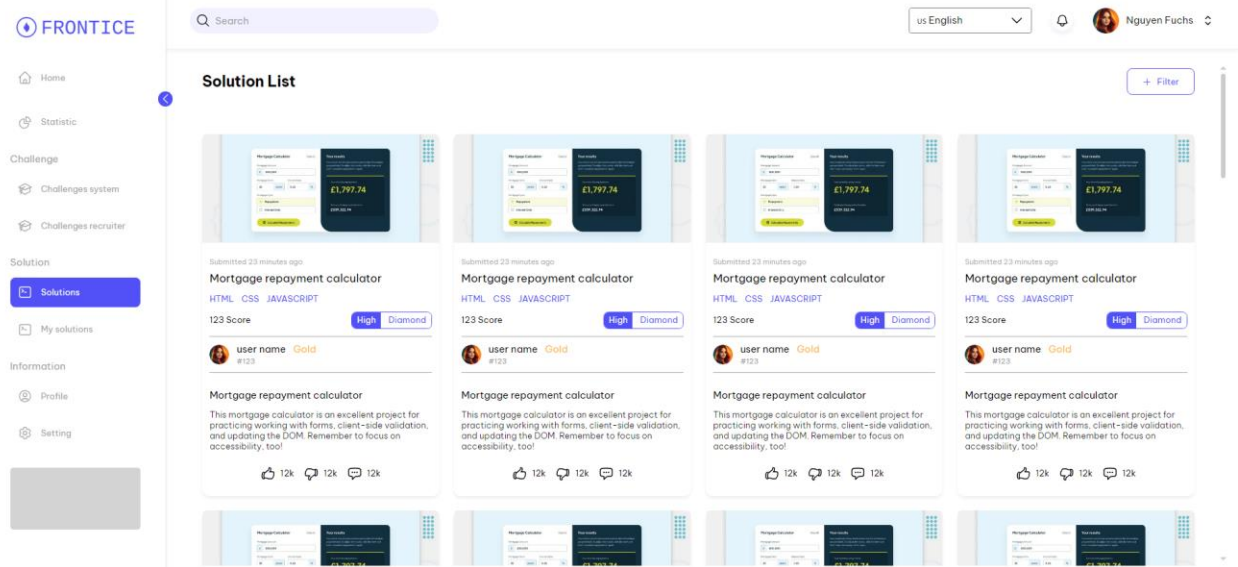


Figure 6 - 33 Solution List Page

Detailed description of the Solution List Page interface:

Table 6 - 10 Description Solution List Page

STT	SIGN	EVENT	DESCRIPTION
1	Window	Initialize	Display the registration page with the main layouts
2	[View Solution]	Click	Go to solution details
3	[Filter]	Click	Show filter to use
4	[Pagination]	Click	Paginate page

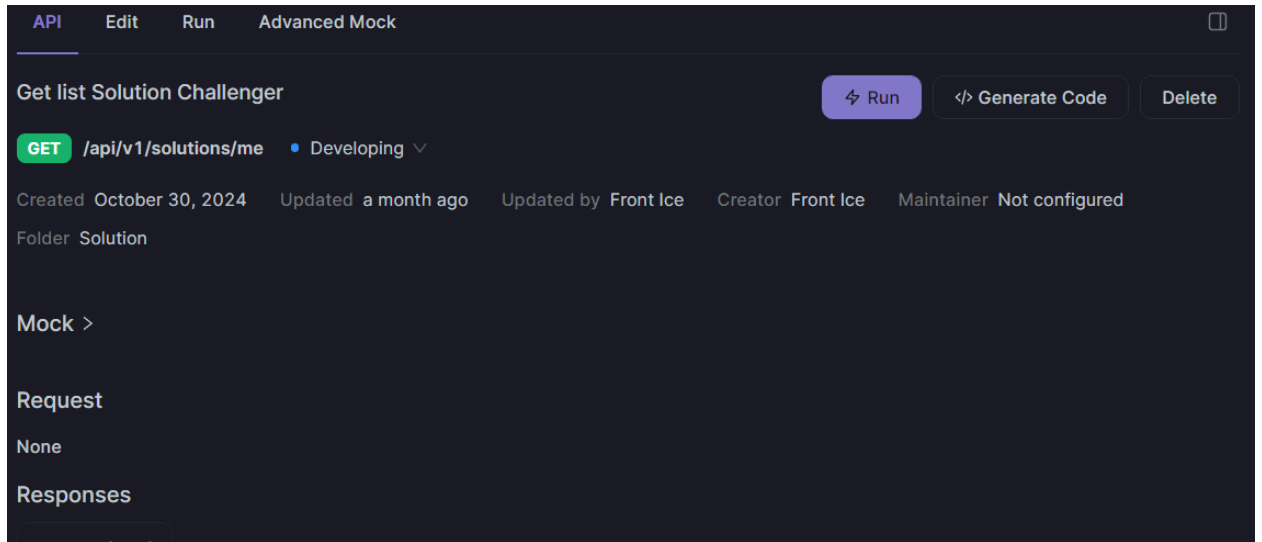


Figure 6 - 34 Solution Challenger API

6.3 Management Page Interface

6.3.1 Login Page

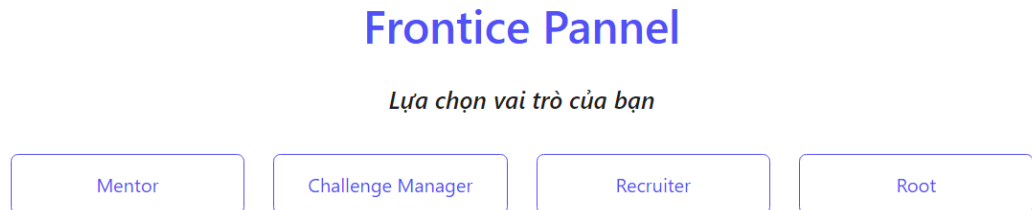


Figure 6 - 35 Login Page Choice Multiple Role

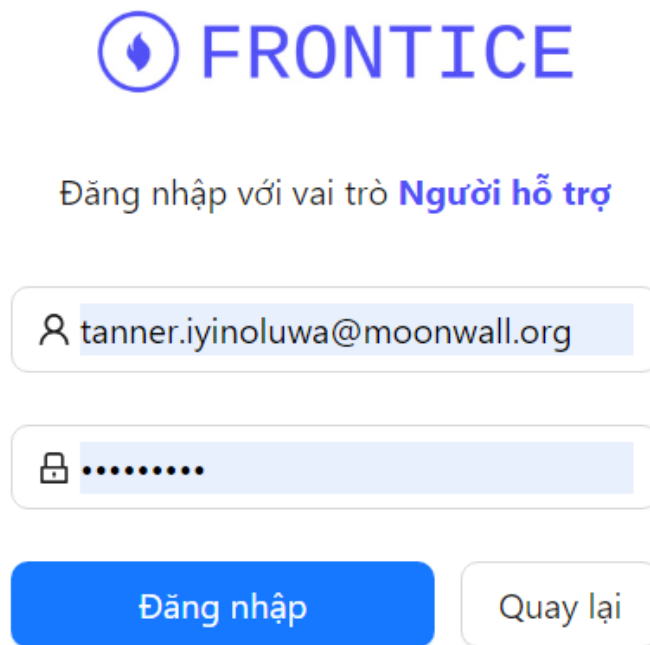


Figure 6 - 36 Login Page1

Detailed description of the Login Page interface:

Table 6 - 11 Description Admin Login Page

STT	SIGN	EVENT	DESCRIPTION
1	Window	Initialize	Display the registration page with the main layouts
2	[Login]	Click	Go to Login

6.3.2 Management Challenge Pages

The screenshot shows the 'Danh sách thử thách' (Challenge List) page in the Frontice application. The page features a sidebar with navigation links: Thử thách, Giải pháp, Nhiệm vụ, and Tài khoản. The main content area displays a table of challenges with columns for STT, Title, CẤP độ, Số điểm, Premium, Tham gia, Hoàn thành, and Hành động. The table lists five challenges, including 'Thử nghiệm', 'Thử thách thẻ tin tức', 'Giao diện thực đơn quán ăn', 'Blog preview card', and 'Mortgage Repayment Calculator'.

STT	Title	CẤP độ	Số điểm	Premium	Tham gia	Hoàn thành	Hành động
1	Thử nghiệm	HTML, Javascript, +1	Bronze	78	×	1	Xem chi tiết
2	Thử thách thẻ tin tức	HTML, CSS, +1	Newbie	29	×	1	Xem chi tiết
3	Giao diện thực đơn quán ăn	HTML, CSS, +1	Newbie	23	×	1	Xem chi tiết
4	Blog preview card	HTML, CSS	Bronze	60	✓	1	Xem chi tiết
5	Mortgage Repayment Calculator	HTML, CSS, +1	Diamond	210	×	0	Xem chi tiết

Figure 6 - 37 Management Challenge Page

This screenshot shows the same 'Danh sách thử thách' page as Figure 6-37, but with a filter overlay on the right side. The filter overlay is titled 'Bộ lọc thử thách' and includes sections for 'Phân loại thử thách', 'Loại thử thách', 'Thời gian đăng tải', and 'Công nghệ sử dụng'. The 'Phân loại thử thách' section has tabs for 'Tất cả', 'Của tôi', and 'Khác'. The 'Loại thử thách' section has a dropdown menu set to 'Tất cả'. The 'Thời gian đăng tải' section has a dropdown menu set to 'Trong khoảng' and input fields for 'Start date' and 'End date'. The 'Công nghệ sử dụng' section has checkboxes for 'JAVASCRIPT', 'HTML', 'CSS', 'JS', and 'CSS'.

Figure 6 - 38 Filter Challenge Page

Detailed description of the Dashboard Page interface:

Table 6 - 12 Description Dashboard Page

STT	SIGN	EVENT	DESCRIPTION
1	Window	Initialize	Display the registration page with the main layouts
2	[Làm mới]	Click	Refresh Page
3	[Tất cả]	Click	Show all of element in page
4	[Tạo thử thách mới]	Click	Create new challenge
5	[Bộ Lọc]	Click	Filter in list challenge
6	[Xem chi tiết]	Click	Move on in Detail Challenge
7	[Phân Trang]	Click	Paginate list challenge
8	[Của tôi]	Click	Show list challenge own account
9	[Người khác]	Click	Show list challenge another account
10	[Xóa]	Click	Delete challenge that account create this challenge.

6.3.3 Create Challenge Page

FRONTICE

Thử thách

Giải pháp

Nhiệm vụ

Tài khoản

Tạo thử thách

Tiêu đề thử thách

Nhập tiêu đề thử thách

Cấp độ thử thách

Chọn cấp độ của thử thách

Mô tả ngắn

Nhập mô tả ngắn cho thử thách

Đăng tải thư mục cung cấp thử thách

Click or drag file to this area to upload

Đăng tải file thiết kế của thử thách

Click or drag file to this area to upload

Support for a single or bulk upload. Strictly prohibited from uploading company data or other banned files.

Đăng tải hình ảnh thử thách

Click or drag file to this area to upload

Support for a single or bulk upload. Strictly prohibited from uploading company data or other banned files.

Đăng tải

Quay lại

Figure 6 - 39 Create Challenge Page

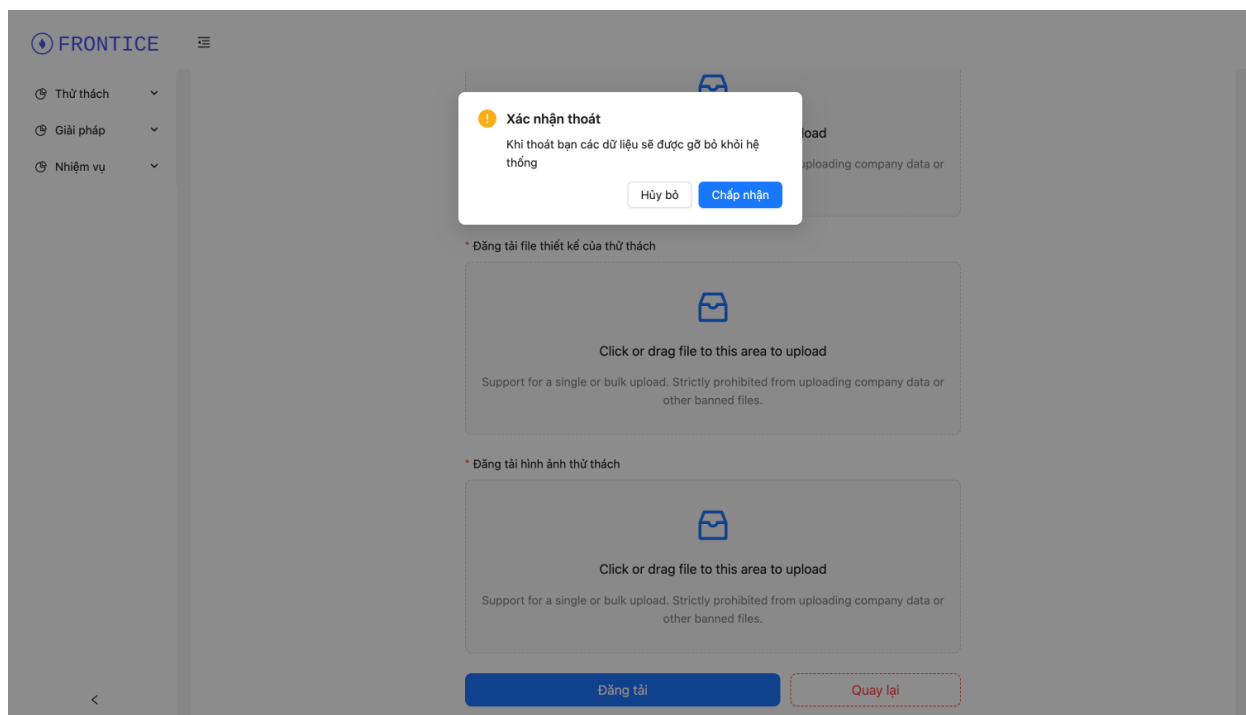


Figure 6 - 40 Exist Button

Detailed description of the Create Challenge Page interface:

Table 6 - 13 Description Create Challenge Page

STT	SIGN	EVENT	DESCRIPTION
1	Window	Initialize	Display the registration page with the main layouts
2	[Đăng tải]	Click	Create Challenge in System
3	[Quay lại]	Click	Cancel the processing of creation

6.3.4 Detail Challenge System Page

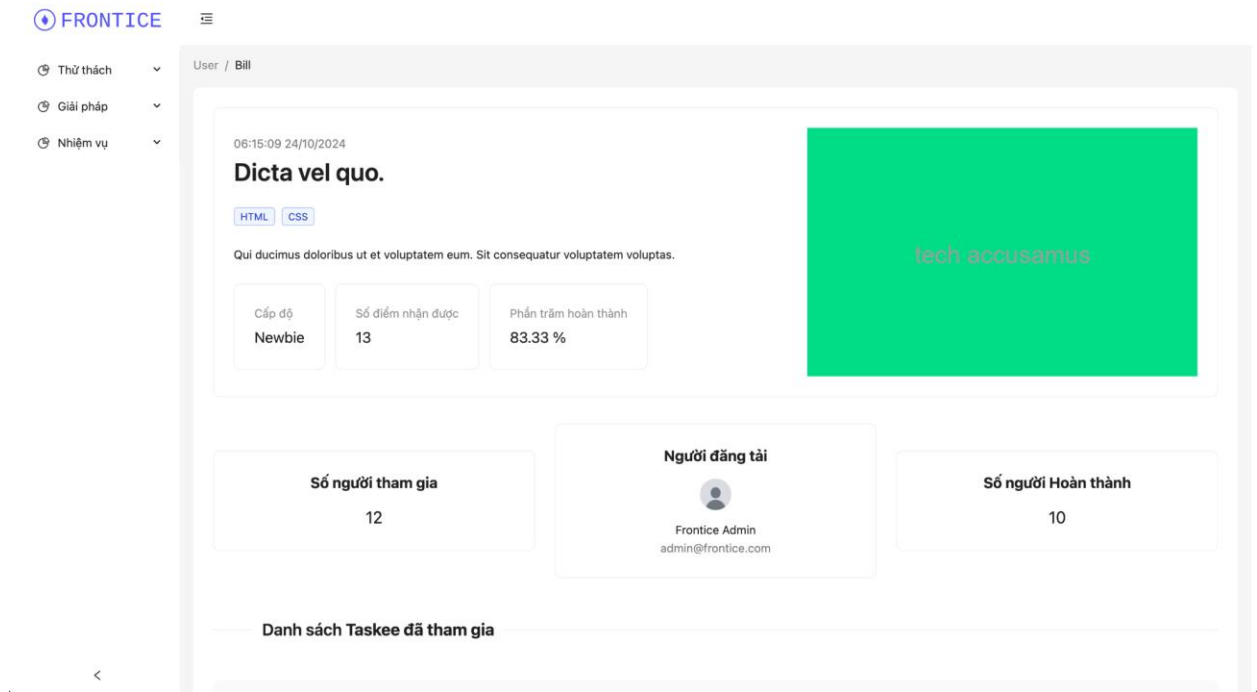


Figure 6 - 41 Detail Challenge System Page

STT	Họ tên	Điểm số	Thông tin	Premium	Hành động
1	Cau Ba Sai gon	13	CV Github profile	X	Xem chi tiết
2	asdasd asdasdasd	13	CV Github profile	✓	Xem chi tiết
3	Khải Nguyễn Văn	13	CV Github profile	X	Xem chi tiết
4	Trần Hoa Nguyễn Văn	13	CV Github profile	✓	Xem chi tiết
5	Trần Hoa Lê Đình	13	CV Github profile	X	Xem chi tiết
6	Khải Đình Văn	13	CV Github profile	✓	Xem chi tiết
7	Trần Hoa Đình	13	CV Github profile	✓	Xem chi tiết
8	Khải Nguyễn Văn	13	CV Github profile	X	Xem chi tiết
9	Trần Hoa Nguyễn Văn	0	CV Github profile	X	Xem chi tiết

Figure 6 - 42 List Challenger attendance

FRONTICE

Thử thách

Giải pháp

Nhiệm vụ

Danh sách Taskee chưa hoàn thành

STT	Họ tên	Điểm số	Thông tin	Premium	Hành động
1	Trần Hoa Nguyễn Văn	0	CV Github profile	×	Xem chi tiết
2	Khải Lê Đình	14	CV Github profile	×	Xem chi tiết

< 1 >

10 / page

Danh sách Taskee đã hoàn thành

STT	Họ tên	Điểm số	Thông tin	Premium	Hành động
1	Cau Ba Sai gon	13	CV Github profile	×	Xem chi tiết
2	asdasd asdasdasd	13	CV Github profile	✓	Xem chi tiết
3	Khải Nguyễn Văn	13	CV Github profile	×	Xem chi tiết
4	Trần Hoa Nguyễn Văn	13	CV Github profile	✓	Xem chi tiết

Figure 6 - 43 List Task not completed

Detailed description of the Details Challenge System Page interface:

Table 6 - 14 Details Challenge System Page

STT	SIGN	EVENT	DESCRIPTION
1	Window	Initialize	Display the registration page with the main layouts
2	[Đăng tải]	Click	Create Challenge in System

6.3.5 Managment Solution Page

FRONTICE

Thử thách

Danh sách

Tạo

Giải pháp

Danh sách

Tổ cáo

Nhiệm vụ

User / Bill

Danh sách giải pháp của Taskee

Làm mới

Bộ lọc

STT	Tiêu đề	Lượt thích	Lượt không thích	Phản hồi của Mentor	Đ	Hành động
1	Quidem delectus quae dolor.	0	0	☐	10:28:	Xem chi tiết
2	Quidem commodi quos et.	0	0	☐	10:28:	Xem chi tiết
3		0	0	☐	10:28:	Xem chi tiết
4	Voluptatem voluptates possimus ut qui.	0	0	☒	06:34:	Xem chi tiết
5	ABCD EFGH IJKLM	0	0	☐	22:02:	Xem chi tiết
6		0	0	☐		Xem chi tiết
7	ABCD EFGH IJKLM	1	0	☐	10:58:	Xem chi tiết

Figure 6 - 44 Management Solution Page

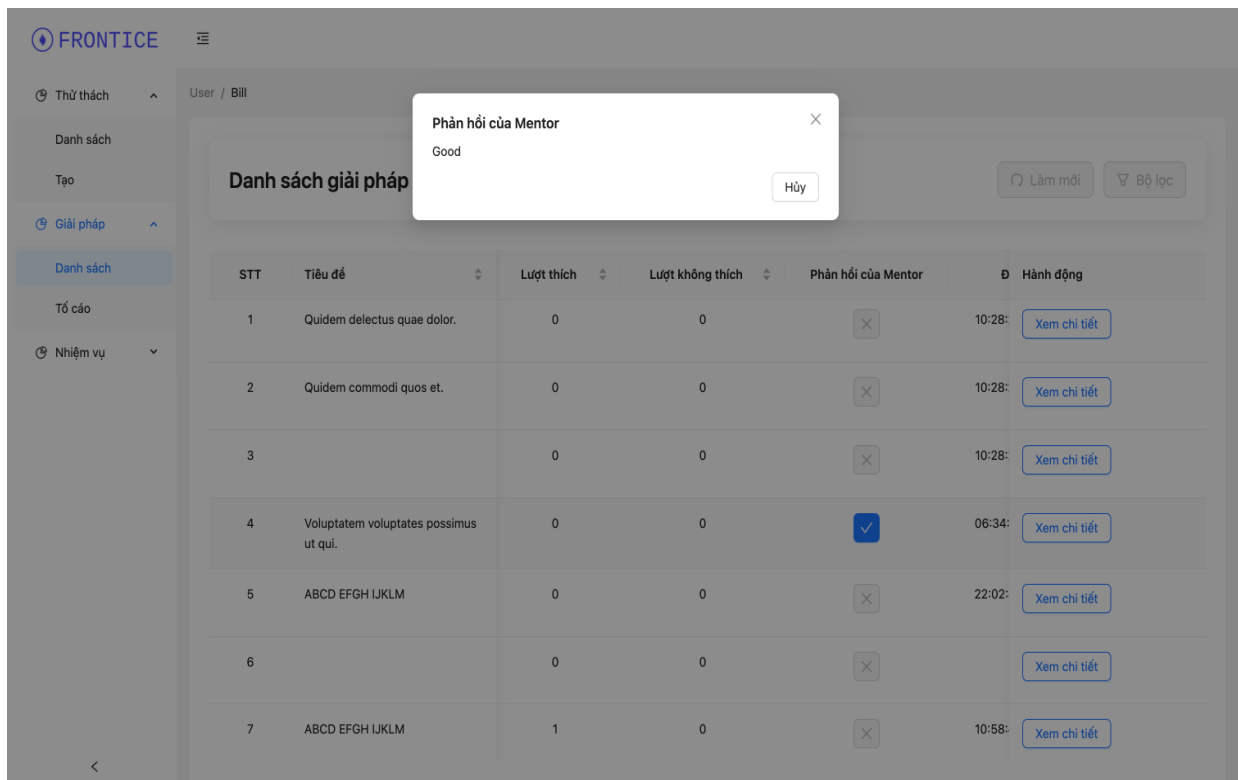


Figure 6 - 45 Inbox of Mentor

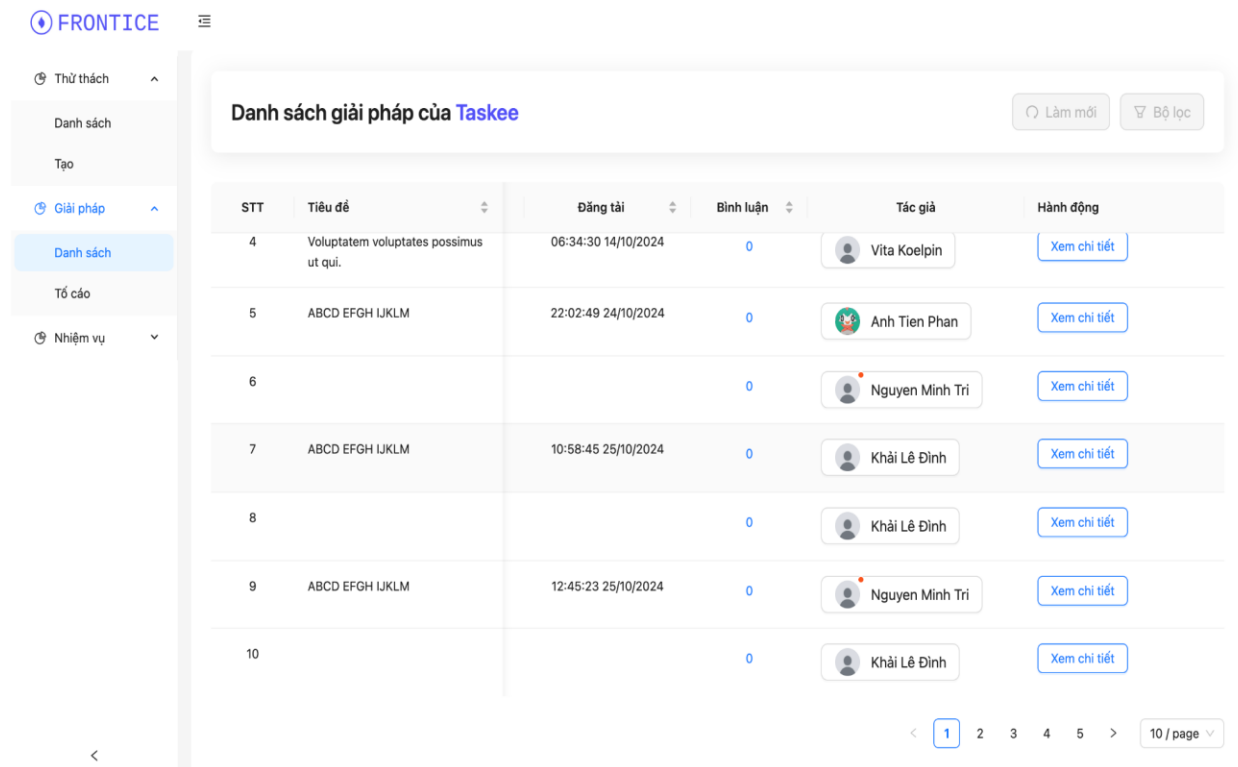


Figure 6 - 46 List of Solution

Detailed description of the Managment Solution Page interface:

Table 6 - 15 Descriptoin Managment Solution Page

STT	SIGN	EVENT	DESCRIPTION
1	Window	Initialize	Display the registration page with the main layouts
2	[Làm mới]	Click	Refresh Page
3	[Tất cả]	Click	Show all of element in page
4	[Bộ Lọc]	Click	Filter in list challenge
5	[Xem chi tiết]	Click	Move on in Detail Challenge
6	[Phân Trang]	Click	Paginate list challenge
7	[Của tôi]	Click	Show list challenge own account
8	[Người khác]	Click	Show list challenge another account
9	[Xóa]	Click	Delete challenge that account create this challenge.

6.3.6 Solution Detail Page

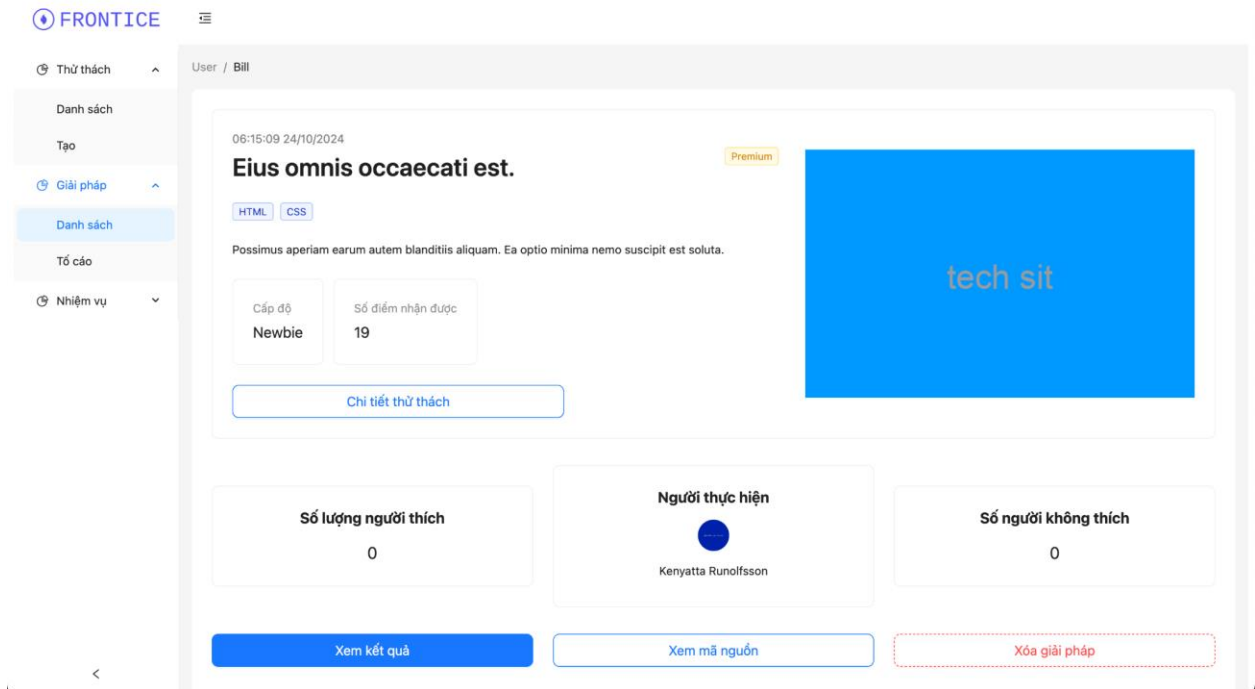


Figure 6 - 47 Solution Details Page

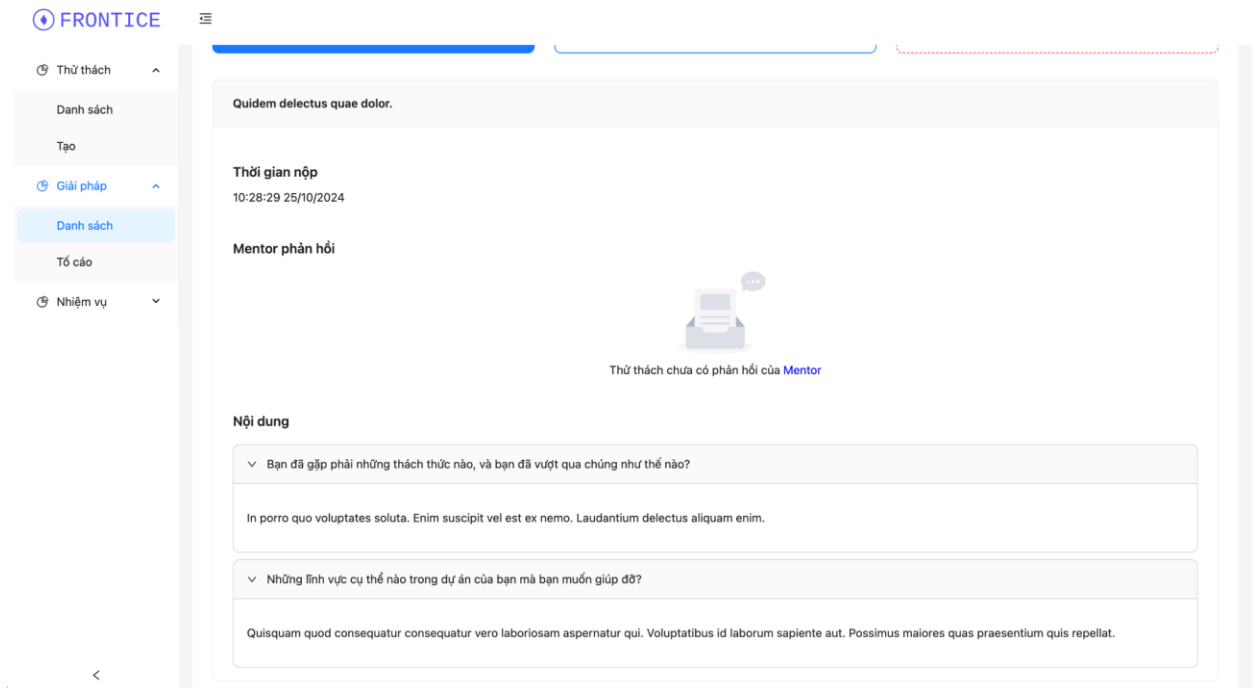


Figure 6 - 48 Solution Details Page

Detailed description of the Detail Solution Page interface:

Table 6 - 16 Description Detail Solution Page

STT	SIGN	EVENT	DESCRIPTION
1	Window	Initialize	Display the registration page with the main layouts
2	[Chi tiết thử thách]	Click	Move on Detail Challenge
3	[Tất cả]	Click	Show all of element in page
4	[Bộ Lọc]	Click	Filter in list challenge
5	[Xem kết quả]	Click	Show result of solution
6	[Phân Trang]	Click	Paginate list solution
7	[Xóa]	Click	Delete solution in system

CHAPTER 7: DEPLOYING IN PRODUCTION

7.1 Platform deploying CI/CD Railway, Gender, Vercel

CI/CD stands for **Continuous Integration (CI)** and **Continuous Deployment (CD)**. This is a software development method that automates the integration of source code, testing, and application deployment. CI/CD improves efficiency, ensures application stability, and minimizes time to release new products

Continuous Integration (CI): This is the process of automatically integrating code from developers into a common repository. Every time a new piece of code is pushed to the repository, the CI system will automatically build and test this code to detect errors early

Continuous Deployment (CD): This is the process of automatically deploying source code that has been successfully tested from the development environment to the production environment. Thanks to CDs, small changes will be put into continuous production without manual intervention

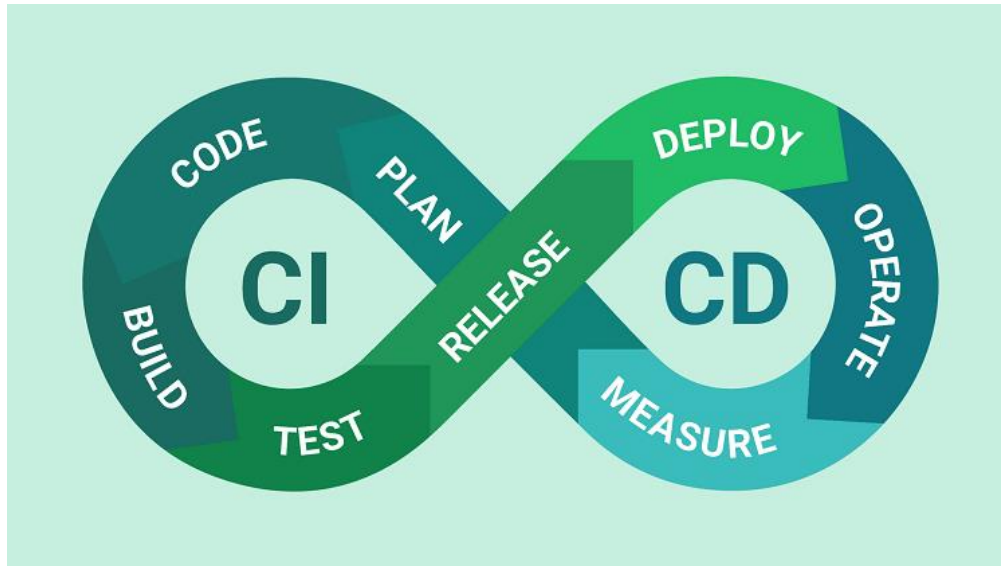


Figure 7 - 1 CI/CD

7.2 Deploy Railway Deployment Environment (Back-end,Database)

Railway is a **Platform as a Service (PaaS) platform**, which allows for easy application deployment and management. Railway supports CI/CD by automatically building and deploying applications when there is a change in the source code from the repository.

Railway supports CI/CD integration through direct connection to the repository, automatically deploying the application every time there is a change.

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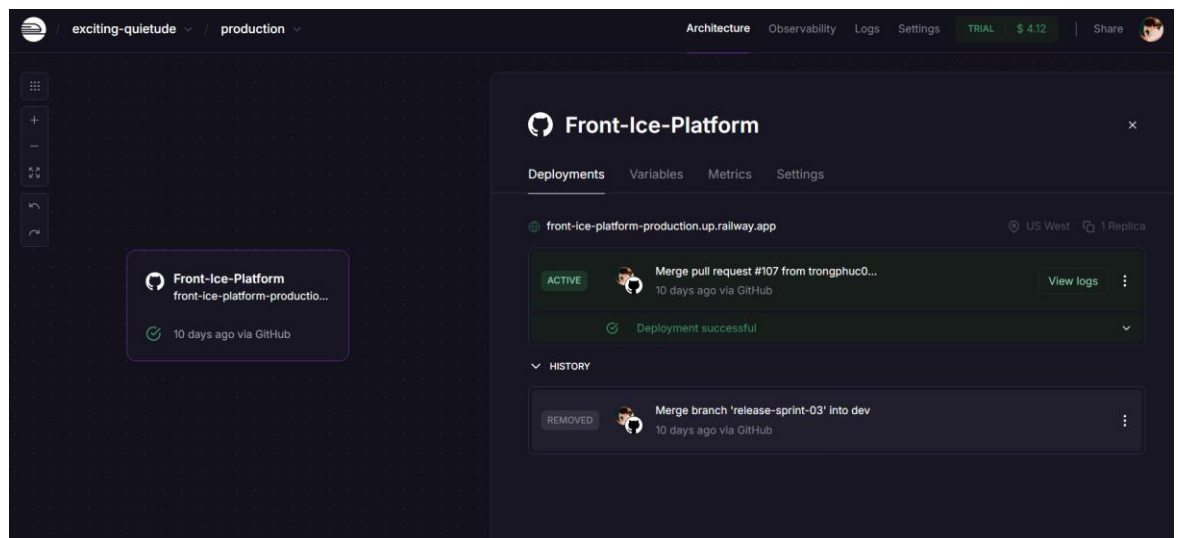


Figure 7 - 2 Railway Deploy Project

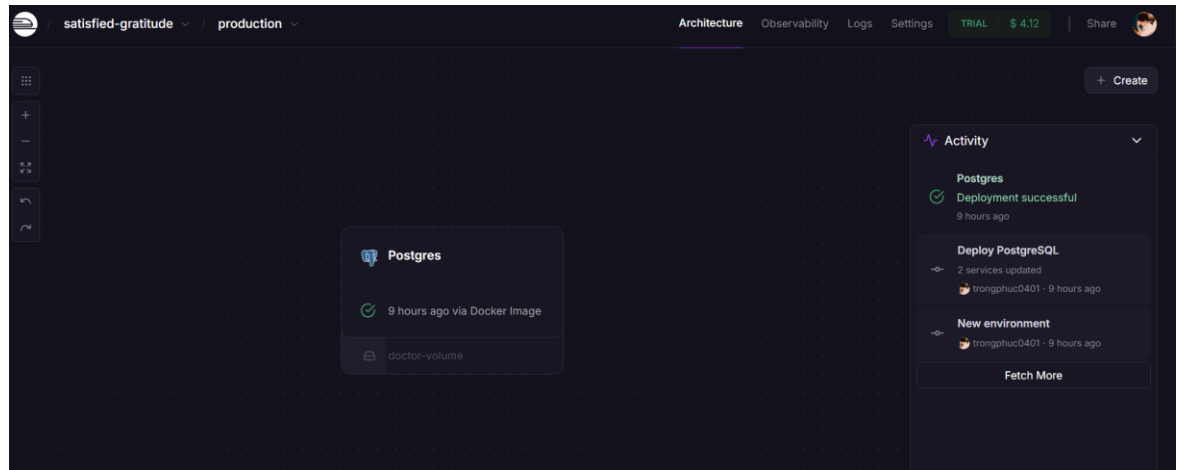


Figure 7 - 3 Railway Deploy Postgres

7.3 Deploy Vercel Deployment Environment (Front-end)

Vercel is a **frontend hosting** platform that stands out for its ability to build and deploy web applications quickly, especially for **Jamstack applications**. Vercel offers CI/CD integration with GitHub, GitLab, and Bitbucket. Every time the source code is pushed to the repository, Vercel will automatically build and deploy a new version of the application

Vercel supports **zero-config** deployment, i.e. no complex configuration, everything is automated.

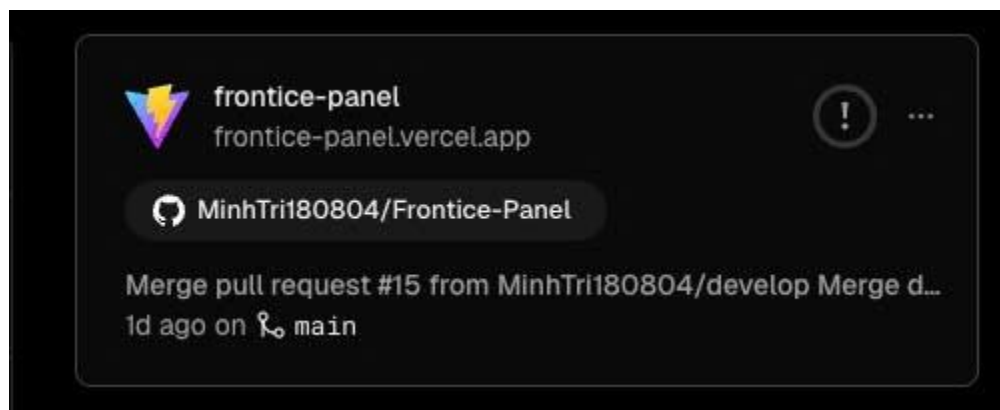


Figure 7 - 4 Deploy Vercel

CHAPTER 8: INCLUDING

1.1 Obtained Result

After the implementation of this Graduation Thesis, the focus has been on implementing the right goal of the project owner, creating an environment for front-end programmers to freely participate and choose the right challenge according to their expertise, language, and specific abilities:

- Successfully building a **BUIDING WEBSITE PROVIDING CHALLENGES AND RECRUITMENT SUPPORT FOR DEVELOPER FRONT-ICE** to meet the requirements set out
- Carefully analyze and understand the business of the system, specifically that this project has been thoroughly analyzed and has FSD (Functional Specification Document)
- Plan and estimate how long a task in the scrum model will take.
- Assign tasks in the entire project using an excel file.
- Build UX/UI in the project.
- Understand how to write APIs in the back end (Spring) and make API calls from the Back-end to the Front-end.
- Understand OOP knowledge to design a system and apply Object Oriented Design to the project
- Write APIs using Spring Framework, test with Postman and API Dog
- Build an interface in a project using Reactjs
- Deploy Back-end, Front-end, Database to the actual environment with Railway, Render
- Collaborate with other programmers in the K-tech Colleague community.

1.2 Development Direction

- Continue to research and develop the website "Front-Ice" to develop more features to help farmers easily access the website
- Building a website is not simply a website but a platform with many devices such as phones and tablets so that developer can easily use it
- Continue to follow the project in the production environment to know how many requests the system can handle during the run.
- Added missing features in excel sheets that were blocked due to time constraints.
- Corrects errors that arise after operation.

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SOURCE CODE

1. Link URL Source Code

Back-end: <https://github.com/trongphuc0401/Front-Ice-Platform>

Front-end:

- Client: <https://github.com/trongphuc0401/Front-Ice-Client>
- Admin: <https://github.com/trongphuc0401/Front-Ice-Management>

2. Link Lively URL

Back-end: <https://front-ice-platform-production.up.railway.app>

Front-end:

- Client: <https://front-ice-client-8zrjhd0e0-trongphuc0401s-projects.vercel.app>
- Admin: <https://frontice-panel.vercel.app>

